Your house: light, sun, heat and wind

Casa Maringá · simulation report · Maringá, PR (lat -23.41, lon -51.93) · 22 October 2026
Weather: typical year from BRA_PR_Maringa.868990_TMYx.2011-2025.epw · annual sun 1,882 kWh/m² · mean temperature 23.0 °C · mean wind 2.3 m/s

How to read this report. Every number comes with a coloured tag telling you whether it is good or bad, on a **level from 1 (bad) to 5 (excellent)**:

Excellent 5/5 Good 4/5 Acceptable 3/5 Poor 2/5 Bad 1/5

Each tag is based on a **Brazilian norm (ABNT)** where one exists, on an **international standard** where Brazil gives no number, or on a clearly labelled **rule of thumb**. The boxes *What counts as good?* at the end of each section show the full scale and the source. The model walls are assumed to be: *Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block)*; roof: *Clay tile + roof void + 12 cm concrete slab ceiling – Old clay tile (heavier, aged)*.
Roof void (attic) ventilation assumed: **5.0 air changes per hour** (common). No measured value for clay-tile roofs over a slab in Brazil was found, so this is an assumption; the table shows how much it matters.

Void air changes/h	Hours too hot (avg room)	Cooling kWh/year
1	1,354	1,782
5	1,210	1,657
10	1,089	1,546

The plan and the window sizes come from the approved municipal drawings (2008, 2012 extension); the materials are estimates, not measurements, and the house as built may differ in details: use the results to compare rooms and options, not as a guarantee.

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1. The short version

Each room is rated on up to nine checks, each on a **level from 1 (bad) to 5 (excellent)**; the OVERALL column is the average level. The next pages explain every check; the cooling column shows what an air conditioner would cost per year:

Room	Daylight (NBR 15575)	Lamps off in daytime	Sun on window	Comfort hours	Hot day	Cold day	Openable window	Cooling cost	OVERALL
Bedroom 1	Superior 5/5	Almost always 5/5	High 5/5	Acceptable 3/5	Intermediate 4/5	Superior 5/5	Meets minimum 3/5	R$ 219/yr Low 4/5	Good 4.3/5
Bathroom 1	Superior 5/5	Almost always 5/5	-	Acceptable 3/5	-	-	-	-	Good 4.3/5
Living room	Superior 5/5	Almost always 5/5	Below minimum 2/5	Good 4/5	Intermediate 4/5	Superior 5/5	Meets minimum 3/5	R$ 325/yr Low 4/5	Good 4.1/5
TV room	Superior 5/5	Almost always 5/5	Below minimum 2/5	Good 4/5	Intermediate 4/5	Superior 5/5	Meets minimum 3/5	R$ 375/yr Low 4/5	Good 4.1/5
Hallway	No requirement 3/5	Almost never 1/5	-	Excellent 5/5	-	-	-	-	Acceptable 3.0/5
Bedroom 2	Superior 5/5	Almost always 5/5	Minimum 3/5	Good 4/5	Intermediate 4/5	Superior 5/5	Comfortable margin 5/5	R$ 146/yr Low 4/5	Good 4.3/5
Bathroom 2	Superior 5/5	Often 4/5	-	Good 4/5	-	-	-	-	Good 4.3/5
Bedroom 3	Superior 5/5	Almost always 5/5	Minimum 3/5	Good 4/5	Intermediate 4/5	Superior 5/5	Comfortable margin 5/5	R$ 142/yr Low 4/5	Good 4.3/5
Kitchen	Superior 5/5	Almost always 5/5	Below minimum 2/5	Acceptable 3/5	-	-	Meets minimum 3/5	R$ 450/yr Moderate 3/5	Good 3.7/5
Outdoor bath	-	Almost always 5/5	-	Acceptable 3/5	-	-	-	-	Good 4.0/5
Outdoor storage	-	Almost always 5/5	-	Acceptable 3/5	-	-	-	-	Good 4.0/5
Whole house								**R$ 1,657/yr**	Good 4.1/5

What works well

- Bedroom 1: 907 lux of daylight at the centre (NBR 15575 superior level)
- Living room: 236 lux of daylight at the centre (NBR 15575 superior level)
- TV room: 549 lux of daylight at the centre (NBR 15575 superior level)
- Bedroom 2: 486 lux of daylight at the centre (NBR 15575 superior level)
- Bedroom 3: 352 lux of daylight at the centre (NBR 15575 superior level)
- Kitchen: 208 lux of daylight at the centre (NBR 15575 superior level)
- Most comfortable room: Hallway (93% of the year in the comfort band; outdoors only 48%)
- External wall heat transmittance U: 2.52 W/m²K meets the limit <= 3.7
- External wall heat capacity: 134.0 kJ/m²K meets the limit >= 130

What deserves attention

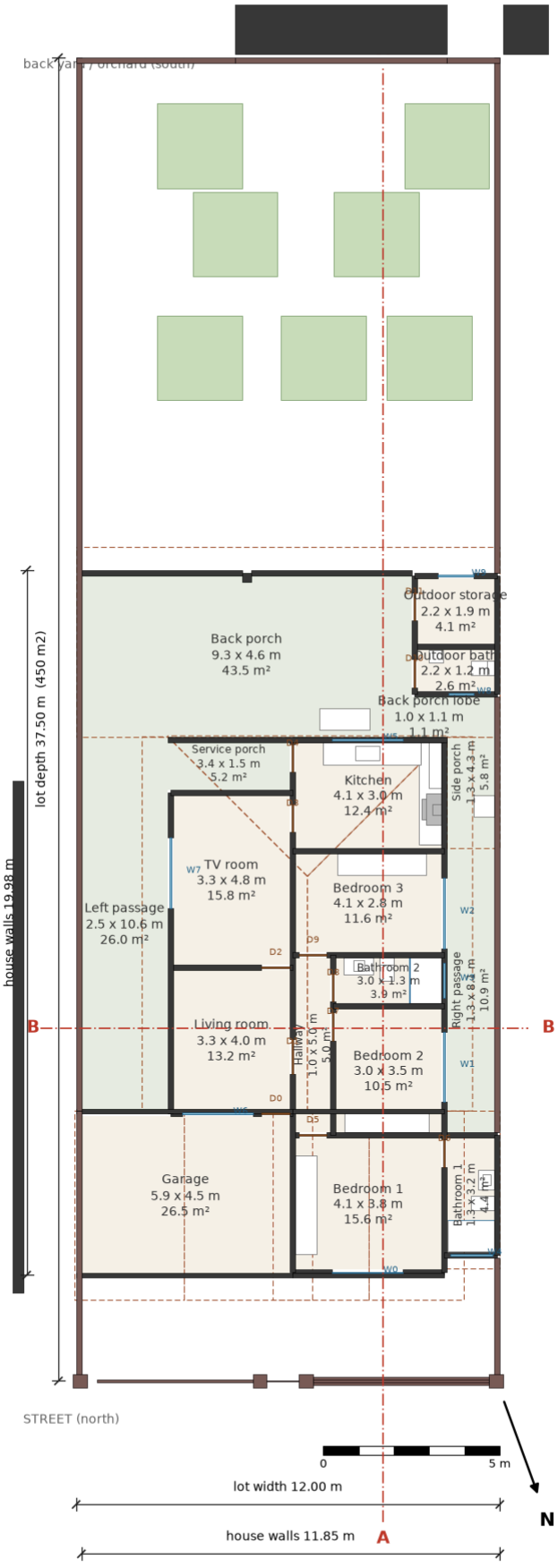
- Least comfortable room: Outdoor bath (68%; 1968 h too hot, 808 h too cold)
- Roof heat transmittance U (indicative): 1.96 W/m²K vs limit <= 1.5

Other studies in this report

Study	Main result	Section
Thermal performance NBR 15575-1:2021 (simulation)	PHFT 54.9% against 47.9% of the reference house; level reached: Minimum.	6.4
Ventilation: air paths, plan changes, hot-air exhaust	Best change: Casement windows + ceiling exhaust (PHFT 58.8% against 54.9%; 8.7 air changes/h in hot hours today).	6.4
Upgrades: insulation, sealing, glass, shading	Largest saving: Rock wool 50 mm; Rubber gaskets + brush seals on frames and doors; Double glazing, low-E; White factory-finished fibre-cement roof (alpha 0.37; DATec 020); Aluminium foil under the tiles (low-emissivity void) (NBR 15220-2); Fixed horizontal brise / overhang 0.6 m; Movable blinds / shutters (closed when the sun is strong) (R$ 1,015 less per year, 6.6 points more comfortable hours).	7
Daylight for green-building rating (LEED / IES LM-83)	sDA300/50% of the house 96%, ASE 13.5%; 3 LEED point(s) from this credit alone.	8
Biophilia: views, 14-pattern matrix, healing index	0 of 6 rooms have a view with sky, vegetation and ground; matrix: 4 patterns in a few areas, 4 only in detail, 1 not included, 5 not evaluated. Healing index 2 of 6 possible (only 3 of 10 components can be calculated).	9.1 - 9.3
Moisture and mould (screening)	With the windows open as modelled, 0 room(s) at risk. In the limit case of windows shut all year without air conditioning: Bedroom 1, Living room, TV room, Bedroom 2, Bedroom 3. Rooms without a moisture source in the model cannot be assessed.	9.4
Suggested materials	Best wall: Double hollow-brick wall (8-hole 10x20x20 cm laid on the larger side) (+889 comfortable h/yr); best roof: Clay tile 2 cm (+306 h/yr). NBR 15220-3 guidelines for the zone included.	6.5

FLOOR PLAN - ground floor

House faces NORTH (street at the bottom) - rooms with name, size and area; sections A-A a



) - sizes from the approved municipal drawings (2008 house, 2012 extension), not a survey of the house as built

Floor plan of the model (north at the bottom).

2. Natural light

Daylight saves electricity and makes rooms pleasant. The Brazilian norm NBR 15575 asks for at least **60 lux** at the centre of living rooms, bedrooms and kitchens (lux = how much light lands on a table). For comparison: a bright overcast day outdoors is thousands of lux.

2.1 Brazilian test: light at the centre of the room, lamps off

Room	Lux at centre	Verdict
Bedroom 1	907	Superior 5/5
Bathroom 1	271	Superior 5/5
Living room	236	Superior 5/5
TV room	549	Superior 5/5
Hallway	0	No requirement 3/5
Bedroom 2	486	Superior 5/5
Bathroom 2	126	Superior 5/5
Bedroom 3	352	Superior 5/5
Kitchen	208	Superior 5/5
Outdoor bath	351	Superior 5/5
Outdoor storage	511	Superior 5/5

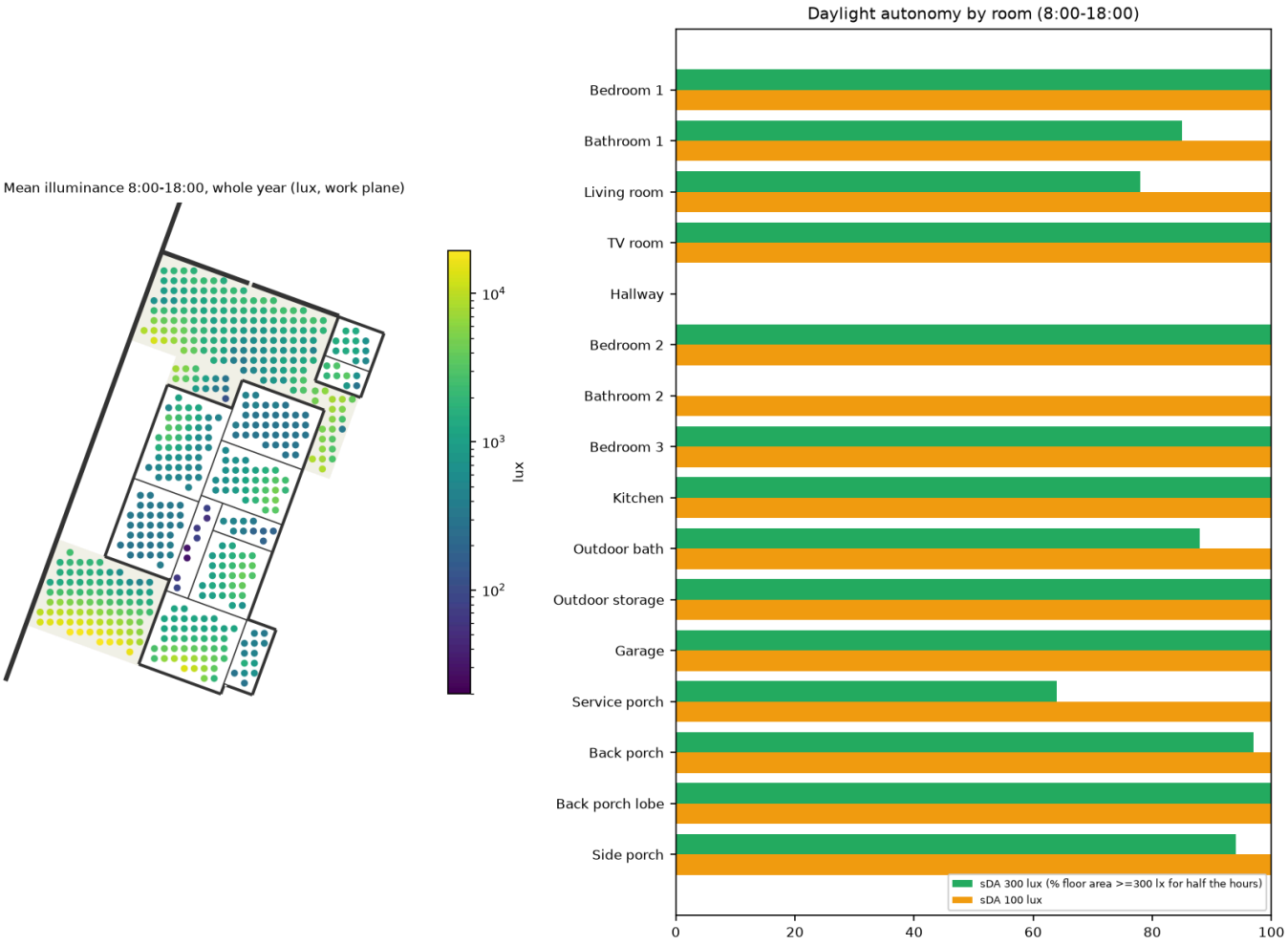
Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10

2.2 How much of the room is bright, and when do I need a lamp?

Room	Goal (lux)	Floor at the goal	Daytime with lamps off	Strong sun (ASE)
Bedroom 1	100	100% Whole room 5/5	98% Almost always 5/5	33% Very high 1/5
Bathroom 1	-	-	88% Almost always 5/5	0% Very low 5/5
Living room	150	100% Whole room 5/5	87% Almost always 5/5	0% Very low 5/5
TV room	150	100% Whole room 5/5	97% Almost always 5/5	7% Acceptable 4/5
Hallway	-	-	1% Almost never 1/5	0% Very low 5/5
Bedroom 2	100	100% Whole room 5/5	97% Almost always 5/5	21% Very high 1/5
Bathroom 2	-	-	71% Often 4/5	0% Very low 5/5
Bedroom 3	100	100% Whole room 5/5	96% Almost always 5/5	19% High 2/5
Kitchen	200	100% Whole room 5/5	88% Almost always 5/5	0% Very low 5/5
Outdoor bath	-	-	94% Almost always 5/5	0% Very low 5/5
Outdoor storage	-	-	98% Almost always 5/5	0% Very low 5/5

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10

Goal = the light level EN 17037 (UK National Annex for dwellings) recommends from daylight: 100 lux bedroom, 150 lux living room, 200 lux kitchen. 'Lamps off' = share of the hours 8:00-18:00 in which the daylight average already reaches the general light level of section 3.

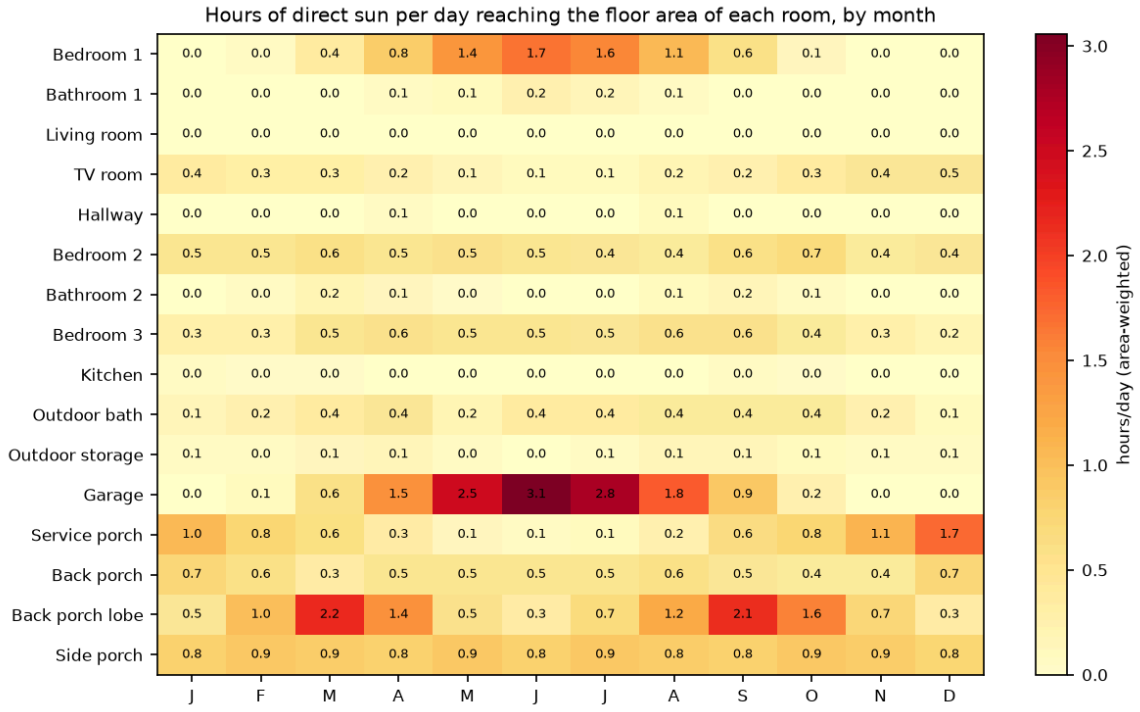


Left: average light on the floor over the year. Right: share of each room that reaches 100 and 300 lux.

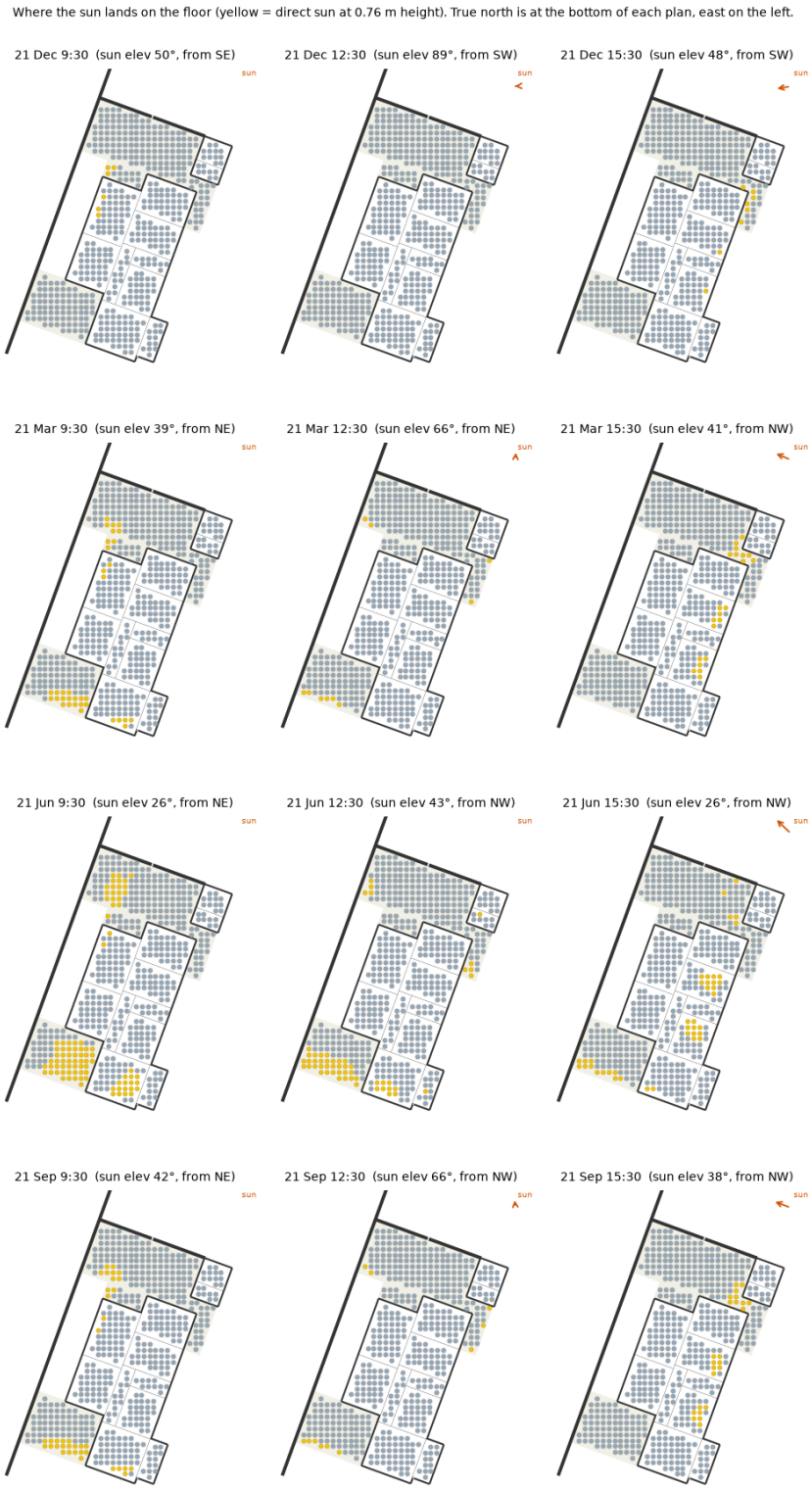
2.3 Sun reaching the windows

Room	Sun on window, hours/day (early spring)	Verdict
Bedroom 1	7.0	High 5/5
Living room	0.0	Below minimum 2/5
TV room	1.0	Below minimum 2/5
Bedroom 2	1.7	Minimum 3/5
Bedroom 3	1.7	Minimum 3/5
Kitchen	0.5	Below minimum 2/5

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10



Hours per day of direct sun that reach the floor of each room, by month (J = January).



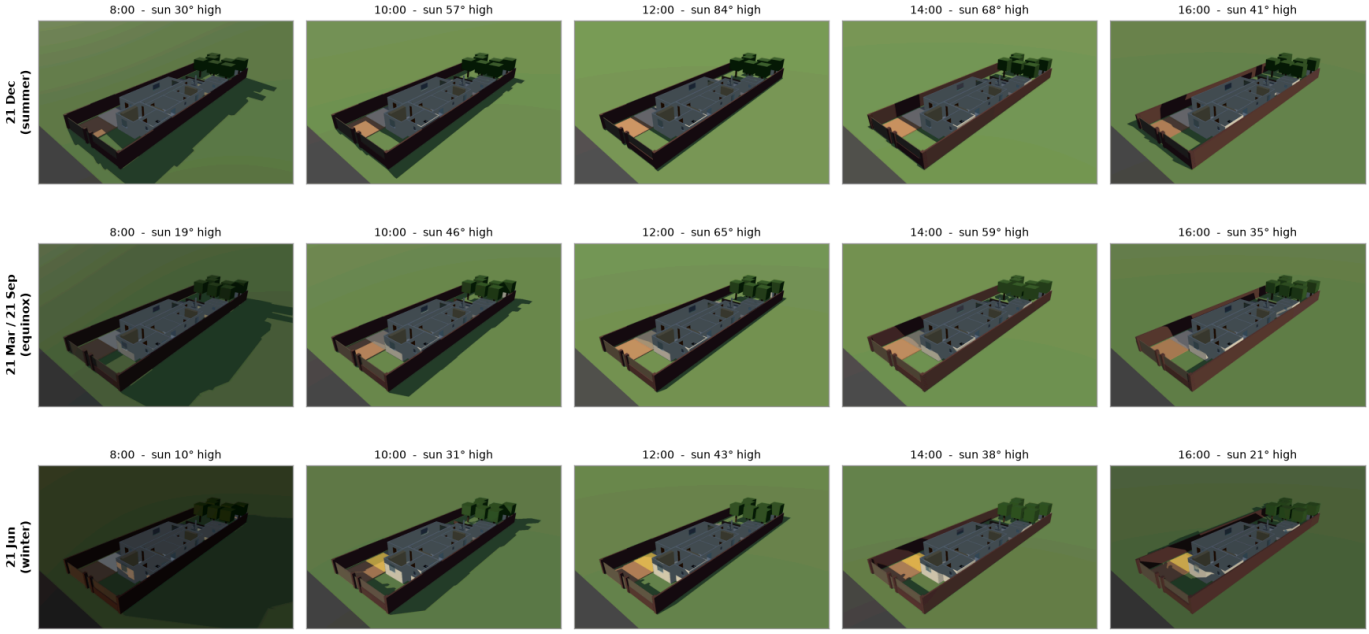
Where the sun lands on the floor on four key dates.

2.4 The sun and its shadows through the day in the 3D model

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10

These pictures come from the same 3D model and sun calculation as the interactive viewer: three key dates, five times of day. Shadows of the house, the roof, the boundary walls and the trees are drawn; the floors that receive direct sun turn yellow/orange. in winter the low sun reaches deep into the rooms that face it, while in summer the sun is high at noon and the roof shades the house.

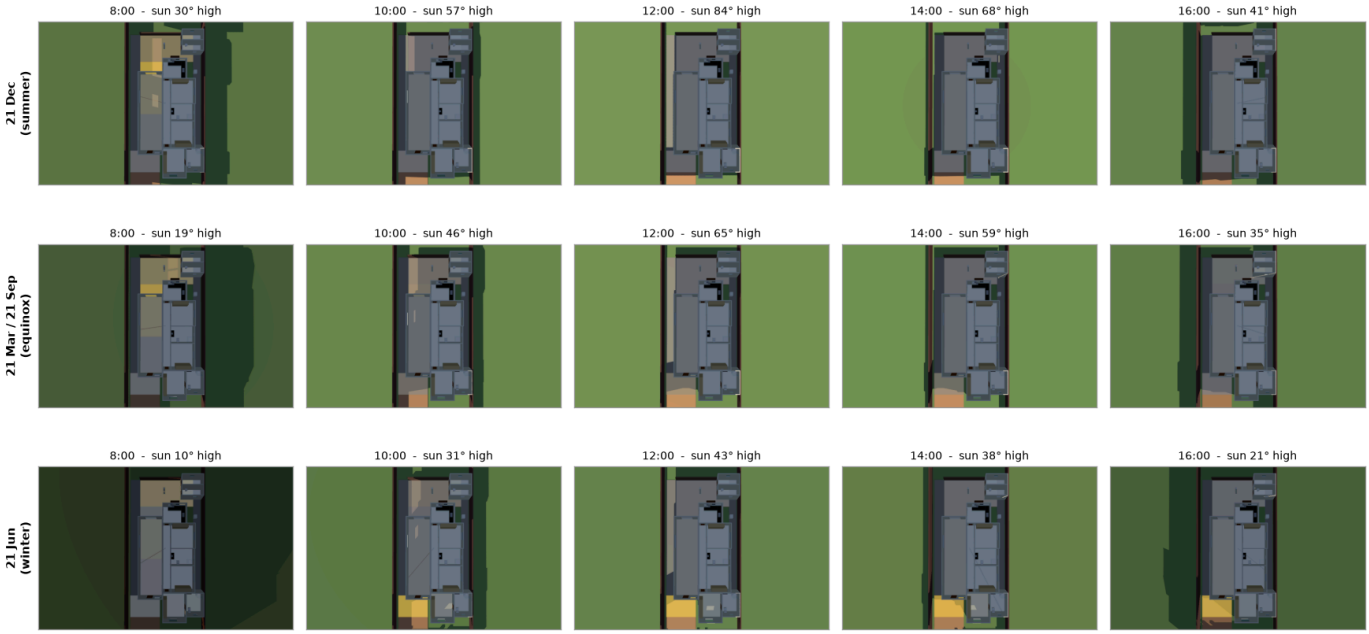
The sun and its shadows through the day on the 3D model (view from the north-west: the north side is nearest to you)



Compass: east is on the left of every picture and west on the right, so the sun rises on the left and sets on the right. Yellow floor = direct sun.

Sun and shadows through the day: 3D view from the north-west (front / street nearest).

The sun and its shadows through the day, seen from above (true north at the bottom)



Compass: east is on the left of every picture and west on the right, so the sun rises on the left and sets on the right. Yellow floor = direct sun.

The same moments seen from above (street / front at the bottom).

What counts as good? (light)

Daylight at the centre of the room (NBR 15575 test) Brazilian norm

Natural light on a table-height plane (0.75 m) at the middle of the room, on 23 April and 23 October at 9:30 and 15:30 with a half-cloudy sky, lamps off. The lowest of the four readings counts. Applies to living rooms, bedrooms, kitchens and service areas.

Fails 1/5	below 60 lux	Below the 60 lux minimum: you need a lamp even at midday.
Minimum 3/5	60 to 90 lux	Meets the legal minimum of the norm.
Intermediate 4/5	90 to 120 lux	Better than the minimum.
Superior 5/5	120 or more lux	Top level of the norm.

This house: Bedroom 1 907 lux Superior 5/5 | Living room 236 lux Superior 5/5 | TV room 549 lux Superior 5/5 | Bedroom 2 486 lux Superior 5/5 | Bedroom 3 352 lux Superior 5/5 | Kitchen 208 lux Superior 5/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements. Our run uses the typical weather of the week around those dates and sky light only (no direct sun), so it is on the safe side.

Daylight at the centre of the room (NBR 15575 test) - bathrooms, hallways, stairs, garages Brazilian norm

Same test as above, but for rooms where people do not stay long.

No requirement 3/5	below 30 lux	The norm does not require daylight here (minimum level: 'not required').
Intermediate 4/5	30 to 45 lux	Meets the intermediate level.
Superior 5/5	45 or more lux	Meets the superior level.

This house: Bathroom 1 271 lux Superior 5/5 | Hallway 0 lux No requirement 3/5 | Bathroom 2 126 lux Superior 5/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements.

Share of the floor that reaches the home target light level International standard

Percentage of the room's floor that gets at least the target light (100 lux bedroom, 150 lux living room, 200 lux kitchen) from daylight alone, for at least half of the daytime hours (8:00-18:00).

Below target 2/5	below 50 % of floor	Less than half of the room reaches the target: lamps will be needed in the daytime.
Meets target 4/5	50 to 95 % of floor	At least half of the floor reaches the target for half of the day.
Whole room 5/5	95 or more % of floor	Practically the entire room reaches the target.

This house: Bedroom 1 100% Whole room 5/5 | Living room 100% Whole room 5/5 | TV room 100% Whole room 5/5 | Bedroom 2 100% Whole room 5/5 | Bedroom 3 100% Whole room 5/5 | Kitchen 100% Whole room 5/5

Source: EN 17037:2018 Daylight in buildings (UK National Annex dwelling targets 100/150/200 lx; sunlight 1.5/3/4 h). EN 17037 asks for the target over at least 50% of the area for half of the daylight hours (+ a lower level over 95%).

Spatial daylight autonomy sDA 300/50% (work-space standard) International standard

Percentage of the floor with at least 300 lux of daylight for at least half of the occupied hours. 300 lux is a WORKING-light level, so this is a strict check: useful for a home office, kitchen or study; bedrooms and hallways are not expected to reach it.

Below recommendation 2/5	below 55 % of floor	Less than 55% of the floor reaches 300 lux for half of the day.
Nominally acceptable 3/5	55 to 75 % of floor	Accepted by the IES / LEED (55% or more).
Preferred 5/5	75 or more % of floor	Preferred level (75% or more).

This house: Bedroom 1 100% Preferred 5/5 | Bathroom 1 85% Preferred 5/5 | Living room 78% Preferred 5/5 | TV room 100% Preferred 5/5 | Hallway 0% Below recommendation 2/5 | Bedroom 2 100% Preferred 5/5 | Bathroom 2 0% Below recommendation 2/5 | Bedroom 3 100% Preferred 5/5 | Kitchen 100% Preferred 5/5 | Outdoor bath 88% Preferred 5/5 | Outdoor storage 100% Preferred 5/5

Source: IES LM-83-12 (sDA and ASE), as used by LEED v4.1.

Annual sunlight exposure ASE (glare and overheating risk) International standard

Percentage of the floor that receives strong direct sun (over 1000 lux) for more than 250 hours a year. High values mean glare, faded furniture and heat.

Very low 5/5	below 3 % of floor	Practically no harsh direct sun.
Acceptable 4/5	3 to 10 % of floor	Within the 10% limit used by IES / LEED.
High 2/5	10 to 20 % of floor	Above the 10% limit: consider curtains, blinds or an eave.
Very high 1/5	20 or more % of floor	Large area with long strong sun: glare and heat likely.

This house: Bedroom 1 33% Very high 1/5 | Bathroom 1 0% Very low 5/5 | Living room 0% Very low 5/5 | TV room 7% Acceptable 4/5 | Hallway 0% Very low 5/5 | Bedroom 2 21% Very high 1/5 | Bathroom 2 0% Very low 5/5 | Bedroom 3 19% High 2/5 | Kitchen 0% Very low 5/5 | Outdoor bath 0% Very low 5/5 | Outdoor storage 0% Very low 5/5

Source: IES LM-83-12 (sDA and ASE), as used by LEED v4.1.

Daytime hours without needing lamps Rule of thumb

Share of the daytime hours in which the room's average daylight already reaches the recommended general light level (see the lamp table), so the lights can stay off.

Almost never 1/5	below 10 % of 8:00-18:00 hours	You will switch the lights on all day.
Rarely 2/5	10 to 25 % of 8:00-18:00 hours	Lamps needed most of the day.
Sometimes 3/5	25 to 50 % of 8:00-18:00 hours	Lamps needed on dull days and mornings / evenings.
Often 4/5	50 to 75 % of 8:00-18:00 hours	Lamps mostly off during the day.
Almost always 5/5	75 or more % of 8:00-18:00 hours	Lamps off nearly the whole day.

This house: Bedroom 1 98% Almost always 5/5 | Bathroom 1 88% Almost always 5/5 | Living room 87% Almost always 5/5 | TV room 97% Almost always 5/5 | Hallway 1% Almost never 1/5 | Bedroom 2 97% Almost always 5/5 | Bathroom 2 71% Often 4/5 | Bedroom 3 96% Almost always 5/5 | Kitchen 88% Almost always 5/5 | Outdoor bath 94% Almost always 5/5 | Outdoor storage 98% Almost always 5/5

Source: Rule of thumb of this study (no norm gives a number); ABNT NBR 5413:1992 (withdrawn in 2013 but still the usual Brazilian reference for HOME lighting levels).

Direct sun on the window in early spring International standard

Hours of direct sun that reach the room's windows on a typical day in early spring: 1 August to 21 September in the southern hemisphere, 1 February to 21 March in the northern one (the EN 17037 dates). Sun in winter / early spring is welcome for health and warmth.

Below minimum 2/5	below 1.5 hours/day	Less than 1.5 h of sun: the room feels sunless.
Minimum 3/5	1.5 to 3 hours/day	Meets the EN 17037 minimum (1.5 h).
Medium 4/5	3 to 4 hours/day	Meets the EN 17037 medium level (3 h).
High 5/5	4 or more hours/day	Meets the EN 17037 high level (4 h).

This house: Bedroom 1 7.0 h High 5/5 | Living room 0.0 h Below minimum 2/5 | TV room 1.0 h Below minimum 2/5 | Bedroom 2 1.7 h Minimum 3/5 | Bedroom 3 1.7 h Minimum 3/5 | Kitchen 0.5 h Below minimum 2/5

Source: EN 17037:2018 Daylight in buildings (UK National Annex dwelling targets 100/150/200 lx; sunlight 1.5/3/4 h). EN 17037 asks this for at least one main room (preferably the living room); bathrooms and hallways are exempt.

3. Lamps: how much light to install (lumens)

Lux is the light that reaches a surface; **lumens** is the light a lamp gives out. To get an average of X lux over a floor of A m² you need X × A ÷ (utilisation × maintenance = 0.45 × 0.8 = 0.36) lumens. The 'minimum' column is the bare lux × m².

3.1 Recommended light levels at home

Kind of room	General light (lux): min / recommended / max	Task light (lux)	Source
Living / dining room	100 / 150 / 200	300 / 500 / 750 - reading, writing, sewing	NBR 5413 (home); EN 12464-1 / NBR 8995-1 for reading tasks
Bedroom	100 / 150 / 200	200 / 300 / 500 - reading in bed, dressing table	NBR 5413 (home)
Kitchen	100 / 150 / 200	200 / 300 / 500 - stove, sink and worktop (EN 12464-1 asks 500)	NBR 5413 (home); EN 12464-1
Bathroom	100 / 150 / 200	200 / 300 / 500 - mirror (shaving, make-up)	NBR 5413 (home)
Hallway / circulation	75 / 100 / 150	-	NBR 5413 (home)
Laundry / storage / outdoor bath	100 / 150 / 200	200 / 300 / 500 - laundry and tools	NBR 5413 (home)
Porch / carport (covered outdoor)	50 / 75 / 100	-	NBR 5413 (garages 50-75-100)

Calculated with: **3D model (IFC) and tables** · norm tables, no calculation · details and reliability in section 10

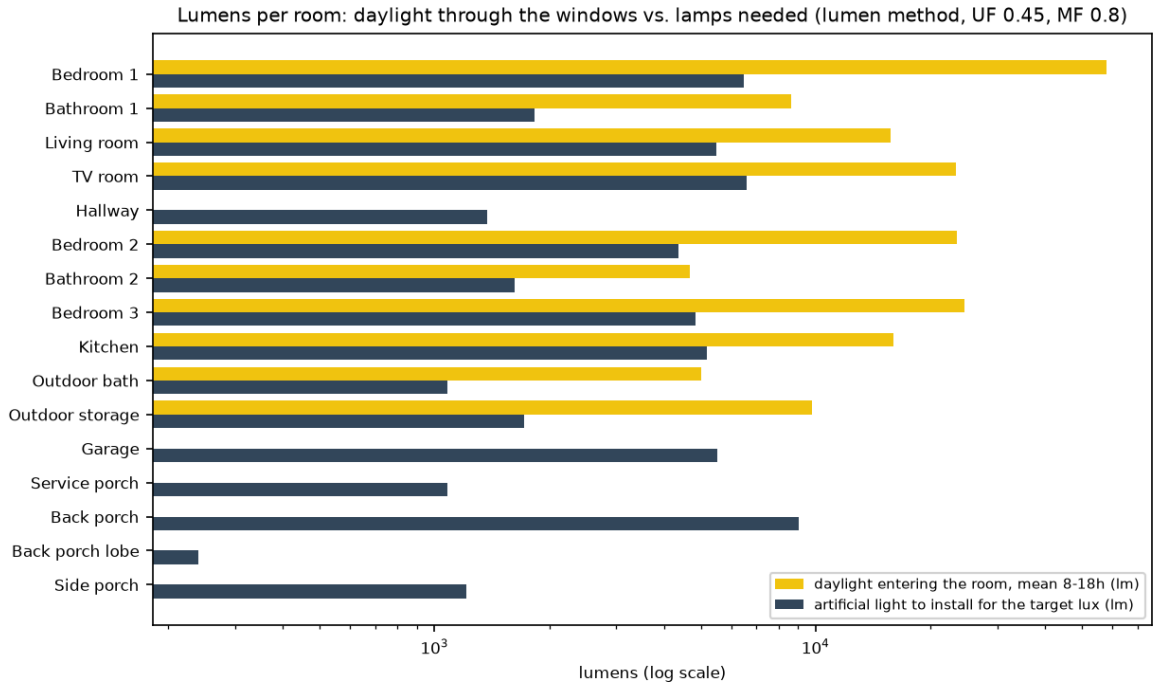
Reading: light the room to the general level (100-200 lux) and add a directional **reading lamp of about 450-800 lumens** (about 4-8 W LED) near the page to reach **300-500 lux** on it (NBR ISO/CIE 8995-1 and EN 12464-1 for reading and writing; NBR 5413 for homes). Reading under only the 100-150 lux of general light strains the eyes. For scale: 100 lm = night light / small candle-type; 450 lm = equivalent of an old 40 W bulb; 800 lm = equivalent of an old 60 W bulb; 1600 lm = equivalent of an old 100 W bulb.

3.2 Lamp lumens for each room of this house

Room	m²	General light goal	Task light (reading etc.)	Lamp lumens: minimum - design	LED watts (design)
Bedroom 1	15.6	150 lux	300 lux (reading in bed, dressing table)	2,334 - 6,484	65
Bathroom 1	4.4	150 lux	300 lux (mirror (shaving, make-up))	658 - 1,828	18
Living room	13.2	150 lux	500 lux (reading, writing, sewing)	1,980 - 5,500	55
TV room	15.8	150 lux	500 lux (reading, writing, sewing)	2,376 - 6,600	66
Hallway	5.0	100 lux	-	495 - 1,375	14
Bedroom 2	10.5	150 lux	300 lux (reading in bed, dressing table)	1,575 - 4,375	44
Bathroom 2	3.9	150 lux	300 lux (mirror (shaving, make-up))	585 - 1,625	16
Bedroom 3	11.6	150 lux	300 lux (reading in bed, dressing table)	1,743 - 4,842	48
Kitchen	12.5	150 lux	300 lux (stove, sink and worktop (EN 12464-1 asks 500))	1,868 - 5,188	52
Outdoor bath	2.6	150 lux	300 lux (laundry and tools)	389 - 1,082	11
Outdoor storage	4.1	150 lux	300 lux (laundry and tools)	617 - 1,714	17

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10

Use the low end for light walls and ceilings and good fittings; the high end for dark colours or recessed spots. LED at 100 lm per watt. Daylight already covers part of the day (section 2.2).



Daylight entering through the windows (yellow) compared with the lamp lumens needed for the target light (dark).

4. Windows, fresh air and wind

NBR 15575-4 asks that the openable part of the windows is at least **7% of the floor area** of each living room and bedroom, and limits glass to 20% of the floor unless it is shaded. NBR 15220-3 recommends wind-driven cross-ventilation as the main summer strategy for this climate.

4.1 Window size

Room	Floor m²	Glass m²	Glass / floor	Openable / floor
Bedroom 1	15.6	3.00	19.3% Ideal 5/5	9.6% Meets minimum 3/5
Bathroom 1	4.4	0.72	16.4%	8.2%
Living room	13.2	2.40	18.2% Ideal 5/5	9.1% Meets minimum 3/5
TV room	15.8	2.40	15.2% Ideal 5/5	7.6% Meets minimum 3/5
Hallway	5.0	0.00	0.0%	0.0%
Bedroom 2	10.5	2.40	22.9% Large 4/5	11.4% Comfortable margin 5/5
Bathroom 2	3.9	0.60	15.4%	7.7%
Bedroom 3	11.6	2.40	20.7% Large 4/5	10.3% Comfortable margin 5/5
Kitchen	12.5	2.40	19.3% Ideal 5/5	9.6% Meets minimum 3/5
Outdoor bath	2.6	0.49	18.9%	9.4%
Outdoor storage	4.1	1.00	24.3%	12.2%

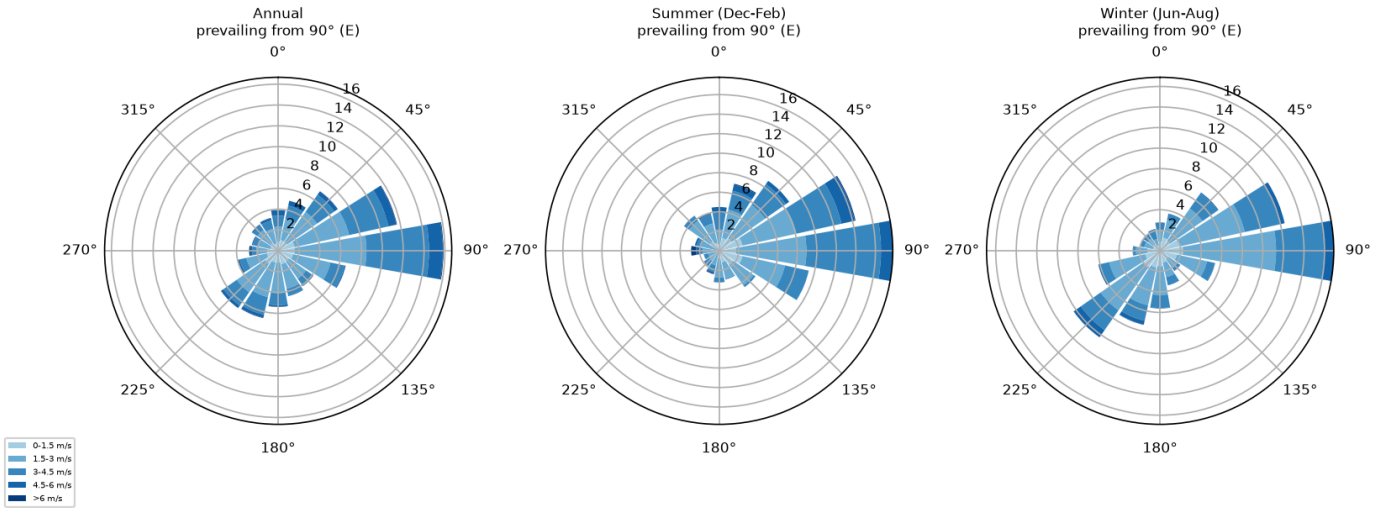
Calculated with: **3D model (IFC) and tables** · sizes read from the IFC model · details and reliability in section 10

Window sizes and heights are those of the approved drawings (bedrooms, living rooms and kitchen 2.00 x 1.20 m with sill 1.20 m, suite 2.00 x 1.50 m with sill 0.90 m, bathrooms 1.00 x 0.60 and 1.20 x 0.60 m; 2012 annex 1.00 x 1.00 and 0.70 x 0.70 m); a sliding window is assumed to open half of its area. If the windows as built differ, change them and rerun the analysis.

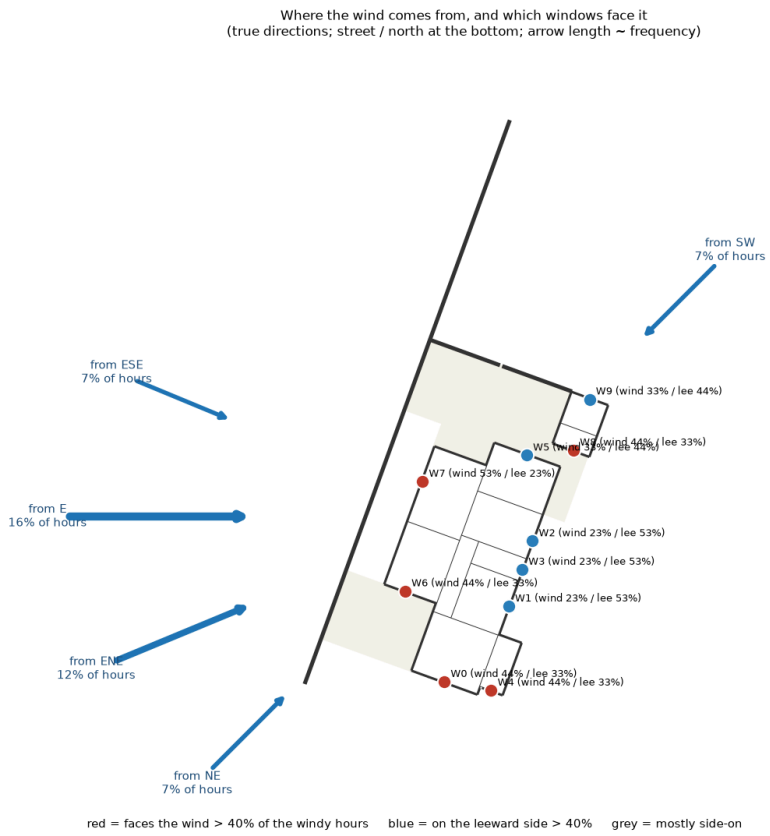
4.2 Cross-ventilation

Room	Windy hours with a through-draught
Bedroom 1	36% Some 3/5
Bathroom 1	0%
Living room	100% Excellent 5/5
TV room	77% Excellent 5/5
Hallway	76%
Bedroom 2	36% Some 3/5
Bathroom 2	36%
Bedroom 3	36% Some 3/5
Kitchen	77% Excellent 5/5
Outdoor bath	0%
Outdoor storage	0%

Calculated with: **Fast engine (Python, this project)** · wind-hours rule from the weather file; no pressure calculation (the pressure-based ventilation is in section 6.4) · details and reliability in section 10



Wind roses: where the wind comes from and how strong, whole year, summer and winter.



Wind directions against the windows of the house (red windows face the wind, blue are on the sheltered side).

What counts as good? (windows and wind)

Openable window area for ventilation Brazilian norm

Area of windows that can really open (glass, frames and fixed leaves are discounted), divided by the floor area of the room. Zones 1-7 need at least 7%.

Fails 1/5	below 7 % of floor area	Less than the 7% minimum of the norm (unless the city's building code says otherwise).
Meets minimum 3/5	7 to 10 % of floor area	Meets the 7% minimum.
Comfortable margin 5/5	10 or more % of floor area	More than the minimum: good air changes.

This house: Bedroom 1 9.6% Meets minimum 3/5 | Living room 9.1% Meets minimum 3/5 | TV room 7.6% Meets minimum 3/5 | Bedroom 2 11.4% Comfortable margin 5/5 | Bedroom 3 10.3% Comfortable margin 5/5 | Kitchen 9.6% Meets minimum 3/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas. We assume a sliding window opens about half of its area; a casement / tilting window opens more.

Glass area compared with the floor Brazilian norm

Window area divided by floor area. NBR 15575-4 limits transparent areas to 20% of the floor (rooms up to 20 m2) unless there is shading or high-performance glass; NBR 15220-3 recommends 15-25% openings for zones 1 to 6 (medium openings).

Very small 2/5	below 7 % of floor area	Too little window for light and air.
Small 3/5	7 to 15 % of floor area	Works, but a little dark or stuffy.
Ideal 5/5	15 to 20 % of floor area	Inside the NBR 15575-4 limit and the NBR 15220-3 recommendation.
Large 4/5	20 to 25 % of floor area	Recommended by NBR 15220-3 if shaded; above the NBR 15575-4 20% unless shaded.
Too large 2/5	25 or more % of floor area	More sun heat than the norms accept without shading.

This house: Bedroom 1 19.3% Ideal 5/5 | Living room 18.2% Ideal 5/5 | TV room 15.2% Ideal 5/5 | Bedroom 2 22.9% Large 4/5 | Bedroom 3 20.7% Large 4/5 | Kitchen 19.3% Ideal 5/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas; ABNT NBR 15220-3: Brazilian bioclimatic zoning and building guidelines (zones and limits depend on the bioclimatic zone of the project).

Cross-ventilation possible Rule of thumb

Share of the windy hours in which the wind hits a window on one side of the connected rooms and another window on the opposite side can let it out (doors assumed open).

Almost none 1/5	below 5 % of windy hours	Air hardly flows through the room.
Weak 2/5	5 to 20 % of windy hours	Cross-ventilation seldom happens.
Some 3/5	20 to 40 % of windy hours	Works in part of the windy hours.
Good 4/5	40 to 60 % of windy hours	Works in many windy hours.
Excellent 5/5	60 or more % of windy hours	A through-draught is possible most of the time.

This house: Bedroom 1 36% Some 3/5 | Living room 100% Excellent 5/5 | TV room 77% Excellent 5/5 | Bedroom 2 36% Some 3/5 | Bedroom 3 36% Some 3/5 | Kitchen 77% Excellent 5/5

Source: Rule of thumb of this study (no norm gives a number).

Average outdoor wind speed International standard

Average wind at the site over the year (weather file, open terrain at 10 m; at the house it is lower).

Calm 2/5	below 1.0 m/s	Little natural ventilation from the wind alone.
Light air 3/5	1.0 to 2.0 m/s	Gentle; openings help but cross-ventilation is weak.
Light to gentle breeze 5/5	2.0 to 5.0 m/s	Ideal range for natural ventilation.
Moderate breeze 4/5	5.0 to 8.0 m/s	Strong enough to need wind protection outdoors.
Fresh breeze or more 2/5	8.0 or more m/s	Wind damage and discomfort risk.

This house: Maringá, PR, whole year 2.3 m/s Light to gentle breeze 5/5

Source: WMO Beaufort wind scale; Rule of thumb of this study (no norm gives a number). Beaufort 1 = 0.3-1.5 m/s, 2 = 1.6-3.3, 3 = 3.4-5.4, 4 = 5.5-7.9 m/s. Indoors ASHRAE-55 accepts air speeds up to about 0.8 m/s.

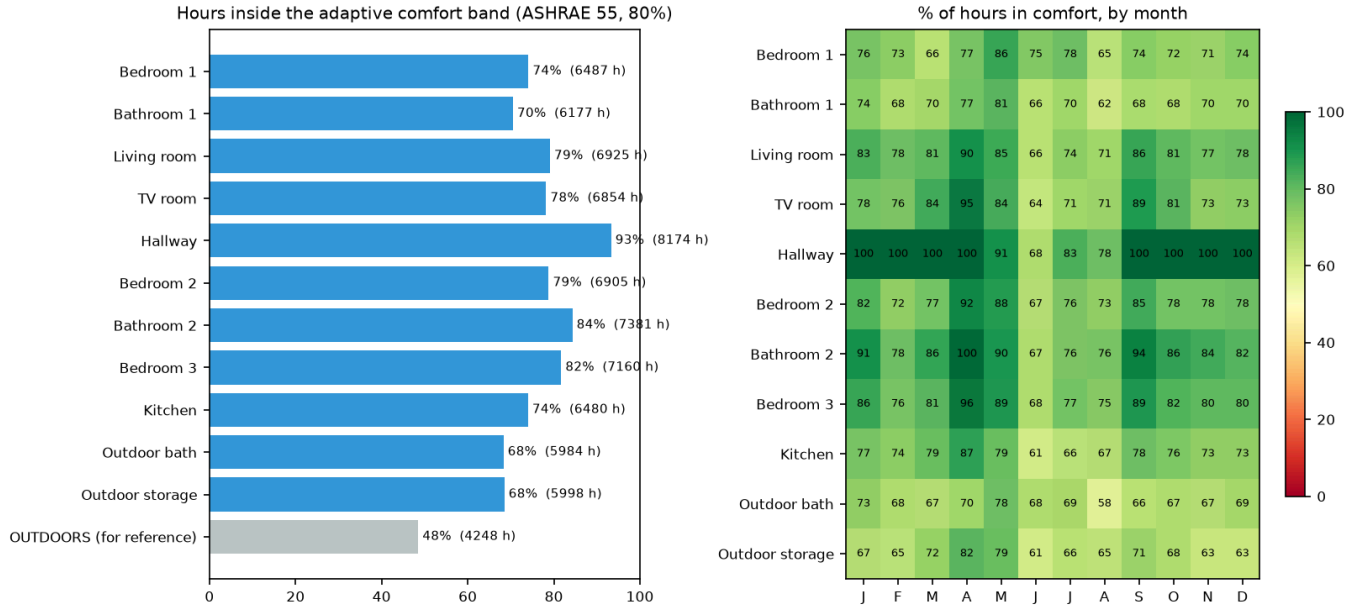
5. Temperature and comfort

Without air conditioning, comfort depends on the walls, roof, windows, shade and ventilation. The simulation computes the indoor temperature of every room for each of the 8,760 hours of a typical year. **Comfortable** means inside the ASHRAE-55 adaptive comfort band (it moves with the weather: people accept warmer rooms when it is warmer outside).

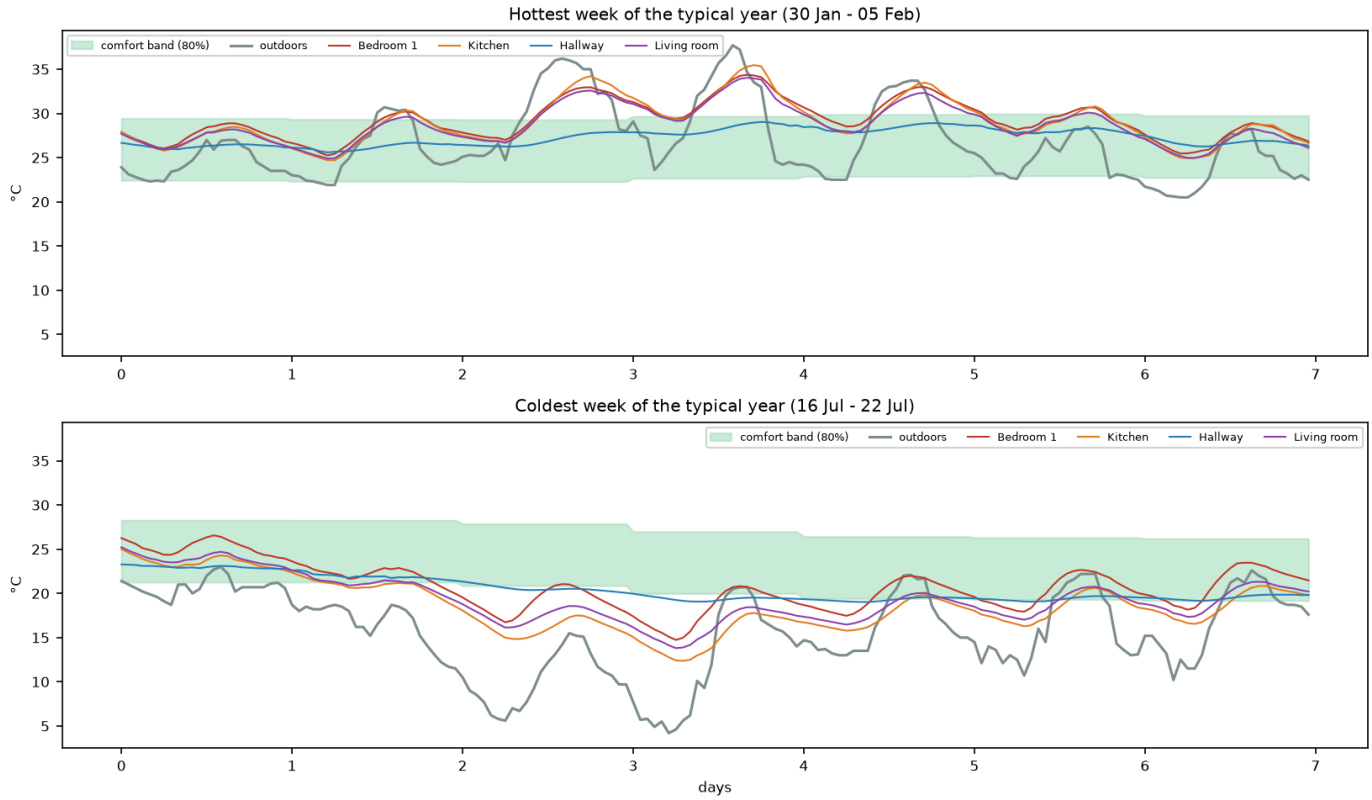
5.1 Hours in the comfortable band

Room	Comfortable hours	Too hot	Too cold	Hottest	Coldest
Bedroom 1	6,487 h = 74% Acceptable 3/5	1,717 h	556 h	34.4 °C Very hot 2/5	13.0 °C Cold 2/5
Bathroom 1	6,177 h = 70% Acceptable 3/5	1,757 h	826 h	36.2 °C Extreme 1/5	11.3 °C Dangerous 1/5
Living room	6,925 h = 79% Good 4/5	1,009 h	826 h	34.0 °C Very hot 2/5	12.7 °C Cold 2/5
TV room	6,854 h = 78% Good 4/5	1,042 h	864 h	34.3 °C Very hot 2/5	12.8 °C Cold 2/5
Hallway	8,174 h = 93% Excellent 5/5	0 h	586 h	29.0 °C Warm 4/5	17.1 °C Cool 3/5
Bedroom 2	6,905 h = 79% Good 4/5	1,164 h	691 h	34.6 °C Very hot 2/5	13.7 °C Cold 2/5
Bathroom 2	7,381 h = 84% Good 4/5	724 h	655 h	32.7 °C Very hot 2/5	14.6 °C Cold 2/5
Bedroom 3	7,160 h = 82% Good 4/5	947 h	653 h	33.9 °C Very hot 2/5	14.2 °C Cold 2/5
Kitchen	6,480 h = 74% Acceptable 3/5	1,239 h	1,041 h	35.5 °C Extreme 1/5	11.6 °C Dangerous 1/5
Outdoor bath	5,984 h = 68% Acceptable 3/5	1,968 h	808 h	35.9 °C Extreme 1/5	11.0 °C Dangerous 1/5
Outdoor storage	5,998 h = 68% Acceptable 3/5	1,744 h	1,018 h	36.7 °C Extreme 1/5	10.8 °C Dangerous 1/5
OUTDOORS (for reference)	4,248 h = 48% Poor 2/5	1,193 h	3,319 h	37.7 °C	4.2 °C

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10



Left: comfortable hours per room. Right: share of comfortable hours by month (green = good, red = bad).



A hot week and a cold week: indoor temperatures against the comfort band and the outdoor temperature.

5.2 Brazilian design-day check (NBR 15575)

Hottest day of the typical year: **02 Feb** (37.7 °C outside). Coldest day: **19 Jul** (4.2 °C outside).

Room	Hottest day: indoor peak	vs. outdoors	Verdict	Coldest day: indoor minimum	vs. outdoors	Verdict
Bedroom 1	34.5 °C	-3.2	Intermediate 4/5	14.5 °C	+10.3	Superior 5/5
Bathroom 1 *(not rated)*	36.3 °C	-1.4		12.3 °C	+8.1	
Living room	34.1 °C	-3.6	Intermediate 4/5	13.6 °C	+9.4	Superior 5/5
TV room	34.4 °C	-3.3	Intermediate 4/5	13.5 °C	+9.3	Superior 5/5
Hallway *(not rated)*	29.0 °C	-8.7		19.1 °C	+14.9	
Bedroom 2	34.9 °C	-2.8	Intermediate 4/5	15.0 °C	+10.8	Superior 5/5
Bathroom 2 *(not rated)*	32.9 °C	-4.8		16.3 °C	+12.1	
Bedroom 3	34.2 °C	-3.5	Intermediate 4/5	15.7 °C	+11.5	Superior 5/5
Kitchen *(not rated)*	35.5 °C	-2.2		12.2 °C	+8.0	
Outdoor bath *(not rated)*	35.9 °C	-1.8		12.1 °C	+7.9	
Outdoor storage *(not rated)*	36.8 °C	-0.9		11.4 °C	+7.2	

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10

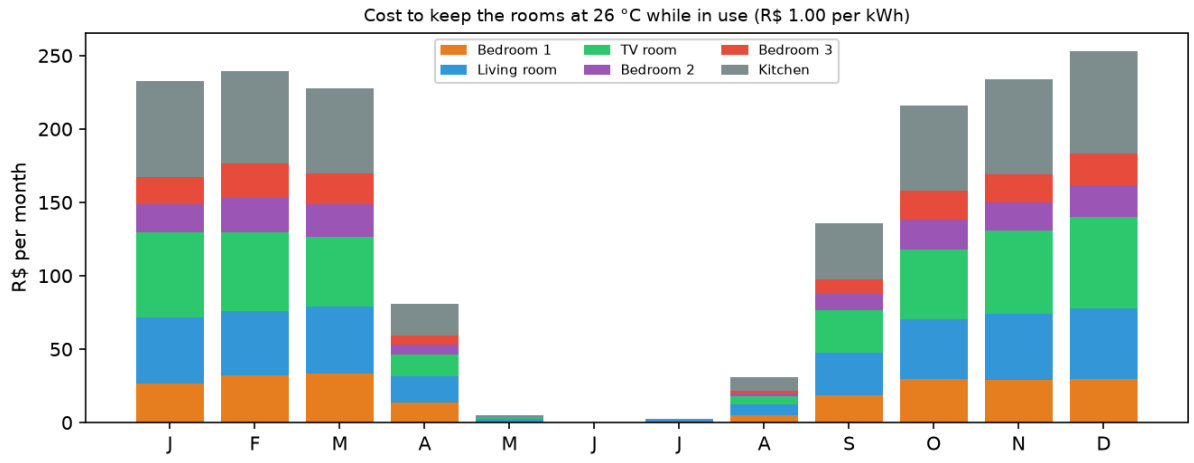
The newest edition (NBR 15575-1:2021) compares the house with a reference house over the whole year; that needs a second simulation, so this report uses the classic absolute design-day criteria, which are easier to understand.

5.3 What it costs to cool the rooms in summer

If a room is too hot you can cool it with an air conditioner. The simulation switches on a split air conditioner whenever the room is above **26 °C** during the hours it is used (bedroom: 21:00-08:00, living: 14:00-23:00, kitchen: 08:00-21:00), with windows closed, and adds up the electricity: heat removed ÷ COP 3.0 (an inverter model does better) × tariff **R$ 1.00 per kWh** (all taxes in; an assumption based on the Copel residential tariff after the +20% adjustment of June 2026; change it with *run_all.bat --tariff 0.95*).

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10

Room	When it is cooled	Hours the A/C runs	Electricity kWh/year	Cost R$/year	Cost Dec-Feb	Worst month	Peak heat to remove	A/C size	Level
Bedroom 1 (15.6 m²)	21:00-08:00	1235	219	**R$ 219**	R$ 89	Mar (R$ 34)	2.2 kW	9,000 BTU/h	Low 4/5
Living room (13.2 m²)	14:00-23:00	1690	325	**R$ 325**	R$ 137	Dec (R$ 48)	2.1 kW	9,000 BTU/h	Low 4/5
TV room (15.8 m²)	14:00-23:00	1653	375	**R$ 375**	R$ 174	Dec (R$ 62)	2.5 kW	12,000 BTU/h	Low 4/5
Bedroom 2 (10.5 m²)	21:00-08:00	1208	146	**R$ 146**	R$ 64	Feb (R$ 24)	1.6 kW	9,000 BTU/h	Low 4/5
Bedroom 3 (11.6 m²)	21:00-08:00	1174	142	**R$ 142**	R$ 64	Feb (R$ 23)	1.6 kW	9,000 BTU/h	Low 4/5
Kitchen (12.5 m²)	08:00-21:00	2216	450	**R$ 450**	R$ 198	Dec (R$ 70)	1.8 kW	9,000 BTU/h	Moderate 3/5
Total			**1,657**	**R$ 1,657**					



Monthly cost of cooling by room.

'Peak heat to remove' is the cooling power needed in the hottest hour; the A/C size adds 10% and rounds up to a catalogue size (1 kW = 3,412 BTU/h). Only sensible heat is counted (no dehumidification), so real bills are a little higher. Rooms with a High or Very high level are the ones where shade, a better roof or better insulation pay back fastest.

What counts as good? (temperature, cooling and heating)

Hours in the comfortable temperature band — Rule of thumb

Share of the 8,760 hours of the year in which the room temperature stays inside the ASHRAE-55 adaptive comfort band (it moves with the weather: warmer outdoors = wider acceptance). Outdoors, for reference, only about half of the hours.

Bad 1/5	below 40 % of the year	Uncomfortable most of the year.
Poor 2/5	40 to 60 % of the year	Uncomfortable much of the year.
Acceptable 3/5	60 to 75 % of the year	Comfortable most of the time, with some hot or cold spells.
Good 4/5	75 to 90 % of the year	Comfortable most of the year.
Excellent 5/5	90 or more % of the year	Comfortable nearly all year without air conditioning or heating.

This house: Bedroom 1 74% Acceptable 3/5 | Bathroom 1 70% Acceptable 3/5 | Living room 79% Good 4/5 | TV room 78% Good 4/5 | Hallway 93% Excellent 5/5 | Bedroom 2 79% Good 4/5 | Bathroom 2 84% Good 4/5 | Bedroom 3 82% Good 4/5 | Kitchen 74% Acceptable 3/5 | Outdoor bath 68% Acceptable 3/5 | Outdoor storage 68% Acceptable 3/5 | OUTDOORS (for reference) 48% Poor 2/5

Source: ASHRAE 55 adaptive comfort model (80% / 90% acceptability); Rule of thumb of this study (no norm gives a number). The norms judge thermal comfort relative to a reference house or on design days; this percentage scale is our own.

Hottest day: indoor peak vs. outdoor peak (NBR 15575) — Brazilian norm

On the hottest day of the typical year, the hottest indoor air temperature compared with the hottest outdoor temperature. Negative means the house is cooler than the outdoors.

Superior 5/5	below -4 degC (indoor max - outdoor max)	Indoor peak at least 4 degC below the outdoor peak.
Intermediate 4/5	-4 to -2 degC (indoor max - outdoor max)	Indoor peak at least 2 degC below the outdoor peak.
Minimum 3/5	-2 to 0.0001 degC (indoor max - outdoor max)	Indoor peak no higher than the outdoor peak.
Fails 1/5	0.0001 or more degC (indoor max - outdoor max)	Indoor peak higher than outdoors: the house overheats.

This house: Bedroom 1 -3.2 °C Intermediate 4/5 | Living room -3.6 °C Intermediate 4/5 | TV room -3.3 °C Intermediate 4/5 | Bedroom 2 -2.8 °C Intermediate 4/5 | Bedroom 3 -3.5 °C Intermediate 4/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements. Applies to bedrooms and living rooms. We use the hottest day of the weather file, not the norm's standard design day.

Coldest day: indoor minimum vs. outdoor minimum (NBR 15575) — Brazilian norm

On the coldest day of the typical year, the coldest indoor air temperature compared with the coldest outdoor temperature. Positive means the house is warmer than the outdoors.

Fails 1/5	below 3 degC (indoor min - outdoor min)	Less than 3 degC warmer than outdoors.
Minimum 3/5	3 to 5 degC (indoor min - outdoor min)	At least 3 degC warmer than outdoors.
Intermediate 4/5	5 to 7 degC (indoor min - outdoor min)	At least 5 degC warmer than outdoors.
Superior 5/5	7 or more degC (indoor min - outdoor min)	At least 7 degC warmer than outdoors.

This house: Bedroom 1 +10.3 °C Superior 5/5 | Living room +9.4 °C Superior 5/5 | TV room +9.3 °C Superior 5/5 | Bedroom 2 +10.8 °C Superior 5/5 | Bedroom 3 +11.5 °C Superior 5/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements. Winter check is required in bioclimatic zones 1 to 5 (the zone of the project is in the report).

Hottest hour of the year inside — Rule of thumb

Highest operative temperature (air + wall surfaces) reached in the room.

Comfortable 5/5	below 28 degC	Never really hot.
Warm 4/5	28 to 30 degC	Warm on the hottest days; a fan is enough.
Hot 3/5	30 to 32 degC	Hot spells; fans help, sleep is affected.
Very hot 2/5	32 to 35 degC	Heat stress risk for children and the elderly.
Extreme 1/5	35 or more degC	Dangerous without cooling.

This house: Bedroom 1 34.4 °C Very hot 2/5 | Bathroom 1 36.2 °C Extreme 1/5 | Living room 34.0 °C Very hot 2/5 | TV room 34.3 °C Very hot 2/5 | Hallway 29.0 °C Warm 4/5 | Bedroom 2 34.6 °C Very hot 2/5 | Bathroom 2 32.7 °C Very hot 2/5 | Bedroom 3 33.9 °C Very hot 2/5 | Kitchen 35.5 °C Extreme 1/5 | Outdoor bath 35.9 °C Extreme 1/5 | Outdoor storage 36.7 °C Extreme 1/5

Source: ASHRAE 55 adaptive comfort model (80% / 90% acceptability); Rule of thumb of this study (no norm gives a number). The adaptive-comfort upper limit in a warm month is about 29 degC (80% acceptability).

Coldest hour of the year inside International standard

Lowest operative temperature reached in the room.

Dangerous 1/5	below 12 degC	Cold stress; risk of heart and breathing problems.
Cold 2/5	12 to 16 degC	Below 16 degC the risk of respiratory problems rises.
Cool 3/5	16 to 18 degC	A jumper is needed; below the WHO 18 degC recommendation.
Good 4/5	18 to 20 degC	Meets the WHO recommendation of at least 18 degC.
Pleasant 5/5	20 or more degC	Never uncomfortably cold.

This house: Bedroom 1 13.0 °C Cold 2/5 | Bathroom 1 11.3 °C Dangerous 1/5 | Living room 12.7 °C Cold 2/5 | TV room 12.8 °C Cold 2/5 | Hallway 17.1 °C Cool 3/5 | Bedroom 2 13.7 °C Cold 2/5 | Bathroom 2 14.6 °C Cold 2/5 | Bedroom 3 14.2 °C Cold 2/5 | Kitchen 11.6 °C Dangerous 1/5 | Outdoor bath 11.0 °C Dangerous 1/5 | Outdoor storage 10.8 °C Dangerous 1/5

Source: WHO Housing and Health Guidelines 2018 (indoor temperature at least 18 C).

Electricity to keep the room cool with an air conditioner Rule of thumb

Electricity a split air conditioner would use in a year to hold the room at 26 C (upper limit of the summer comfort zone of NBR 16401-2) during the hours the room is used (bedrooms 21:00-08:00, living 14:00-23:00, kitchen 08:00-21:00). A room that overheats in summer costs more here. Cost = kWh x tariff.

Very low 5/5	below 10 kWh per m2 per year	Rarely needs cooling.
Low 4/5	10 to 25 kWh per m2 per year	A few weeks of cooling a year.
Moderate 3/5	25 to 45 kWh per m2 per year	Cooling needed through most of the summer.
High 2/5	45 to 70 kWh per m2 per year	Expensive to keep cool: the room overheats a lot.
Very high 1/5	70 or more kWh per m2 per year	Very expensive: shade, insulation or a better roof are worth more than a bigger air conditioner.

This house: Bedroom 1 14.1 kWh/m^2 Low 4/5 | Living room 24.6 kWh/m^2 Low 4/5 | TV room 23.7 kWh/m^2 Low 4/5 | Bedroom 2 13.9 kWh/m^2 Low 4/5 | Bedroom 3 12.2 kWh/m^2 Low 4/5 | Kitchen 36.1 kWh/m^2 Moderate 3/5

Source: Rule of thumb of this study (no norm gives a number); ABNT NBR 16401-2 (air conditioning, indoor comfort): summer zone about 22.5-25.5 C at 65% RH and 23-26 C at 35% RH. Assumes COP 3.0 (Inmetro class A split of about 2012), windows closed while the air conditioner runs, sensible heat only (no dehumidification) so real bills are somewhat higher. Tariff is an assumption, see the report.

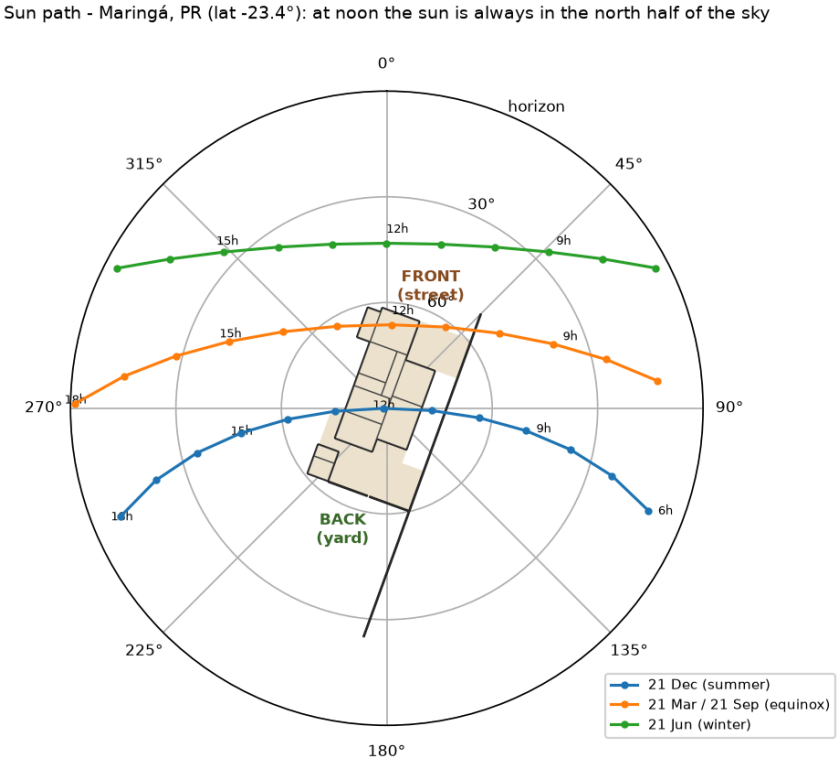
6. Sun on the outside, walls and roof

At this latitude the midday sun is always in the north half of the sky, so the walls that face north get the winter sun (good) and little summer sun, while east and west walls take strong summer sun (bad).

6.1 Sun energy on each surface (kWh per m², no shading)

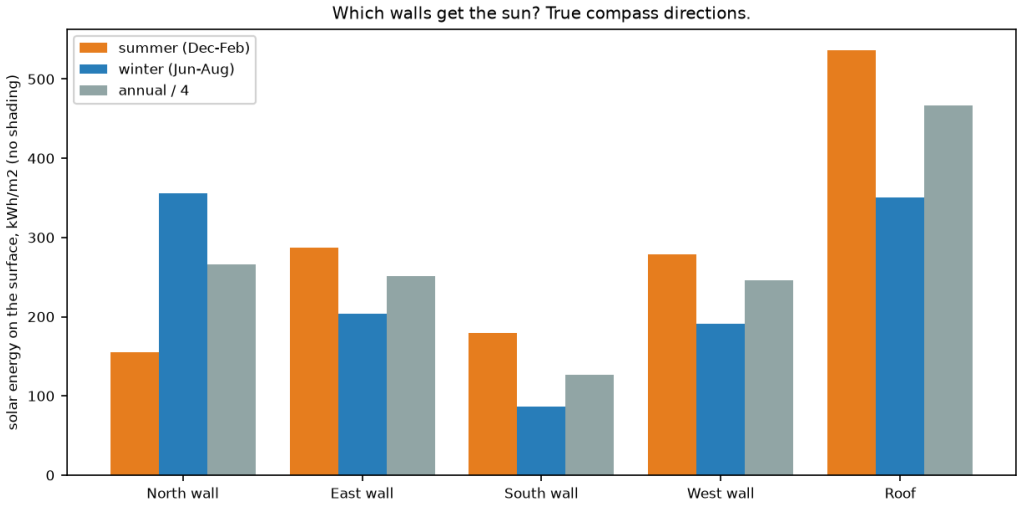
Surface	Whole year	Summer (Dec-Feb)	Winter (Jun-Aug)
North wall	1066	155 Moderate 4/5	356 High 5/5
East wall	1004	287 High 3/5	204 Moderate 3/5
South wall	507	179 Moderate 4/5	86 Low 2/5
West wall	982	279 High 3/5	191 Moderate 3/5
Roof (horizontal)	1865	536	351

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10



The house plan is drawn at the centre in its real orientation; its size is not to scale with the sky angles.

Path of the sun over the year with the house drawn at the centre in its real orientation: the front (street) faces north, the back faces south.



Solar energy on each wall by season.

6.2 Do the walls and roof meet NBR 15575?

NBR 15220-3 bioclimatic zone: **3**. NBR 15220-3:2005 lists Maringá in bioclimatic zone 3. A non-official article (UGREEN, 2026) says the 2024 revision puts it in zone 4B; that text was not checked. The NBR 15575 limit tables use zones 1 to 8, so zone 3 is used here.

Item	This house	Zone	Limit (NBR 15575)	Result
External wall heat transmittance U Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block) (paint / roof absorptance 0.45)	2.52 W/m²K	3	≤ 3.7	meets
External wall heat capacity Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block) (paint / roof absorptance 0.45)	134 kJ/m²K	3	≥ 130	meets
Roof heat transmittance U (indicative) Clay tile + roof void + 12 cm concrete slab ceiling (paint / roof absorptance 0.70)	1.96 W/m²K	3	≤ 1.5	does not meet

Part	Construction	Thermal lag φ	Solar factor FSo	Absorptance α	Weight
Wall	Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block)	3.0 h	4.5 %	0.45	light
Roof + ceiling	Clay tile + roof void + 12 cm concrete slab ceiling	6.4 h	5.5 %	0.70	light

φ = thermal lag in hours, FSo = 4 × U × α in % (NBR 15220-2 and 15220-3). They are not limits of NBR 15575; the NBR 15220-3 recommendations for them are in section 6.5.

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10

6.3 What if the materials change?

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10

An existing house rarely gets new walls, but the roof / ceiling can often be improved, so the two are shown separately (each with the other part kept as it is today). Roof upgrades such as wool insulation are in section 7.

Walls

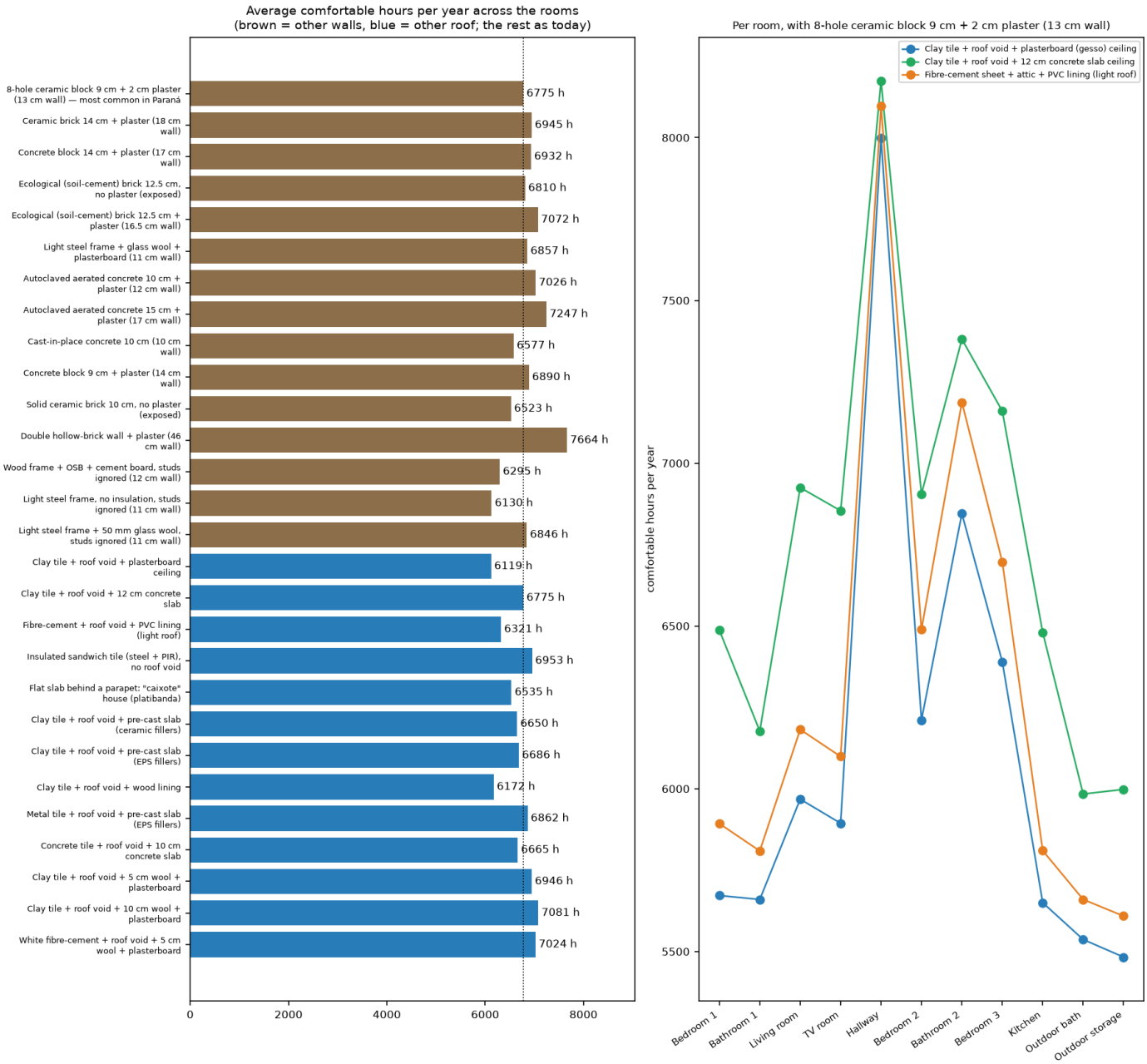
Roof / ceiling kept: Clay tile + roof void + 12 cm concrete slab ceiling

Walls	Avg. comfortable hours	Too hot	Too cold	Cooling bill	vs. current
Double hollow-brick wall (8-hole 10x20x20 cm laid on the larger side) + 2.5 cm plaster both sides = 46 cm wall	7,664 h = 87% Good 4/5	576 h	520 h	R$ 970/yr	R$ -687/yr
Autoclaved aerated concrete block 15 cm + 1 cm plaster both sides = 17 cm wall	7,247 h = 83% Good 4/5	865 h	648 h	R$ 1,041/yr	R$ -616/yr
Soil-cement (ecological) brick 12.5 cm + 2 cm plaster both sides = 16.5 cm wall	7,072 h = 81% Good 4/5	1,021 h	667 h	R$ 1,687/yr	+R$ 30/yr
Autoclaved aerated concrete block 10 cm + 1 cm plaster both sides = 12 cm wall	7,026 h = 80% Good 4/5	1,028 h	706 h	R$ 1,198/yr	R$ -459/yr
Ceramic hollow brick 14 cm + 2 cm plaster both sides = 18 cm wall (thicker, 'tijolo 6 furos em pe')	6,945 h = 79% Good 4/5	1,088 h	727 h	R$ 1,451/yr	R$ -206/yr
Concrete block 14 cm + 1.5 cm plaster both sides = 17 cm wall	6,932 h = 79% Good 4/5	1,110 h	717 h	R$ 1,730/yr	+R$ 73/yr
Hollow concrete block 9 cm + 2.5 cm plaster both sides = 14 cm wall	6,890 h = 79% Good 4/5	1,138 h	732 h	R$ 1,733/yr	+R$ 76/yr
Light steel frame: cement board 10 mm + 90 mm glass wool cavity + plasterboard 12.5 mm = 11.3 cm wall	6,857 h = 78% Good 4/5	1,175 h	728 h	R$ 968/yr	R$ -689/yr
Light steel frame, 50 mm glass wool (studs ignored): cement board 10 mm + wool 50 mm + 40 mm air cavity + plasterboard 12.5 mm = 11.3 cm wall	6,846 h = 78% Good 4/5	1,184 h	730 h	R$ 963/yr	R$ -694/yr
Soil-cement (ecological) brick 12.5 cm, no plaster, exposed = 12.5 cm wall	6,810 h = 78% Good 4/5	1,197 h	753 h	R$ 1,877/yr	+R$ 220/yr
Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block) **(current)**	6,775 h = 77% Good 4/5	1,210 h	775 h	R$ 1,657/yr	-
Cast-in-place concrete 10 cm, no plaster = 10 cm wall	6,577 h = 75% Good 4/5	1,367 h	816 h	R$ 2,326/yr	+R$ 669/yr
Solid ceramic brick 10 cm, no plaster, exposed = 10 cm wall	6,523 h = 74% Acceptable 3/5	1,385 h	852 h	R$ 2,063/yr	+R$ 406/yr
Wood frame (studs ignored): cement board 8 mm + OSB 9.5 mm + 90 mm air cavity + plasterboard 12.5 mm = 12 cm wall	6,295 h = 72% Acceptable 3/5	1,521 h	944 h	R$ 1,535/yr	R$ -122/yr
Light steel frame without insulation (studs ignored): cement board 10 mm + 90 mm air cavity + plasterboard 12.5 mm = 11.3 cm wall	6,130 h = 70% Acceptable 3/5	1,617 h	1,013 h	R$ 1,636/yr	R$ -21/yr

Roof / ceiling

Walls kept: Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block)

Roof / ceiling	Avg. comfortable hours	Too hot	Too cold	Cooling bill	vs. current
Clay tile 2 cm + roof void + 10 cm wool + plasterboard ceiling	7,081 h = 81% Good 4/5	919 h	760 h	R$ 1,064/yr	R$ -593/yr
White fibre-cement 6 mm + roof void + 5 cm wool + plasterboard ceiling	7,024 h = 80% Good 4/5	871 h	865 h	R$ 1,037/yr	R$ -620/yr
Insulated sandwich tile (steel + 30 mm PIR core), no attic	6,953 h = 79% Good 4/5	942 h	865 h	R$ 1,068/yr	R$ -589/yr
Clay tile 2 cm + roof void + 5 cm wool + plasterboard ceiling	6,946 h = 79% Good 4/5	1,004 h	810 h	R$ 1,159/yr	R$ -498/yr
Metal tile + roof void + pre-cast slab with EPS fillers	6,862 h = 78% Good 4/5	978 h	919 h	R$ 1,290/yr	R$ -367/yr
Clay tile + roof void + 12 cm concrete slab ceiling **(current)**	6,775 h = 77% Good 4/5	1,210 h	775 h	R$ 1,657/yr	-
Clay tile + roof void + pre-cast slab with EPS fillers	6,686 h = 76% Good 4/5	1,246 h	827 h	R$ 1,565/yr	R$ -92/yr
Concrete tile + roof void + 10 cm concrete slab	6,665 h = 76% Good 4/5	1,290 h	805 h	R$ 1,752/yr	+R$ 95/yr
Clay tile + roof void + pre-cast slab with ceramic fillers	6,650 h = 76% Good 4/5	1,286 h	825 h	R$ 1,671/yr	+R$ 14/yr
Flat waterproofed concrete slab behind a parapet (platibanda), no attic	6,535 h = 75% Acceptable 3/5	1,438 h	787 h	R$ 1,936/yr	+R$ 279/yr
Fibre-cement sheet + attic + PVC lining (light roof)	6,321 h = 72% Acceptable 3/5	1,452 h	987 h	R$ 1,626/yr	R$ -31/yr
Clay tile + roof void + wood lining	6,172 h = 70% Acceptable 3/5	1,601 h	987 h	R$ 1,853/yr	+R$ 196/yr
Clay tile + roof void + plasterboard (gesso) ceiling	6,119 h = 70% Acceptable 3/5	1,641 h	1,000 h	R$ 1,920/yr	+R$ 263/yr



Comfortable hours with different walls and roofs.

What counts as good? (sun and construction)

Summer sun on a wall Rule of thumb

Solar energy on a vertical wall with no shading in the three summer months (Dec-Feb in the south, Jun-Aug in the north). Lower is better: it is heat you do not want.

Low 5/5	below 150 kWh/m2	Little summer heat from the sun.
Moderate 4/5	150 to 225 kWh/m2	Some heat; an eave or a light colour handles it.
High 3/5	225 to 300 kWh/m2	Hot wall: needs shading, light colour, or insulation.
Very high 2/5	300 or more kWh/m2	Very hot wall.

This house: North wall 155 Moderate 4/5 | East wall 287 High 3/5 | South wall 179 Moderate 4/5 | West wall 279 High 3/5

Source: Rule of thumb of this study (no norm gives a number). Roughly 1.7 / 2.5 / 3.3 kWh/m2 per day. The norm handles wall heat through limits on U-value and paint colour.

Winter sun on a wall Rule of thumb

Solar energy on a vertical wall in June-August. Higher is better in a cool winter: free heat (NBR 15220-3 recommends winter solar heating in zones 1 to 3).

Low 2/5	below 120 kWh/m2	Almost no winter sun.
Moderate 3/5	120 to 220 kWh/m2	Some winter sun.
Good 4/5	220 to 320 kWh/m2	Good winter sun.
High 5/5	320 or more kWh/m2	A lot of free winter heat.

This house: North wall 356 High 5/5 | East wall 204 Moderate 3/5 | South wall 86 Low 2/5 | West wall 191 Moderate 3/5

Source: Rule of thumb of this study (no norm gives a number); ABNT NBR 15220-3: Brazilian bioclimatic zoning and building guidelines (zones and limits depend on the bioclimatic zone of the project).

Wall heat transmittance U (NBR 15575-4) Brazilian norm

How fast heat flows through the wall (lower = better insulated). The limits depend on the bioclimatic zone and on the outside paint (solar absorptance); see the report. Bands below: limits of zones 3-8 (3.7 if the paint is light, absorptance up to 0.6; 2.5 if darker).

Meets both limits 5/5	below 2.5 W/m2K	Meets the limit even with dark paint.
Light paint only 4/5	2.5 to 3.7 W/m2K	Meets the limit only with light-coloured paint (absorptance 0.6 or less).
Fails 1/5	3.7 or more W/m2K	Above the limit: the simplified method is not met.

This house: External wall heat transmittance U 2.52 (limit <= 3.7) Light paint only 4/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas.

Wall heat capacity (NBR 15575-4) Brazilian norm

How much heat the wall stores (higher = calmer indoor temperature). The limit depends on the bioclimatic zone (the minimum of 130 applies in zones 1-7); see the report.

Fails 1/5	below 130 kJ/m2K	Less heat storage than the minimum.
Meets 5/5	130 or more kJ/m2K	Meets the minimum of 130.

This house: External wall heat capacity 134 (limit >= 130) Meets 5/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas.

Roof heat transmittance U (NBR 15575-5) Brazilian norm

Heat flow through roof + ceiling. The limits depend on the bioclimatic zone and on the roof colour (solar absorptance); see the report. Bands below: limits of zones 3-6 (2.3 if the roof is light, absorptance up to 0.6; 1.5 if darker, e.g. red clay tile).

Meets both limits 5/5	below 1.5 W/m2K	Meets the limit even for a dark roof.
Light roof only 4/5	1.5 to 2.3 W/m2K	Meets the limit only if the roof surface is light.
Fails 1/5	2.3 or more W/m2K	Above the limit.

This house: Roof heat transmittance U (indicative) 1.96 (limit <= 1.5) Light roof only 4/5

Source: ABNT NBR 15575-5:2021 amendment draft: simplified method for roofs. Our roof U is indicative: a ventilated attic is not modelled by the standard calculation.

6.4 NBR 15575-1:2021 thermal performance by simulation (PHFT)

Calculated with: **EnergyPlus + AirflowNetwork + OpenFOAM (CFD)** · EnergyPlus solves the heat and air flow, OpenFOAM gives the wind pressure on the openings · details and reliability in section 10

The 2021 edition of NBR 15575-1 judges the thermal performance of a dwelling by computer simulation: the house as designed (**real model**) is compared with the same house built with reference materials (**reference model**: 100 mm walls and floors, tile + air chamber + 100 mm slab roof, windows of 17% of the floor area, no fixed shading). The main indicator is the **PHFT**, the percentage of occupied hours with an operative temperature between 18 and 26 °C (the range for climates with an annual mean below 25 °C: this weather file has 23.0 °C), over the bedrooms and the living room, with windows opened only while the room is occupied and the indoor air is at least 19 °C and warmer than outdoors. The standard also limits the hottest and coldest operative temperature and, for the intermediate and superior levels, asks for a reduction of the annual heating and cooling load (ideal air conditioner at 23 °C / 21 °C while occupied).

Room	PHFT real	PHFT reference	Hot hours %	Cold hours %	T_{max} / T_{min} real
Bedroom 1	70.6%	66.0%	27.3	2.1	32.1 / 15.5 °C
Living room	37.2%	22.5%	61.4	1.4	35.4 / 16.2 °C
TV room	38.2%	33.3%	60.2	1.7	35.4 / 15.8 °C
Bedroom 2	66.3%	60.2%	32.0	1.6	32.6 / 16.3 °C
Bedroom 3	62.2%	57.4%	35.0	2.8	32.9 / 16.2 °C

Result for the house as it is: PHFT of the dwelling = 54.9% against 47.9% of the reference (minimum level needs more than 90% of the reference: **yes**). Hottest operative temperature 35.4 °C against 37.6 °C allowed up to +2 °C: **yes**. Coldest 15.5 °C against 13.9 °C allowed down to −1 °C: **yes**. Total load 6,346 kWh/yr against 8,821 kWh/yr of the reference (28.1% lower). Level reached: **Minimum**.

Package	PHFT real	ΔPHFT (needed 17.2)	T_{max}	T_{min}	Load reduction	Level
Base	54.9%	7.0	35.4	15.5	28.1%	**Minimum**
Rock wool 50 mm	62.3%	14.4	33.1	16.7	49.5%	**Minimum**
Glass wool 50 mm	62.5%	14.6	33.1	16.8	50.1%	**Minimum**
PET wool 50 mm	62.5%	14.6	33.0	16.8	50.1%	**Minimum**
Sheep wool 50 mm	62.5%	14.6	33.0	16.8	50.1%	**Minimum**
Rock wool 100 mm	63.7%	15.8	32.8	16.9	52.9%	**Minimum**
Draught sealing	54.9%	7.0	35.4	15.5	28.1%	**Minimum**
Double glazing	55.0%	7.1	35.3	15.8	28.8%	**Minimum**
Low-E double glazing	56.6%	8.7	35.0	15.8	33.7%	**Minimum**
Fixed brise 0.6 m	55.4%	7.5	35.4	15.4	29.4%	**Minimum**
Movable blinds	56.2%	8.3	35.3	15.3	31.6%	**Minimum**
All together	71.0%	23.1	32.1	16.9	65.7%	**Superior**
Light wall paint	55.3%	7.4	35.3	15.4	29.3%	**Minimum**
Light roof	61.8%	13.9	33.8	15.4	47.4%	**Minimum**
Radiant barrier under the tiles	60.2%	12.3	33.8	15.9	44.1%	**Minimum**

The intermediate level needs ΔPHFT of at least 45 − 0.58 × PHFT of the reference (single-family house) and the load reduction of table 20; the superior level needs the same ΔPHFT and a reduction of 35% (reference load below 100 kWh/m²·yr) or 55% (above), or a PHFT of 95% or more. Packages with draught sealing, glazing and shading barely move the indicators; roof insulation is what moves the house towards the higher levels (needed ΔPHFT 17.2; 50 mm wools reach 14.6). Levels reached: All together (Superior). A package that lands exactly on the needed value is borderline: the result moves with small changes of the model.

The levels of the standard and what they mean for living in the house

Level	What the standard asks	What living in the house is like	In this house
Not met	The PHFT is 90% of the reference house or less, or the hottest / coldest operative temperature is more than 2 °C / 1 °C worse than the reference.	The house is clearly worse than the standard reference: more hot afternoons and cold nights indoors, and the air conditioner has to carry the comfort most of the year.	No package reaches it.
Minimum	PHFT above 90% of the reference house; hottest operative temperature at most 2 °C above the reference and coldest at most 1 °C below it.	About as comfortable as a house built with the reference materials (100 mm walls, tile and slab roof). On hot afternoons and cold dawns the rooms leave the 18-26 °C band; expect to use the fan or the air conditioner on many days and to wear a jacket on cold mornings.	**Current house**: PHFT 54.9% against 47.9% of the reference (ΔPHFT 7.0), load reduction 28.1%.
Intermediate	Everything of the minimum level, plus a PHFT gain over the reference house (ΔPHFT) of at least 45 − 0.58 × PHFT of the reference (single-family house) and a reduction of the annual heating and cooling load (table 20).	Noticeably more hours inside the comfortable band without any equipment, rooms that stay cooler in the afternoon and warmer at dawn, and a smaller air-conditioning bill. The air conditioner becomes a help for the hottest days, not a daily need.	No package reaches it.
Superior	Everything of the intermediate level, with a load reduction of 35% (reference load below 100 kWh/m²·year) or 55% (above), or a PHFT of 95% or more.	The house holds a comfortable temperature almost by itself: few hot or cold hours, the air conditioner is rarely needed and, when it is, a small unit is enough. This is the level that pays back in the bill.	Reached by **All together**: PHFT 71.0%, ΔPHFT 23.1, load reduction 65.7%.

The 'In this house' column uses the results of this simulation; the description of life in the house is our reading of the numbers, not text of the standard.

Cross ventilation (wind-driven)

Natural ventilation is solved by the EnergyPlus AirflowNetwork: every window, open door and crack is a flow path whose pressure comes from the wind (pressure coefficients of every opening, calculated by the OpenFOAM CFD for 24 wind directions), from the temperature difference and from the height of the opening. Air changes per hour through the exterior windows and doors, base house:

Calculated with: **EnergyPlus + AirflowNetwork + OpenFOAM (CFD)** · air flow by the AirflowNetwork with CFD pressure coefficients · details and reliability in section 10

Room	Mean ACH	90th percentile	Hours above 5 ACH
Bedroom 1	4.4	11.6	3,393
Bathroom 1	4.2	5.8	2,783
Living room	2.0	8.3	1,964
TV room	1.5	6.9	1,594
Bedroom 2	5.6	15.8	3,438
Bathroom 2	4.3	6.0	2,979
Bedroom 3	3.6	9.9	3,385
Kitchen	0.2	0.4	0

How the air moves through the house

Calculated with: **EnergyPlus + AirflowNetwork + OpenFOAM (CFD)** · EnergyPlus airflow network with the pressure coefficients of the OpenFOAM CFD · details and reliability in section 10

The airflow network gives, for every hour, the volume of air that crosses each window, door and opening. The figure shows the average of those flows on the plan, for all the hours with the windows open and for the hot hours (operative temperature above 26 °C); arrows are blue where fresh air comes in, red where air leaves the house, orange for air moving between rooms, and the thickness follows the flow in m³/h. This is the flow between rooms (one air node per room), not a velocity field inside each room: it shows which openings feed the house and where the air leaves, not the draught at a given spot.

How the air moves through the house (net flow of every opening) — m³/h



Opening	Type	Air goes from	to	m³/h
window 1 · Bedroom 1	window	outdoors	Bedroom 1	161
door 5 · Bedroom 1	door	Bedroom 1	Hallway	161
window 1 · Bedroom 2	window	Bedroom 2	outdoors	157
door 7 · Hallway	door	Hallway	Bedroom 2	157

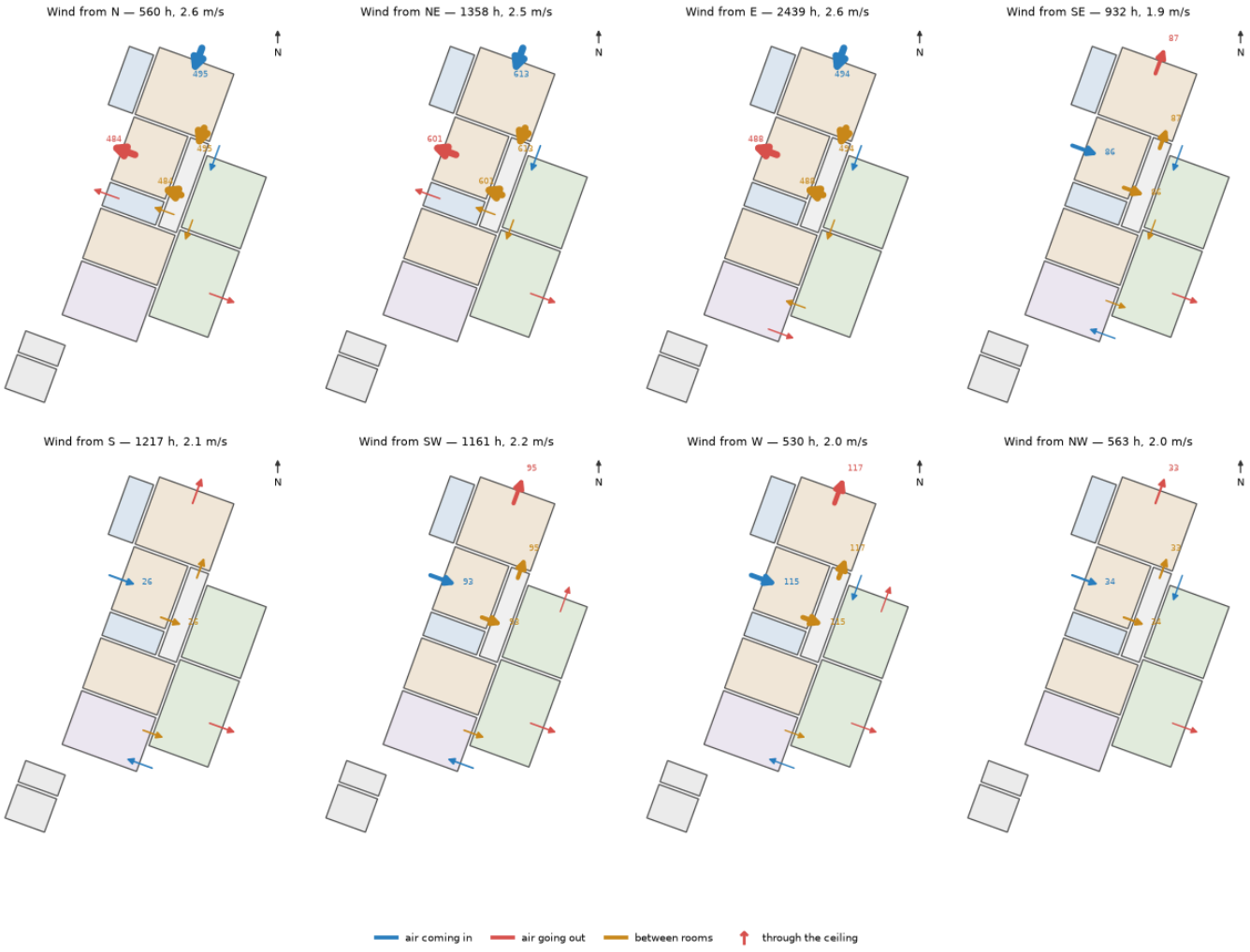
Main air paths of the house as it is in the hot hours (flows below 20 m³/h omitted).

Circulation for each wind direction

> Calculated with: **EnergyPlus + AirflowNetwork + OpenFOAM (CFD)** · EnergyPlus airflow network with the pressure coefficients of the OpenFOAM CFD · details and reliability in section 10

Same flows, separated by the direction the wind comes from (eight sectors of 45°, wind of the weather file). The pressure on each opening comes from the 24 CFD cases, so the pattern changes with the wind: openings on the windward side feed the house and those on the leeward side or under suction let the air out.

Air flow for each wind direction (hours with the windows open) — m³/h

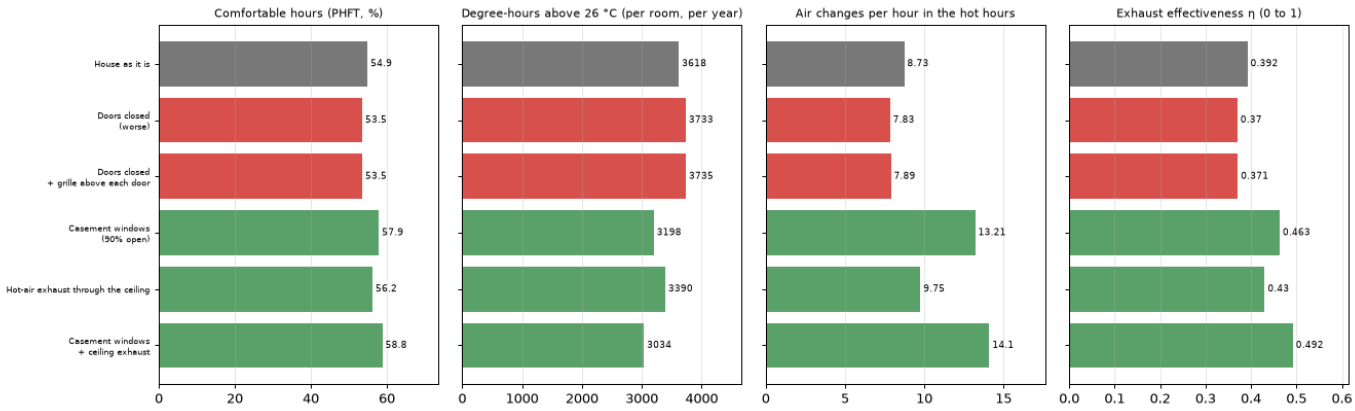


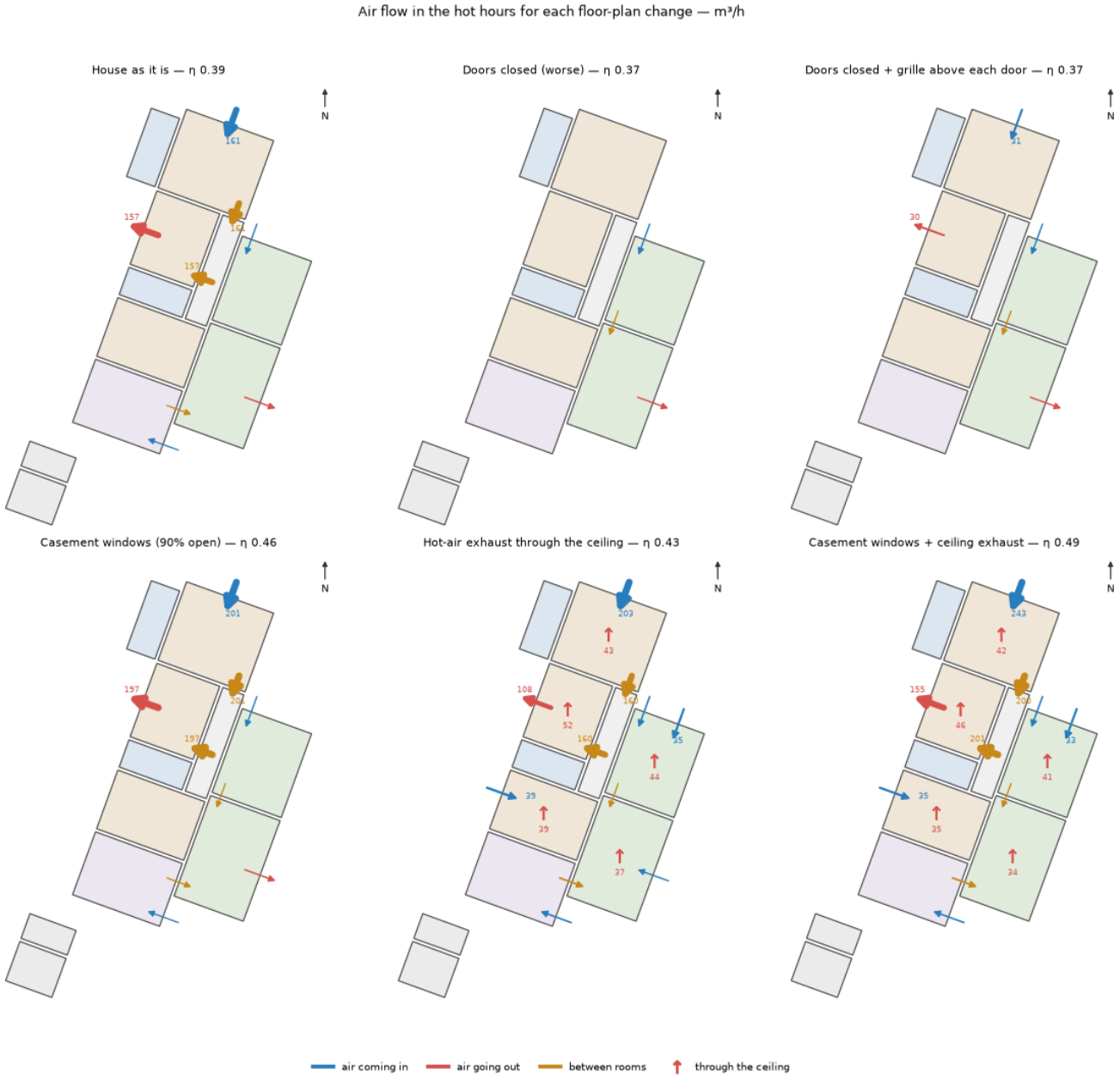
Plan changes that improve or worsen ventilation

Calculated with: **EnergyPlus + AirflowNetwork + OpenFOAM (CFD)** · EnergyPlus airflow network with the pressure coefficients of the OpenFOAM CFD · details and reliability in section 10

Each change is a new full-year run of the same model, with the same weather, occupancy and CFD pressures; only the opening in question changes. The table compares comfort and the exhaust of hot air, using the occupied hours of the bedrooms and the living room.

Change	PHFT	Hottest °C	Degree-hours above 26 °C	Air changes per hour in hot hours	Exhaust effectiveness η	Heat carried out (kWh/year)
House as it is Sliding windows (50% open), doors of the bedrooms and the living room open while the room is occupied.	54.9%	35.4	3,618	8.7	0.39	7,810
Doors closed (worse) The doors between the bedrooms / living room and the rest of the house stay closed: air has no way through.	53.5% (-1.4)	35.5	3,733 (+115)	7.8 (-0.9)	0.37 (-0.02)	7,194 (-616)
Doors closed + grille above each door Same closed doors, with a permanent 0.5 x 0.4 m grille above each one (the usual Brazilian transom grille).	53.5% (-1.4)	35.5	3,735 (+117)	7.9 (-0.8)	0.37 (-0.02)	7,326 (-484)
Casement windows (90% open) Windows that open 90% of the glazed area (casement or tilt) instead of the 50% of sliding windows; same size and frames.	57.9% (+3.0)	35.3	3,198 (-420)	13.2 (+4.5)	0.46 (+0.07)	8,883 (+1073)
Hot-air exhaust through the ceiling A 0.2 m2 grille in the ceiling of each bedroom and of the living room, and eave openings of the attic 5 times larger (2% of the floor area instead of 0.4%).	56.2% (+1.3)	35.2	3,390 (-228)	9.8 (+1.0)	0.43 (+0.04)	10,424 (+2614)
Casement windows + ceiling exhaust Casement windows (90% open) together with the ceiling grilles and the larger eave openings.	58.8% (+3.9)	35.1	3,034 (-584)	14.1 (+5.4)	0.49 (+0.10)	11,494 (+3684)





Reading: the best change for comfort is **Casement windows + ceiling exhaust** (PHFT 58.8% against 54.9%, exhaust effectiveness 0.49 against 0.39); the worst is **Doors closed (worse)** (PHFT 53.5%). The reference house has PHFT 47.9%.

Hot-air exhaust effectiveness

Calculated with: **EnergyPlus + AirflowNetwork** · temperatures and flows of the airflow network · details and reliability in section 10

There is no single standard coefficient for this, so the report uses three measurable indicators: (1) the **air changes per hour in the hot hours**; (2) the **exhaust effectiveness η** = (T closed − T ventilated) ÷ (T closed − T outdoor), computed with the operative temperatures of each room in the occupied hours when the house with the windows closed is above 26 °C and more than 1 °C above the outdoor air (0 = the ventilation does not cool the room, 1 = the room follows the outdoor air); (3) the **heat carried out** by the air that leaves the house, ρ·cp·V·(T zone − T outdoor), in kWh per year. Age of air and the removal of a plume of hot air at ceiling level need an interior CFD, which this study does not do.

The wind pressure coefficients (Cp) of every window, door and eave opening come from a **CFD simulation with OpenFOAM** (steady RANS, k-epsilon, atmospheric boundary layer over suburban roughness, 12 wind directions (every 30°), 1.2 to 2.0 million cells each; pressure read 0.3 m in front of each opening and expressed as a coefficient of the dynamic pressure of the EnergyPlus wind profile at the height of the opening). Against the generic Swami & Chandra coefficients the CFD gives stronger suction and overpressure at the openings (largest windward Cp 0.31 against 0.58; mean difference per opening 0.44). Effect on the result: PHFT of the dwelling 54.9% with CFD against 56.0% with the generic coefficients, reference 47.9% against 48.9% (ΔPHFT 7.0 against 7.1): the verdict does not depend on the choice.

The house in the CFD follows the real geometry of the IFC model: the two roofs with their valley and ridge (up to 4.1 m), the eaves and the carport and porch roofs as thin plates, with windows and doors closed. A first version used rooms extruded to a flat 3.5 m top; the real roof shape changes the coefficients by - on average per opening (root mean square over the 12 wind directions both versions have), mostly at the attic openings and the windows under the eaves.

CFD limits: the lot walls and the neighbouring buildings are included but their heights are estimates from photographs, and there are no trees; windows and doors are closed in the CFD (only the outside is solved); a steady RANS model gives mean pressures only (no gusts, no unsteady wake). Use it as a better-than-generic estimate, not a wind-tunnel result.

Source of the procedure: section 11.4 of ABNT NBR 15575-1:2021 (full text, copy kept in norms/source_docs) and the public amendment draft of March 2021 (LabEEE); tables 3, 4, 20 and 21 agree in both, the equation of figure 4 (ΔPHFTmín = 45 − 0.58 × PHFT of the reference, single-family) was read from the figure of the draft. Bioclimatic zone: NBR 15220-3:2005 lists Maringá in bioclimatic zone 3. A non-official article (UGREEN, 2026) says the 2024 revision puts it in zone 4B; that text was not checked. The NBR 15575 limit tables use zones 1 to 8, so zone 3 is used here. Assumptions of ours: the kitchen and the hallway are transient rooms (not counted in the PHFT); interior doors between a bedroom or the living room and a transient room stay open; the 18-26 °C band and the 23/21 °C set points apply because the annual mean temperature is below 25 °C;the wind pressure coefficients come from the CFD (the generic Swami & Chandra formula is kept only as a comparison); the lot walls and neighbouring buildings are in the CFD with estimated heights but not in the shadow and daylight models; ground temperature is the annual mean of the weather file.

6.5 Suggested materials and the NBR 15220-3 guidelines

Calculated with: **Fast engine (Python, this project)** · catalogue values (NBR 15220-2 tables, published constructions); ranking by comfortable hours of the fast engine · details and reliability in section 10

Each construction below replaces only the wall (the roof stays as it is) or only the roof (the wall stays as it is) in the year-long simulation of the fast engine; the three with the most comfortable hours that beat the current one are listed. These are a ranking of options, not a recommendation to rebuild: changing walls is rarely practical in a finished house.

Walls (with the current roof)

Construction	Δ comfortable h/yr	Δ cooling R$/yr	U	CT	φ	FSo	NBR 15575	Source
Double hollow-brick wall (8-hole 10x20x20 cm laid on the larger side) + 2.5 cm plaster both sides = 46 cm wall	**+889**	-687	1.00	368	10.7	1.8	meets	typical Sorgato (2009) Fig. 45: double wall of 8-hole bricks 10x20x20 on the larger side, 2.5 cm plaster both sides (46 cm), thermal lag 10.8 h
Autoclaved aerated concrete block 15 cm + 1 cm plaster both sides = 17 cm wall	**+472**	-616	0.93	115	6.1	1.7	does not meet	typical NBR 15220-2 mortar, via LabEEE catalogue v4 Tab. A.1; sim_house.py: AAC about 500 kg/m3, k 0.17 (NBR 15220)
Soil-cement (ecological) brick 12.5 cm + 2 cm plaster both sides = 16.5 cm wall	**+297**	+30	2.91	292	4.4	5.2	meets	typical NBR 15220-2 mortar, via LabEEE catalogue v4 Tab. A.1; sim_house.py: hollow, about 1900 kg/m3 solid; equivalent k 0.9 is the NBR 15220 order of magnitude

Roofs (with the current wall)

Construction	Δ comfortable h/yr	Δ cooling R$/yr	U	CT	φ	FSo	NBR 15575	Source
Clay tile 2 cm + roof void + 10 cm wool + plasterboard ceiling	**+306**	-593	0.37	11	3.4	0.9	meets	assumption NBR 15220-2 ceramic tile, via LabEEE catalogue v4 Tab. A.2 (used for the NBR roof U only; the simulation keeps C_cover); DATec 020 (Tecverde) lambda; (...)
White fibre-cement 6 mm + roof void + 5 cm wool + plasterboard ceiling colder winters: +90 cold h	**+249**	-620	0.64	10	2.4	0.9	meets	assumption NBR 15220-2 fibre-cement, via LabEEE catalogue v4 Tab. A.2 (used for the NBR roof U only; the simulation keeps C_cover); DATec 020 (Tecverde) lambda; (...)
Insulated sandwich tile (steel + 30 mm PIR core), no attic colder winters: +90 cold h	**+178**	-589	0.66	3	1.4	1.1	meets	assumption sim_house.py: sandwich panel with 30 mm PIR core; typical manufacturer value; House3D default (to be checked against NBR 15220-2)

Δ = change against the current construction (cooling and heating: negative = saving). U in W/m²K, CT in kJ/m²K, φ in hours, FSo in %. NBR 15575 = simplified-method limits for the zone of the site (walls: U and CT; roofs: U). Confidence is the weakest of the layer materials (verified, typical or assumption); the source column is shortened, the full list per material is in norms/MATERIALS_SOURCES.md.

> **Winter warning:** a lighter construction stores less heat, so the nights and the cold months get colder (DELOS 2026). The warning appears when the hours that are too cold rise by more than 10% against the current construction. If cold nights matter more to you than hot afternoons, prefer an option without the warning or add heating.

NBR 15220-3 guidelines for the zone (recommendation, not mandatory)

Zone 3: medium openings (15-25 % of floor area), allow winter sun; shading: allow winter sun. Recommended wall: light reflective wall; recommended roof: light insulated roof.

Item	This house	Guideline	Meets?	What to change
Wall: U (W/m²K)	2.52	≤ 3.6	meets	-
Wall: Thermal lag φ (h)	3.00	≤ 4.3	meets	-
Wall: FSo (%)	4.5	≤ 4.0	does not meet	lower FSo: lighter paint (lower absorptance) or a lower U
Roof: U (W/m²K)	1.96	≤ 2.0	meets	-
Roof: Thermal lag φ (h)	6.40	≤ 3.3	does not meet	lower thermal lag: lighter roof and ceiling
Roof: FSo (%)	5.5	≤ 6.5	meets	-

These are **recommendations, not mandatory**: NBR 15220-3:2005 has no normative character (source: Lamberts, UFSC ECV 5161, lecture 12). Sorgato (2009) tested the upper limits of thermal lag by simulation and found them contradicted (heavier walls and roofs did better in zone 3), so a miss on φ is a hint, not a defect. The mandatory limits are those of NBR 15575 in section 6.2.

7. Upgrades: insulation, sealing, glazing and shading

Each upgrade below is simulated on the whole year and compared with the house as it is today. **Roof wool** lies on top of the ceiling (attic side); **sealing** means rubber gaskets and brush seals on the frames and doors; **double glazing** replaces the single glass; a **fixed brise** is a horizontal projection above each window; **movable blinds** close by themselves in the model when the sun on the window is strong. The last row combines the main measures.

The packages **Light wall paint** (aged absorptance 0.39, white paint after 3 years), **Light roof** (aged 0.37) and **Radiant barrier under the tiles** (foil) are assumptions to be checked on site. The radiant barrier is modelled as a bright foil face (emissivity 0.05) in the roof void: the fast engine adds 0.40 m²K/W to the cover resistance (air layer 0.61 against 0.21 m²K/W, NBR 15220-2) and EnergyPlus gives the foil its own emissivity; this resistance is an assumption, and NBR 15575-1:2021 (section 11.2) asks radiant barriers to pass an ageing test (salt spray, ASTM D5894) before their benefit can be counted. A light roof cuts the hot hours but can raise the cold hours in winter (less stored heat), so read the comfortable hours, not only the cooling bill.

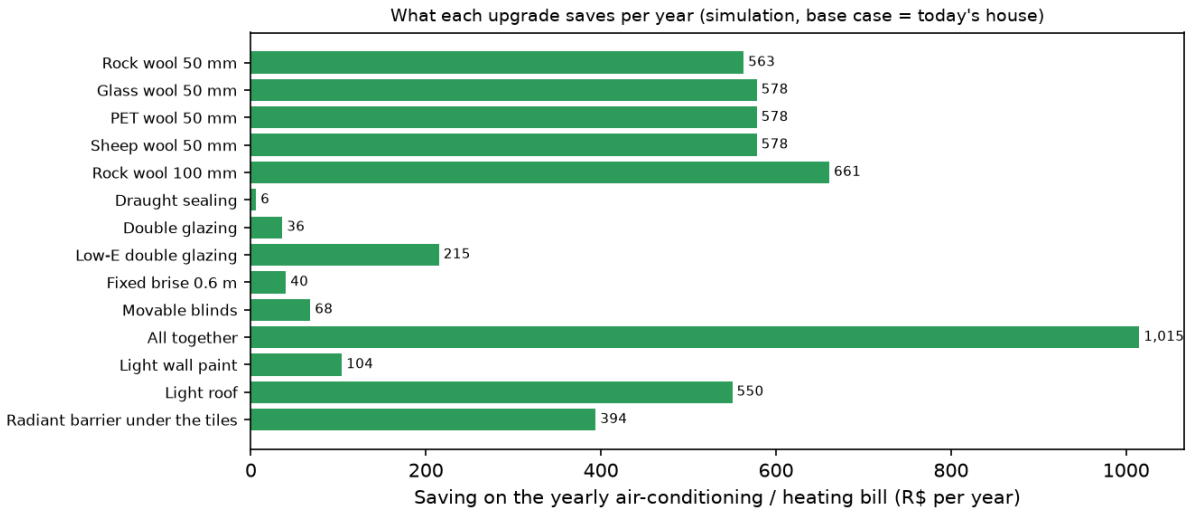
7.1 Results by upgrade

Upgrade	What it is	Comfortable hours	Hours too hot	Hottest hour	Cooling R$/yr	Change in bill	Daylight
Base	Existing house	82.0% Good 4/5	788 h	32.9 °C	R$ 1,657	R$ 0/yr	100%
Rock wool 50 mm	Rock wool 50 mm	86.8% Good 4/5	521 h	31.4 °C	R$ 1,094	R$ -563/yr	100%
Glass wool 50 mm	Glass wool 50 mm	86.9% Good 4/5	515 h	31.3 °C	R$ 1,079	R$ -578/yr	100%
PET wool 50 mm	PET wool (recycled bottles) 50 mm	86.9% Good 4/5	516 h	31.3 °C	R$ 1,079	R$ -578/yr	100%
Sheep wool 50 mm	Sheep wool 50 mm	86.9% Good 4/5	516 h	31.3 °C	R$ 1,079	R$ -578/yr	100%
Rock wool 100 mm	Rock wool 100 mm	87.5% Good 4/5	491 h	31.1 °C	R$ 996	R$ -661/yr	100%
Draught sealing	Rubber gaskets + brush seals on frames and doors	82.2% Good 4/5	785 h	32.8 °C	R$ 1,651	R$ -6/yr	100%
Double glazing	Double glazing, clear	82.5% Good 4/5	783 h	32.7 °C	R$ 1,621	R$ -36/yr	89%
Low-E double glazing	Double glazing, low-E	83.6% Good 4/5	656 h	32.4 °C	R$ 1,442	R$ -215/yr	80%
Fixed brise 0.6 m	Fixed horizontal brise / overhang 0.6 m	82.5% Good 4/5	729 h	32.8 °C	R$ 1,617	R$ -40/yr	100%
Movable blinds	Movable blinds / shutters (closed when the sun is strong)	82.6% Good 4/5	711 h	32.7 °C	R$ 1,589	R$ -68/yr	100%
All together	Rock wool 50 mm; Rubber gaskets + brush seals on frames and doors; Double glazing, low-E; White factory-finished fibre-cement roof (alpha 0.37; DATec 020); Aluminium foil under the tiles (low-emissivity void) (NBR 15220-2); Fixed horizontal brise / overhang 0.6 m; Movable blinds / shutters (closed when the sun is strong)	88.6% Good 4/5	337 h	30.3 °C	R$ 642	R$ -1,015/yr	80%
Light wall paint	White paint, aged 3 years (alpha 0.39; NBR 15575-1:2021 Annex A)	82.8% Good 4/5	689 h	32.6 °C	R$ 1,553	R$ -104/yr	100%
Light roof	White factory-finished fibre-cement roof (alpha 0.37; DATec 020)	83.0% Good 4/5	492 h	31.8 °C	R$ 1,107	R$ -550/yr	100%
Radiant barrier under the tiles	Aluminium foil under the tiles (low-emissivity void) (NBR 15220-2)	84.6% Good 4/5	579 h	32.0 °C	R$ 1,263	R$ -394/yr	100%

Calculated with: **Fast engine (Python, this project)** · EnergyPlus cross-check in section 7.4 · details and reliability in section 10

Where the numbers of the table above come from:

Column	How it is calculated	Engine	How far to trust it
Comfortable hours	Share of the 8,760 hours of the year in which the operative temperature of the room (mean of air and wall-surface temperature) is inside the 80% acceptability band of the ASHRAE 55 adaptive model; average of the rooms. The air conditioner runs on its schedule in this calculation, so the percentage is higher than the PHFT of section 6.4 (house without air conditioning).	Fast engine. Cross-checked with EnergyPlus in section 7.4.	Medium: good for ranking the upgrades, the absolute percentage is approximate.
Hours too hot	Hours of the year with the operative temperature above the upper limit of the same band; average per room.	Fast engine.	Medium, same as above.
Hottest hour	Highest operative temperature of the year in each room, averaged over the rooms. It comes from one hour of the typical weather year, so real heat waves will be hotter.	Fast engine.	Medium-low: a single extreme hour is sensitive to the weather file.
Cooling R$/yr	Heat removed by an air conditioner that holds 26 °C in the hours each room kind is used, divided by the efficiency (COP 3.0) and multiplied by the tariff (R$ 1.00/kWh, an assumption set in the project file).	Fast engine (energy); tariff and COP are assumptions.	Low: use it as an order of magnitude and to compare upgrades. EnergyPlus gives roughly half of this energy (section 7.4).
Change in bill	Cooling (and heating, when enabled) cost of the upgrade minus the cost of the house as it is. Negative = saving.	Fast engine.	Low-medium: the difference between upgrades is more reliable than the absolute value.
Daylight	Visible transmittance of the new glass divided by that of the existing glass. Only the glass is counted: blinds and shading are not applied here, and no light engine is run for this column.	Catalogue value (no engine); the daylight of the house itself is in the daylight sections (Radiance).	Medium: a catalogue value, not a calculation.



7.2 Which wool and how thick? (change in yearly bill and average comfortable hours)

Wool	25 mm	50 mm	75 mm	100 mm	150 mm
Glass wool λ 0.040 W/m·K	R$ -453 86.1%	R$ -578 86.9%	R$ -637 87.3%	R$ -670 87.5%	R$ -709 87.8%
Rock wool λ 0.044 W/m·K	R$ -434 85.9%	R$ -563 86.8%	R$ -625 87.2%	R$ -661 87.5%	R$ -702 87.6%
PET wool (recycled bottles) λ 0.040 W/m·K	R$ -453 86.0%	R$ -578 86.9%	R$ -637 87.3%	R$ -672 87.5%	R$ -710 87.7%
Sheep wool λ 0.040 W/m·K	R$ -453 86.0%	R$ -578 86.9%	R$ -638 87.3%	R$ -672 87.5%	R$ -710 87.6%

Calculated with: **Fast engine (Python, this project)** · details and reliability in section 10

Negative = saving per year. More wool always helps, with diminishing returns: the first 50 mm give most of the benefit. Wools differ little in conductivity; the choice is usually decided by price, fire behaviour, handling and humidity.

7.3 Values used and how reliable they are

Measure	Value used	Reliability of the value	Source
Glass wool	λ 0.040 W/m·K, ρ 15 kg/m³	typical	range 0.030-0.046 W/mK (Saint-Gobain, wool insulation comparison); mid-range value
Rock wool	λ 0.044 W/m·K, ρ 40 kg/m³	verified	Rockwool datasheet (panel, 0.044 W/mK at 50 C, EN 12667)
PET wool (recycled bottles)	λ 0.040 W/m·K, ρ 30 kg/m³	assumption	typical manufacturer range 0.037-0.041 W/mK; check the product you buy
Sheep wool	λ 0.040 W/m·K, ρ 25 kg/m³	assumption	typical range 0.038-0.042 W/mK; check the product you buy
Single glass (existing)	U 5.8 W/m²K, g 0.85, VT 0.88	verified	5.8 W/m2K is the single-glass limit of EN 673 / ISO 10077
Double glazing, clear	U 2.8 W/m²K, g 0.76, VT 0.78	typical	typical catalogue values for 4-12-4 clear double glazing (EN 673 / NFRC ranges)
Double glazing, low-E	U 1.8 W/m²K, g 0.40, VT 0.70	typical	typical catalogue values for low-E (solar control) double glazing
Rubber gaskets + brush seals on frames and doors	closed-house air changes −25%	assumption	AIVC: weatherstripping alone cuts house air change 10-14.7%, a full retrofit up to 27.5%; 25% used for frames and doors
Fixed horizontal brise / overhang	depth set per package	assumption	analytic profile-angle shading of the window (sun-path geometry, pvlib); depth in metres from project.json
Movable blinds / shutters (closed when the sun is strong)	closed gain x0.3	assumption	external blinds let through about 10-30% of solar gain when closed; closed when direct sun on the window exceeds 250 W/m2 (08-18 h)

Calculated with: **3D model (IFC) and tables** · catalogue values, not calculated · details and reliability in section 10

The upgrade prices are not included: with a quotation you can divide it by the yearly saving to get the payback time. Daylight column: share of the daylight that still enters (glass transmittance only; blinds and brise are not applied to daylight).

7.4 Cross-check with a reference engine (EnergyPlus)

Calculated with: **EnergyPlus + AirflowNetwork** · details and reliability in section 10

Every package above was recalculated with **EnergyPlus** (US Department of Energy), the open-source engine validated by ASHRAE Standard 140 and used inside DesignBuilder and OpenStudio. It solves the heat balance of every surface with conduction transfer functions, shares walls between rooms, models the attic above each room, shades windows with the real eaves and uses wind- and buoyancy-driven window ventilation (windows 50% open when the outside air is cooler and above 18 °C). The air conditioner keeps the same set point, schedules and COP as the fast engine. There are no internal gains (people, lamps, appliances) in either engine.

Package	Comfort % without A/C (EnergyPlus)	A/C kWh/yr fast	A/C kWh/yr EnergyPlus	Saving/yr EnergyPlus
Base	77.7	1,657	671	-
Rock wool 50 mm	83.1	1,094	286	R$ 385
Glass wool 50 mm	83.1	1,079	278	R$ 393
PET wool 50 mm	83.1	1,079	278	R$ 393
Sheep wool 50 mm	83.1	1,079	278	R$ 393
Rock wool 100 mm	83.4	996	235	R$ 436
Draught sealing	78.0	1,651	669	R$ 2
Double glazing	78.5	1,621	651	R$ 20
Low-E double glazing	78.5	1,442	588	R$ 83
Fixed brise 0.6 m	77.5	1,617	654	R$ 17
Movable blinds	77.3	1,589	627	R$ 44
All together	79.1	642	102	R$ 569
Light wall paint	77.9	1,553	647	R$ 24
Light roof	76.3	1,107	316	R$ 355
Radiant barrier under the tiles	80.8	1,263	331	R$ 340

Result with EnergyPlus: base case: 77.7% of the hours in the comfort zone (fast engine 77.3%), 594 hot hours (fast 1211), 1356 cold hours (fast 775); air-conditioning 671 kWh/yr (fast 1,657). Both engines agree on the ranking: **roof insulation is by far the best measure**, glazing and shading change little.

Differences between the engines are expected: EnergyPlus gives roughly half the air-conditioning energy of the fast engine and more cold hours (it couples the floor to the ground and loses heat through the attic more freely). For decisions, prefer the EnergyPlus figures; the fast engine remains useful for quick what-if runs. Not yet modelled: internal gains, door opening, the slope of the roof (a flat roof with the same attic volume is used) and the neighbours' shade.

8. Sustainable-building daylight classification (LEED / IES LM-83)

Calculated with: **Fast engine (Python, this project)** · Radiance cross-check in 8.1 · details and reliability in section 10

LEED is the most widely used green-building certification in the world. Its daylight credit (v4.1, Option 1) is awarded from the simulated **sDA300/50%** (share of the regularly occupied floor that gets 300 lux for at least half of the occupied hours) and the **ASE1000,250** (share of floor with more than 1,000 lux of direct sun for over 250 hours a year, a glare and overheating risk), both defined in the IES standard LM-83 (current edition LM-83-23, February 2023). Steps: average sDA of at least 40% = 1 point, 55% = 2 points, 75% = 3 points; an ASE above 10% must be justified with a glare strategy.

Room	Area m²	sDA300/50%	ASE1000,250	LEED step if alone
Bedroom 1	15.6	100% 3 points 5/5	33% Very high 1/5	3 point(s)
Living room	13.2	78% 3 points 5/5	0% Very low 5/5	3 point(s)
TV room	15.8	100% 3 points 5/5	7% Acceptable 4/5	3 point(s)
Bedroom 2	10.5	100% 3 points 5/5	21% Very high 1/5	3 point(s)
Bedroom 3	11.6	100% 3 points 5/5	19% High 2/5	3 point(s)
Kitchen	12.5	100% 3 points 5/5	0% Very low 5/5	3 point(s)
Whole house (area-weighted)	79.2	96% 3 points 5/5	14% High 2/5	3 point(s)

Result: area-weighted sDA300/50% = 96.3% and ASE1000,250 = 13.5%: **3 of 3 daylight point(s)** (ASE above 10%: glare strategy needed). Rooms below 40%: none.

This is an indicative pre-assessment: LEED requires a certified assessor, furniture and blind modelling, a 0.6 m grid and the full LM-83 procedure (our grid is 0.5 m, with 8:00-18:00 as occupied hours and blinds open). LEED v4.1 formally cites the 2012 edition of LM-83; the 2023 edition keeps the same two metrics (we could not verify the detailed changes). Homes are normally certified under LEED for Homes, whose daylight rules differ.

LEED daylight (sDA300/50%, area-weighted over the regularly occupied rooms) International standard

Average spatial daylight autonomy of the living rooms, bedrooms and kitchens (floor-area weighted): the share of floor that gets 300 lux for at least half of the occupied hours. LEED v4.1 gives 1 point from 40%, 2 points from 55% and 3 points from 75%. Indicative pre-assessment only: a real LEED credit needs a certified assessor and the full LM-83 method.

No points 2/5	below 40 % of floor	Below the 40% needed for the first point.
1 point 3/5	40 to 55 % of floor	Meets the first LEED daylight step.
2 points 4/5	55 to 75 % of floor	Meets the second LEED daylight step.
3 points 5/5	75 or more % of floor	Meets the top LEED daylight step.

This house: Whole house 96% 3 points 5/5

Source: USGBC LEED v4.1, Daylight credit, Option 1 (simulation): sDA300/50% average of 40% / 55% / 75% = 1 / 2 / 3 points; ASE1000,250 above 10% needs a glare strategy (thresholds read on the LEEDuser v4.1 credit page); ANSI/IES LM-83-23 (published Feb 2023): spatial daylight autonomy (sDA) and annual sunlight exposure (ASE); LEED v4.1 still cites the 2012 edition, same two metrics. Also needed: ASE1000,250 of at most 10% (otherwise explain the glare strategy). Wet rooms, halls and outdoor areas are not 'regularly occupied'.

8.1 Cross-check with a reference engine (Radiance)

Calculated with: **Radiance** · details and reliability in section 10

The numbers above come from our fast engine (ray-traced sun + a uniform-sky split-flux estimate). To test them, the same rooms and sensor points were recalculated with **Radiance** (Lawrence Berkeley National Laboratory), the open-source engine validated by CIE 171 and used by ClimateStudio and Honeybee: Perez all-weather sky from the same weather file, 4 light bounces, annual daylight-coefficient method, and the LM-83 rule that closes the blinds when direct sun hits more than 2% of the room (shade transmittance 5%). A sensor above the roof reproduces the weather file's illuminance within 1%, so the sky is calibrated.

Room	Area m²	sDA300/50% fast	sDA300/50% Radiance	ASE Radiance	Mean lux fast → Radiance
Bedroom 1	15.6	100%	49% 1 point 3/5	47%	3,222 → 2,211
Living room	13.2	78%	0% No points 2/5	0%	323 → 174
TV room	15.8	100%	18% No points 2/5	16%	1,062 → 761
Bedroom 2	10.5	100%	28% No points 2/5	48%	1,880 → 1,385
Bedroom 3	11.6	100%	25% No points 2/5	38%	1,689 → 1,265
Kitchen	12.5	100%	0% No points 2/5	0%	316 → 102

Result with Radiance: area-weighted sDA300/50% = 20.6% (fast engine: 96.3%), ASE1000,250 = 24.4%: **0 of 3 daylight point(s)**. The fast engine is optimistic deep inside the rooms; prefer the Radiance figures for decisions.

Not yet modelled in the Radiance run: furniture, frames and mullions (glass transmittance 0.80 is used for the whole opening), the glass-door lattice, the neighbours' buildings and the tree shade. Blinds follow the simplified per-room LM-83 rule. Windows and brise fixed as in the base case.

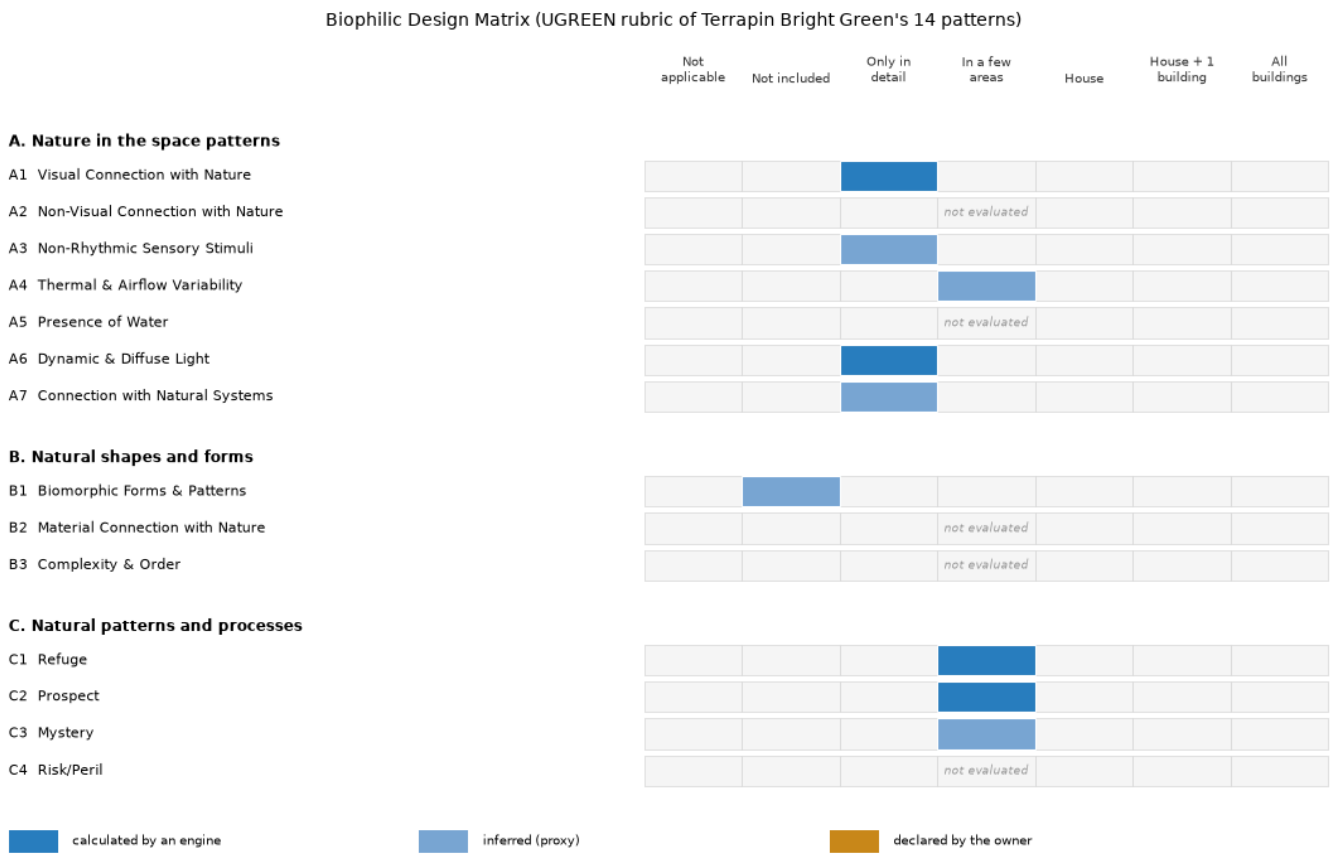
9. Biophilia: views and plan

Calculated with: **Biophilia analysis (ray cast and plan isovist, this project)** + **3D model (IFC) and tables** · rays cast from inside each room through its windows; the engines of sections 2 to 8 supply daylight and ventilation · details and reliability in section 10

Biophilia is the human need to be connected to nature. This section scores how much of it the house delivers, with two published frameworks: the **Biophilic Design Matrix** of 14 patterns (Terrapin Bright Green; the seven-column rubric comes from the UGREEN video) and the **Biophilic Healing Index** of Salingaros (ten components, 0 to 20). It looks at views and surfaces, not only at the plan, because people do not perceive a plan. Every pattern says which engine produced it and how far to trust it; what no engine can measure (water, sounds and smells, materials, complexity) stays *not evaluated* until the owner declares it.

Indicative only: Indicative only: the matrix and the Biophilic Healing Index are not clinically validated; the author of the index calls its link with healing a conjecture. Potted plants indoors do not purify the air in real rooms.

9.1 Biophilic Design Matrix: 14 patterns



A pattern is placed in a column from the share of the counted rooms (occupied rooms other than bathrooms, hall and outdoor areas) that meet a strong criterion: none = not included, a weak form only = only in detail, up to half = in a few areas, more than half = house. The two columns beyond *house* need other buildings and are not used. These thresholds are this tool's own, not a published rule, and the matrix has no official total score, so none is invented. Patterns resting on a weak indicator are capped at *in a few areas*.

Pattern	Column	How	Trust	Evidence
A1 Visual Connection with Nature	**Only in detail**	calculated ray cast (biophilia_views.py)	medium	0 of 6 counted rooms see sky, vegetation (middle layer) and ground through a window, or at least 15% green: -. A view of walls, a street or neighbouring houses is not counted as nature. Trees come from the IFC (boxes at estimated positions).
A2 Non-Visual Connection with Nature	**Not evaluated**	not evaluated -	—	Not evaluated: no engine measures sounds of water and fauna, aromatic plants, textures of real wood and stone. Declare it in project.json (biophilia.matrix.A2).
A3 Non-Rhythmic Sensory Stimuli	**Only in detail**	inferred (proxy) ray cast (biophilia_views.py)	low	Only a proxy: moving leaves and shadows need vegetation in view; sound, smell and touch are not evaluated. Some green is visible from a window.
A4 Thermal & Airflow Variability	**In a few areas**	inferred (proxy) fast engine (cross-ventilation hours) + EnergyPlus ventilation study	low	Proxy: 5 of 6 counted rooms have good air movement: Bedroom 1, Living room, Bedroom 2, Bedroom 3, Kitchen. The pattern asks for variability of the air and temperature, which is not measured: this only shows how much air moves. Rooms with an EnergyPlus result use 5 or more air changes per hour in the hot hours as the criterion (house: 8.7, exhaust effectiveness 0.39); the others use the fast engine's cross-ventilation hours.
A5 Presence of Water	**Not evaluated**	not evaluated -	—	Not evaluated: no engine measures water seen and heard (mirror, pool, fountain). Declare it in project.json (biophilia.matrix.A5).
A6 Dynamic & Diffuse Light	**Only in detail**	calculated window orientations (IFC) + Radiance daylight	medium	0 of 6 counted rooms have windows facing at least two directions: -; 6 reach 100 lux or more on average (8-18 h, Radiance). sDA alone is not used: it measures light for work, not variation.
A7 Connection with Natural Systems	**Only in detail**	inferred (proxy) ray cast + trees of the IFC	low	Only a proxy: vegetation that follows the seasons seen from the rooms; the sun path and weather are not linked to the views yet.
B1 Biomorphic Forms & Patterns	**Not included**	inferred (proxy) IFC geometry	low	0 of 158 building elements of the IFC have curved geometry. The model is built from boxes, so this may only reflect how it was modelled, not the real house.
B2 Material Connection with Nature	**Not evaluated**	not evaluated -	—	Not evaluated: no engine measures natural materials in sight (wood, stone, clay) instead of synthetic laminates; the model only has the wall and roof constructions of project.json, not the finishes. Declare it in project.json (biophilia.matrix.B2).
B3 Complexity & Order	**Not evaluated**	not evaluated -	—	Not evaluated: no engine measures organised complexity: rich but coherent detail. Declare it in project.json (biophilia.matrix.B3).
C1 Refuge	**In a few areas**	calculated plan isovist (biophilia_views.py)	low	6 of 6 counted rooms have a point with at least 60% of the directions blocked within 2 m (protected back) and a sight line of 6 m or more. The criterion is generous in small rooms.
C2 Prospect	**In a few areas**	calculated ray cast + plan isovist (biophilia_views.py)	medium	3 of 6 counted rooms have a window whose nearest obstacle is 6 m or more away (or an open view): Bedroom 1, Living room, Kitchen. Nearest obstacle: median distance in the horizontal band of the opening.
C3 Mystery	**In a few areas**	inferred (proxy) plan isovist (biophilia_views.py)	low	Proxy: 3 of 6 counted rooms show two or more separate groups of long views from their best point (views revealed gradually). Curved walls and brises are not detected.
C4 Risk/Peril	**Not evaluated**	not evaluated -	—	Not evaluated: no engine measures controlled risk: cantilevers, terraces, high edges. Declare it in project.json (biophilia.matrix.C4).

Patterns per column: Not included 1, Only in detail 4, In a few areas 4, House 0; not evaluated 5.

9.2 Biophilic Healing Index (Salingaros)

Calculated with: **Biophilia analysis (ray cast and plan isovist, this project)** · sunlight, life and curves from the model; the rest from the owner · details and reliability in section 10

The index adds ten components, each rated 0 (none), 1 (some) or 2 (a large amount), for a total from 0 to 20 with equal weights. The author calls the link between the index and healing a **conjecture still to be tested**; treat it as a rough, comparative number. Here only 3 of the 10 components can be estimated from the model, so the number below is partial, not a score out of 20.

Component	0 / 1 / 2	How	Evidence
Sunlight (several directions)	**1**	inferred (proxy) direct-sun hours of the fast engine + window orientations (IFC)	4 of 6 counted rooms get direct sun on their windows (0.05 h/day or more, mean over the year), 2 get 0.5 h/day or more; the sunny rooms face 3 of the 4 compass quadrants. The index asks for sunlight from several directions.
Life	**1**	inferred (proxy) trees of the IFC + ray cast	7 tree elements in the IFC; 0% of the counted rooms see 15% or more green. Animals and people are not counted.
Curves	**0**	inferred (proxy) IFC geometry	Curved elements of the IFC (the model is boxes: may be only the modelling).
Color	**—**	not evaluated -	Not evaluated: an observer's estimate (0, 1 or 2) is needed; declare it in project.json (biophilia.bhi).
Gravity	**—**	not evaluated -	Not evaluated: an observer's estimate (0, 1 or 2) is needed; declare it in project.json (biophilia.bhi).
Fractals	**—**	not evaluated -	Not evaluated: an observer's estimate (0, 1 or 2) is needed; declare it in project.json (biophilia.bhi).
Detail	**—**	not evaluated -	Not evaluated: an observer's estimate (0, 1 or 2) is needed; declare it in project.json (biophilia.bhi).
Water	**—**	not evaluated -	Not evaluated: an observer's estimate (0, 1 or 2) is needed; declare it in project.json (biophilia.bhi).
Representations of nature	**—**	not evaluated -	Not evaluated: an observer's estimate (0, 1 or 2) is needed; declare it in project.json (biophilia.bhi).
Organized complexity	**—**	not evaluated -	Not evaluated: an observer's estimate (0, 1 or 2) is needed; declare it in project.json (biophilia.bhi).

Result: 2 of 6 possible in the 3 components the model can estimate; with the owner's ratings 2 of 6 in 3 of 10 components (complete index needs all ten).

9.3 View out of each room

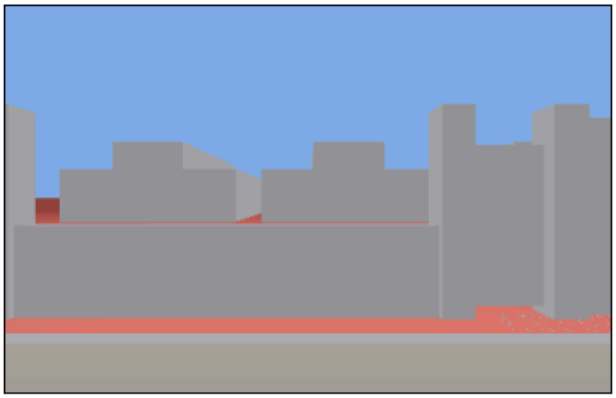
Calculated with: **Biophilia analysis (ray cast and plan isovist, this project)** · ray cast through each window; trees from the IFC at estimated positions · details and reliability in section 10

Each picture is seen from inside the room, looking through one window, seated (1.2 m) and standing (1.6 m). Colours: sky blue, vegetation green, ground red, walls, roofs and neighbours grey. A restorative view has sky, a middle layer of vegetation and ground (each at least 5% of the window), with the nearest obstacle at 6 m or more (rule of the UGREEN video, applied here as this tool's criterion; a wall, a street or the neighbours' houses are not counted as nature). Distances are measured from the eye, 1 m behind the window, so 6 m is 5 m from the glass. The glass is treated as fully transparent, the trees are boxes at estimated positions and the table uses the standing eye.

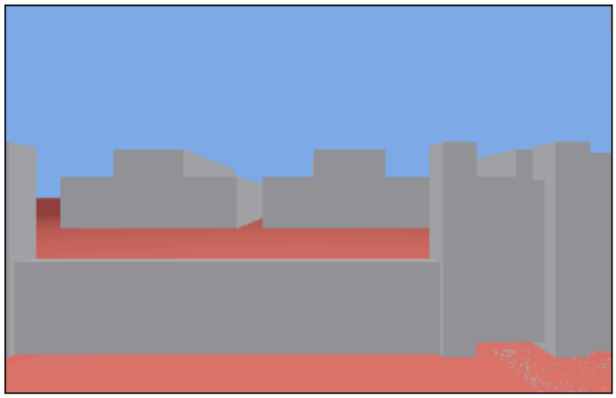
Room	Windows	Green in view (best window)	Sky + middle + ground	Nearest obstacle 6 m or more	Longest sight line (plan)
Bedroom 1	1	0%	no	yes	60 m
Bathroom 1 (service)	1	0%	no	no	5 m
Living room	1	0%	no	yes	33 m
TV room	1	0%	no	no	31 m
Bedroom 2	1	0%	no	no	10 m
Bathroom 2 (service)	1	0%	no	no	4 m
Bedroom 3	1	0%	no	no	15 m
Kitchen	1	4%	no	yes	23 m
Outdoor bath (service)	1	0%	no	no	12 m
Outdoor storage (service)	1	17%	yes	yes	19 m

View out of Bedroom 1

Window 0 — seated (1.2 m)
green 0% · nearest obstacle: 20.1 m · 3 layers: no



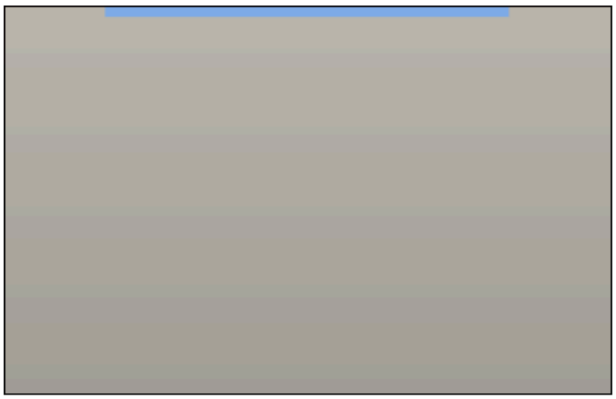
Window 0 — standing (1.6 m)
green 0% · nearest obstacle: 20.2 m · 3 layers: no



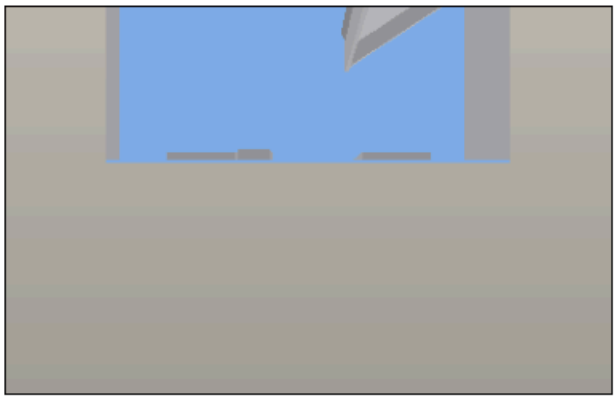
sky vegetation ground built (walls, roofs, neighbours)

View out of Bathroom 1

Window 4 — seated (1.2 m)
green 0% · nearest obstacle: — · 3 layers: no



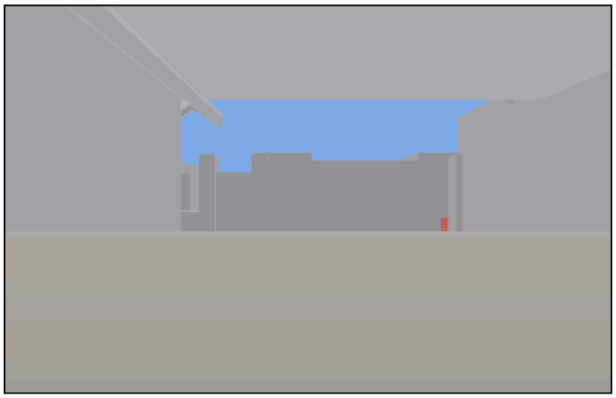
Window 4 — standing (1.6 m)
green 0% · nearest obstacle: — · 3 layers: no



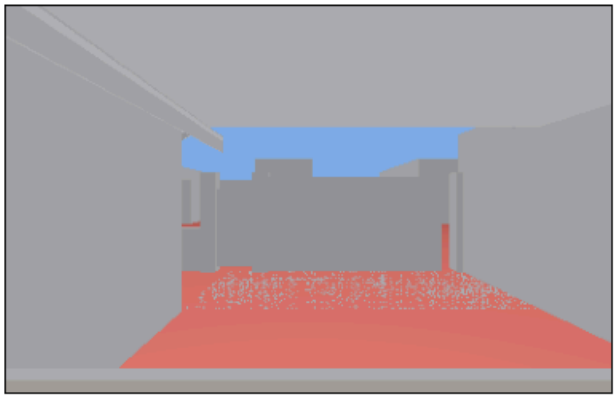
sky vegetation ground built (walls, roofs, neighbours)

View out of Living room

Window 6 — seated (1.2 m)
green 0% · nearest obstacle: 7.5 m · 3 layers: no



Window 6 — standing (1.6 m)
green 0% · nearest obstacle: 7.2 m · 3 layers: no



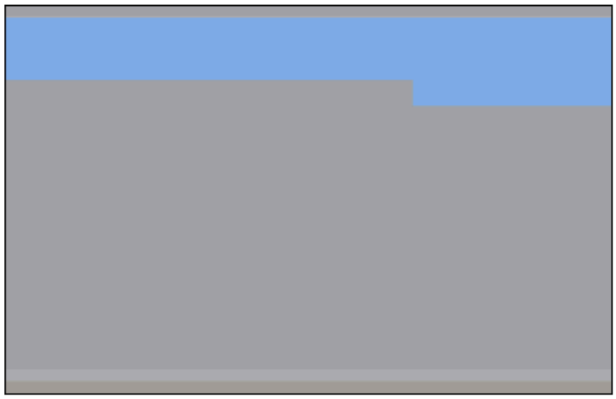
sky vegetation ground built (walls, roofs, neighbours)

View out of TV room

Window 7 — seated (1.2 m)
green 0% · nearest obstacle: 3.9 m · 3 layers: no



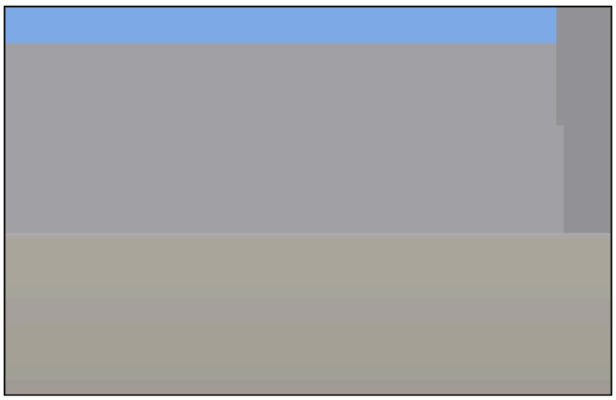
Window 7 — standing (1.6 m)
green 0% · nearest obstacle: 3.9 m · 3 layers: no



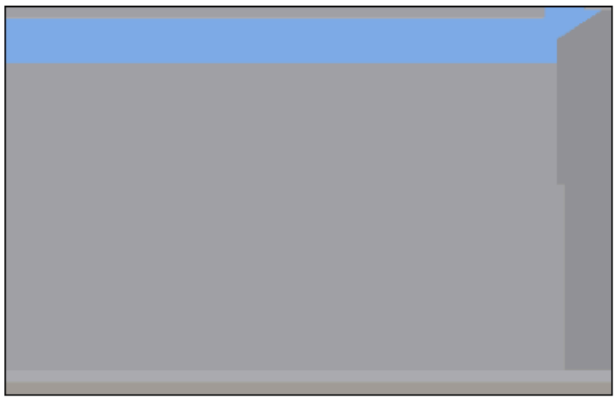
sky vegetation ground built (walls, roofs, neighbours)

View out of Bedroom 2

Window 1 — seated (1.2 m)
green 0% · nearest obstacle: 2.7 m · 3 layers: no



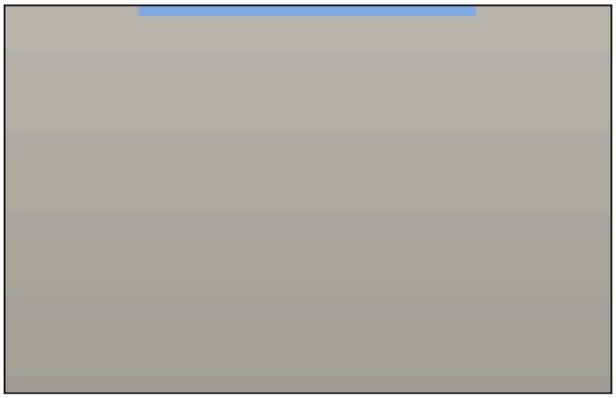
Window 1 — standing (1.6 m)
green 0% · nearest obstacle: 2.9 m · 3 layers: no



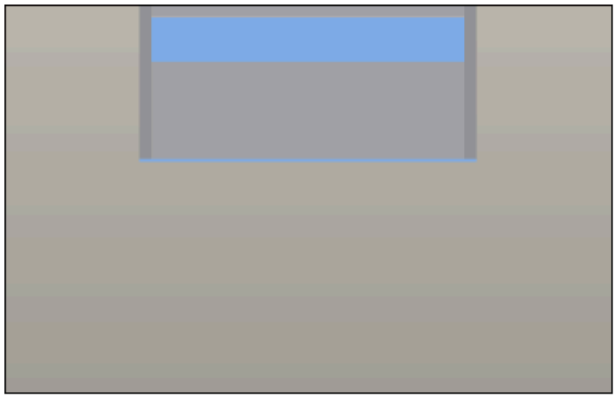
sky vegetation ground built (walls, roofs, neighbours)

View out of Bathroom 2

Window 3 — seated (1.2 m)
green 0% · nearest obstacle: — · 3 layers: no



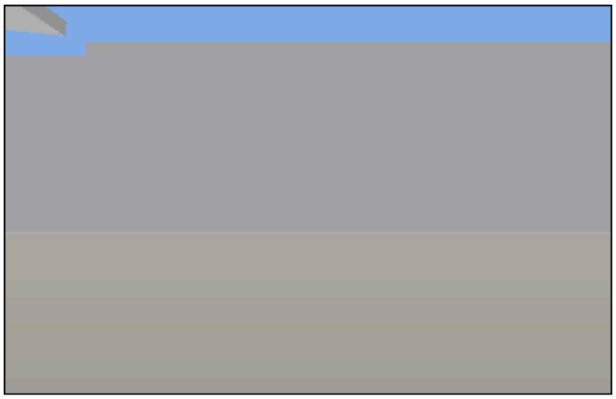
Window 3 — standing (1.6 m)
green 0% · nearest obstacle: — · 3 layers: no



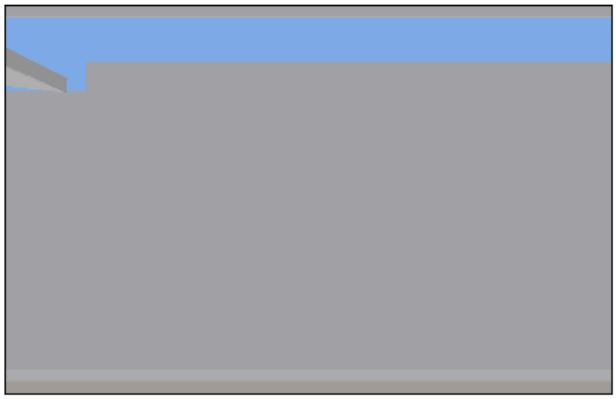
sky vegetation ground built (walls, roofs, neighbours)

View out of Bedroom 3

Window 2 — seated (1.2 m)
green 0% · nearest obstacle: 2.7 m · 3 layers: no



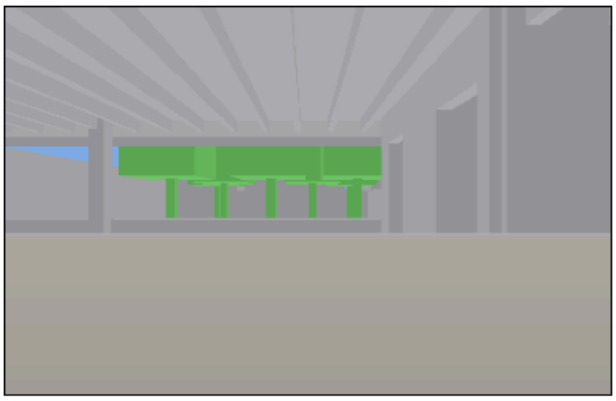
Window 2 — standing (1.6 m)
green 0% · nearest obstacle: 2.9 m · 3 layers: no



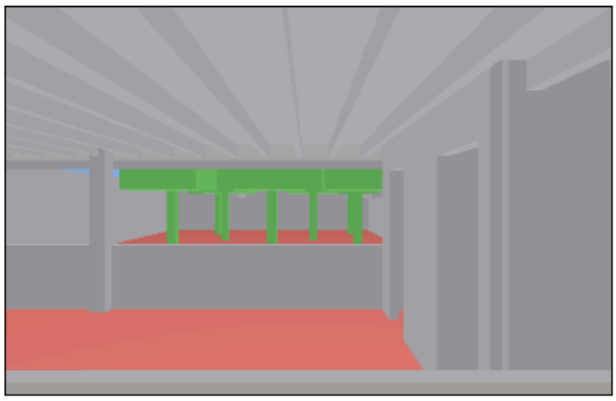
sky vegetation ground built (walls, roofs, neighbours)

View out of Kitchen

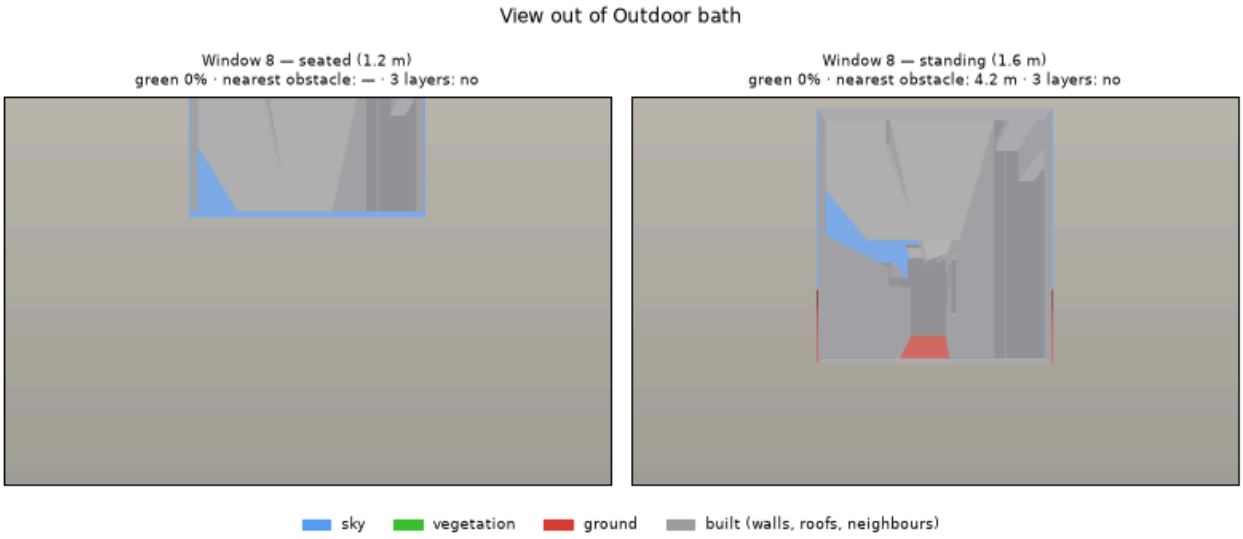
Window 5 — seated (1.2 m)
green 8% · nearest obstacle: 6.5 m · 3 layers: no

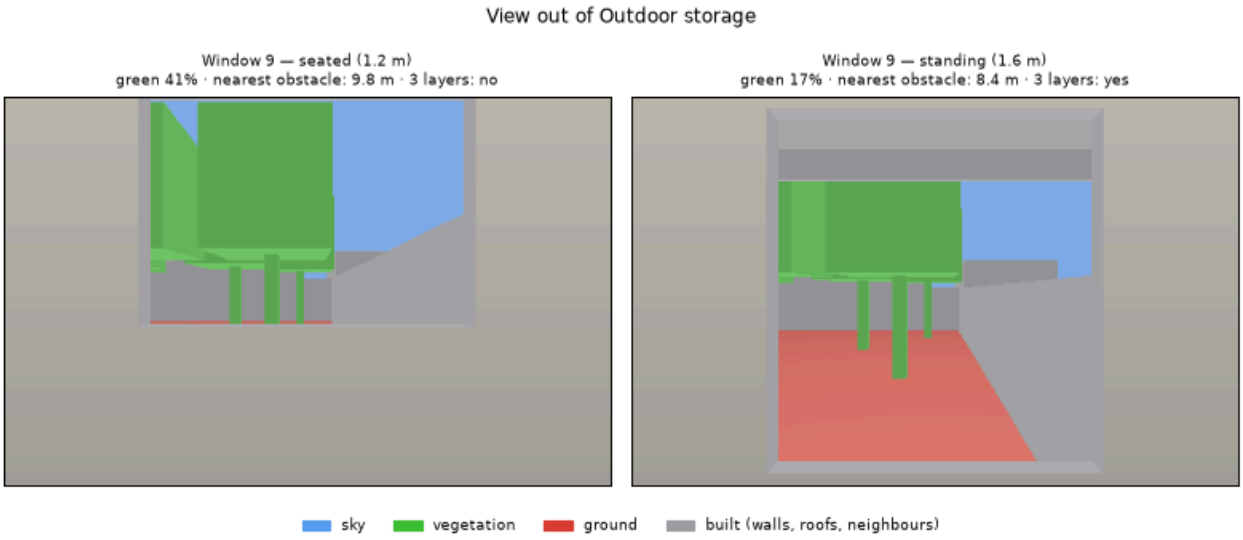


Window 5 — standing (1.6 m)
green 4% · nearest obstacle: 10.7 m · 3 layers: no



sky vegetation ground built (walls, roofs, neighbours)





9.4 Moisture and mould (screening)

Calculated with: **EnergyPlus + AirflowNetwork + OpenFOAM (CFD)** · zone humidity and inside surface temperatures; AirflowNetwork with the CFD pressures · details and reliability in section 10

Little direct sun and weak ventilation, in a humid climate, favour mould. EnergyPlus gives the humidity and the temperature of the coldest wall or ceiling of every room for the whole year; the surface relative humidity follows from them. The screening uses the simplified test of ASHRAE Standard 160: the 30-day running average of the surface relative humidity should stay below 80% while the 30-day average temperature is between 5 and 40 °C. ASHRAE addenda replace this flat limit by a mould-index model because it is conservative (check the current edition), and the WHO (2009) gives no numerical limit, so read it as a screening, not a diagnosis. Two cases are run: **open** = the house as it is, with the occupants opening the windows by the usual rule, and **closed** = a limit case with the windows shut the whole year, no air conditioning and the occupants inside, which is not how the house is used. Outdoors the air is above 80% for 2,058 hours of the year (mean 66%).

Room	Direct sun h/day	Cross-vent. %	Hours air above 80%	Highest 30-day surface RH, open / closed	Year above the limit, open / closed	Reading
Bedroom 1	0.64	36	611	71% / 91%	0.0% / 79.4%	only in the closed-up limit case
Bathroom 1 (service)	0.05	0	890	74% / 74%	0.0% / 0.0%	not assessable: no moisture source in the model · little sun
Living room	0.00	100	559	72% / 88%	0.0% / 69.5%	only in the closed-up limit case · little sun
TV room	0.27	77	633	74% / 94%	0.0% / 89.4%	only in the closed-up limit case · little sun
Hallway (service)	0.01	76	531	72% / 82%	0.0% / 2.2%	not assessable: no moisture source in the model · little sun
Bedroom 2	0.51	36	536	70% / 93%	0.0% / 100.0%	only in the closed-up limit case
Bathroom 2 (service)	0.06	36	586	72% / 70%	0.0% / 0.0%	not assessable: no moisture source in the model · little sun
Bedroom 3	0.42	36	511	68% / 97%	0.0% / 100.0%	only in the closed-up limit case
Kitchen	0.01	77	722	75% / 81%	0.0% / 2.0%	not assessable: no moisture source in the model · little sun
Outdoor bath (service)	0.31	0	743	73% / 73%	0.0% / 0.0%	not assessable: no moisture source in the model
Outdoor storage (service)	0.08	0	703	71% / 71%	0.0% / 0.0%	not assessable: no moisture source in the model · little sun

Rooms to watch: Bedroom 1, Living room, TV room, Bedroom 2, Bedroom 3

Limits: in the model people are the only source of moisture (bedrooms and living room); the kitchen and the bathrooms have none (no cooking, no showers), the walls do not store moisture and there is no rain or rising damp. So the table shows how the climate and the ventilation press on each room; a bathroom that is steamed up every day would be worse than shown. The surface temperature is that of the coldest wall or ceiling of the room, not of corners and behind furniture, where thermal bridges make it worse.

9.5 Suggestions, by effort

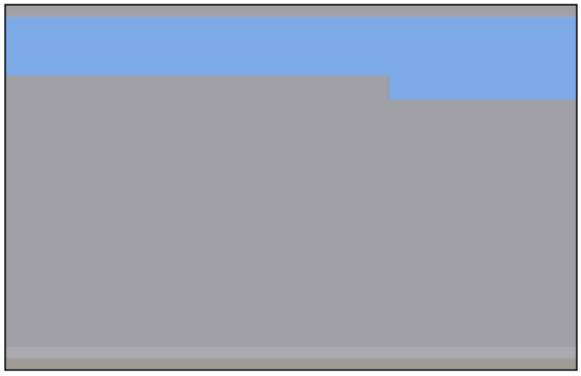
Each suggestion names the patterns it can raise. Only the ventilation changes (EnergyPlus, section 6.4) and the planted wall (measured on the views) carry numbers; the others are recommendations with no invented gain. Shade-tree scenarios (sun, comfort) are not simulated yet; trees only enter the views, never the base-case numbers.

Effort	Suggestion	Patterns	Where / measured
Low effort	Climbing plants or a planter wall on the boundary wall these windows face (the view today is a wall a few metres away).	A1, A3, A7	TV room: green in view 0% → 0% (up to 14%); Bedroom 2: green in view 0% → 0% (up to 39%); Bedroom 3: green in view 0% → 0% (up to 39%) · ray cast (idealised planted slab)
Low effort	Plant a deciduous shade tree on the west side and trees on the axis of the main windows: green in view, moving leaves and shadows, shade in summer. To be simulated as a scenario (shade and views).	A1, A3, A6, A7	Bedroom 1, Living room, TV room, Bedroom 2, Bedroom 3
Low effort	Pergola with a climber over a porch or the west-facing facade: filtered light that changes through the day.	A1, A3, A6	-
Low effort	A small water feature with sound (moving water), preferably on the east side so the morning light reflects indoors.	A5, A2	-
Low effort	Aromatic plants along the circulation (lemongrass, mint, basil) as sensory landmarks.	A2	-
Low effort	Real wood or stone on the surfaces people touch (handles, benches, counters) instead of synthetic laminates.	B2, A2	-
Low effort	Textiles, tiles or ornament inspired by nature (organic shapes, fractal patterns, colours of the landscape) to break flat monochrome surfaces.	B1, B3	-
Low effort	Mark large glass (glazed doors) so birds see it, and keep lights pointing down.	A1	-
Low effort	Do not keep these rooms closed for weeks: with the windows open the screening finds no risk, but a bedroom shut all year with its occupants goes over the limit for months. Keep a window ajar or fit a vent / extractor fan, and use an anti-mould paint on the coldest walls.	A4	Bedroom 1, Living room, TV room, Bedroom 2, Bedroom 3
Medium effort	Replace sliding windows by casement windows that open about 90% of the glazed area (the base case assumes sliding windows that open about half).	A4, A6	PHFT 57.9% (now 54.9%), degree-hours 3,198 (3,618), η 0.46 (0.39) · EnergyPlus (ventilation study)
Medium effort	Add a grille of about 0.2 m² per room in the ceiling and bigger eave openings so the hot air leaves through the attic.	A4	PHFT 56.2% (now 54.9%), degree-hours 3,390 (3,618), η 0.43 (0.39) · EnergyPlus (ventilation study)
Medium effort	Both together: wider casement windows plus ceiling exhaust.	A4	PHFT 58.8% (now 54.9%), degree-hours 3,034 (3,618), η 0.49 (0.39) · EnergyPlus (ventilation study)
Medium effort	Prefer neutral-tint glass: dark glass distorts the colours of the landscape. The model assumes a clear glass (transmittance 0.8, no colour), so this is advice only, not simulated.	A1	-
High effort	Larger window or glass door looking at the best landscape of the lot (a garden or trees), with the nearest obstacle at 6 m or more (restorative view: sky, vegetation and ground).	A1, C2	-
High effort	Light well, winter garden or sunken garden next to rooms with no direct daylight; best decided before building or in a renovation.	A1, A4, A6, C1	-

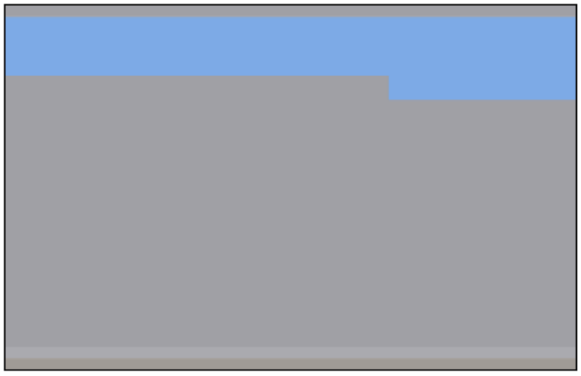
Planted-wall scenario: an idealised slab of foliage 0.4 m thick and 4 m wide is placed 0.2 m in front of the obstacle each window faces (windows whose nearest obstacle is under 4 m away and covers 60% or more of the opening). The figure and the first number use a slab 1.8 m high (a climbing wall as tall as the lowest boundary wall); the number after 'up to' is an upper bound with a slab 3.4 m high that covers the whole obstacle. It is measured only on the views, with the same ray cast; the base-case numbers do not change, and it is not a planting design.

Planted wall in front of the obstacle (idealised slab 0.4 × 4 × 1.8 m)

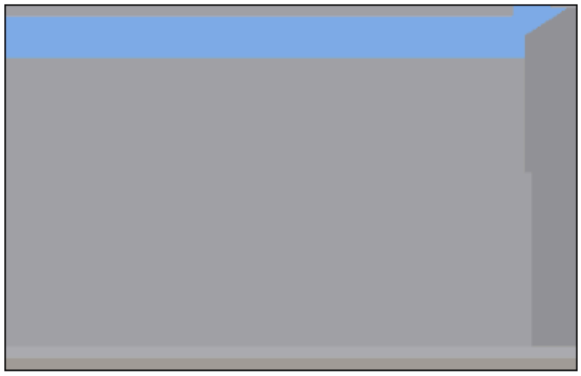
TV room · Window 7 — today
green 0%



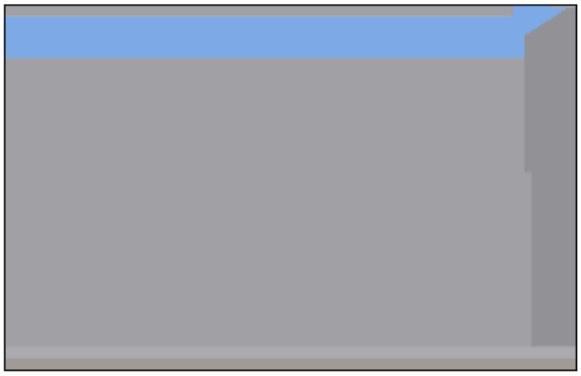
TV room · Window 7 — with a planted wall
green 0%



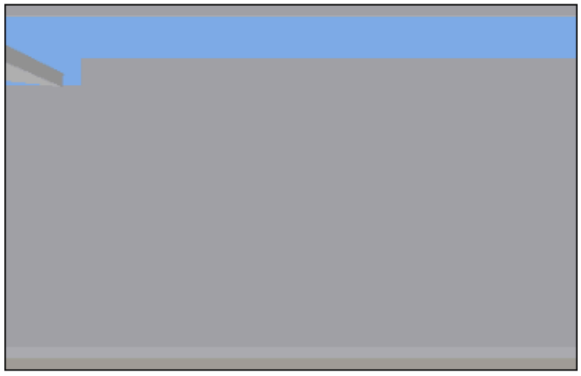
Bedroom 2 · Window 1 — today
green 0%



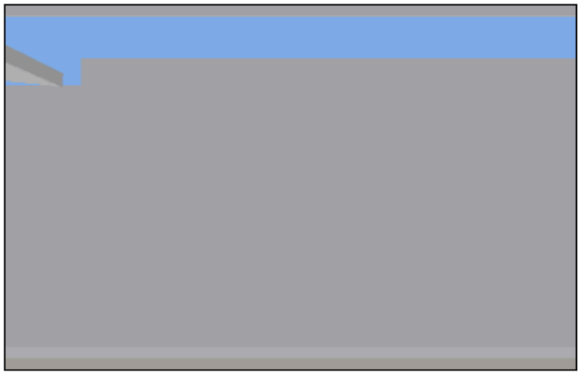
Bedroom 2 · Window 1 — with a planted wall
green 0%



Bedroom 3 · Window 2 — today
green 0%



Bedroom 3 · Window 2 — with a planted wall
green 0%



10. How this was calculated and how far to trust it

This section tells you how the numbers were produced, so you can judge how far to trust them. The fast engine is open-source Python code written for this project; it is cross-checked with three open-source reference engines: Radiance (daylight), EnergyPlus (heat, energy and NBR 15575) and OpenFOAM (wind pressure). The fast sub-models are standard in building physics but simplified. Nothing here has been measured on the real building.

> Validation status: the code is checked by regression tests (the base case must not change when a feature is added) and by sanity checks of sun geometry and energy balance; it has NOT been validated against measurements, and the reference engines were run on a simplified version of the model (see their limits). For a permit, certification or investment decision, confirm with a simulation by a qualified professional (e.g. EnergyPlus, Radiance).

Reference engines

RADSITE | radiance-online.org

Radiance 6.0.2
Daylight: sDA, ASE, LEED (LM-83)

EnergyPlus 26.2.0
Heat, energy and NBR 15575 (PHFT)

OpenFOAM®

OpenFOAM v2312 (Docker)
Wind CFD: pressure on the openings

Logos belong to their projects (Radiance: LBNL; EnergyPlus: US DOE / NREL; OpenFOAM: OpenCFD / ESI Group) and are shown only to credit the software used.

The engines in detail: parameters, formulas and references

Fast engine (Python, this project)

Used for: Sun and shadows, daylight (lux, lumens, sDA, ASE), window sizes and cross-ventilation hours, indoor temperature, comfort, air-conditioning / heating energy and cost, materials and upgrade scenarios (sections 2 to 7).

Parameters:

- 8,760 hourly steps of the typical weather year; sensor grid 0.5 m at 0.76 m height
- Daylight: luminous efficacy 100 lm/W (direct) and 115 lm/W (diffuse), mean reflectance 0.55, utilisation factor 0.45, maintenance factor 0.8
- Thermal: surface film resistances 0.04 (outside), 0.13 (wall), 0.17 (ceiling and floor) m^2K/W; outside film coefficient 17 W/m^2K; wall solar absorptance 0.45; glass U, g and VT and all layer properties from materials/catalog.json (NBR 15220-2 U, CT, thermal lag and solar factor; zone-dependent NBR 15575-4/-5 limits)
- Comfort: ASHRAE 55 adaptive band, 80% and 90% acceptability; running-mean outdoor temperature limited to 10-33.5 °C
- Cooling: set point and schedules per room type (project.json), COP and tariff as stated in section 5.3; windows closed while the equipment runs

Formulas:

- Lumens to install = lux × area ÷ (utilisation 0.45 × maintenance 0.8)
- Comfort temperature = 0.31 × T(outdoor running mean) + 17.8 °C; comfortable band = ± 3.5 °C (80%) or ± 2.5 °C (90%)
- Operative temperature = (air temperature + mean surface temperature) ÷ 2
- Electricity = heat removed ÷ COP; yearly cost = electricity × tariff
- Fixed brise: shadow depth = overhang × tan(profile angle); movable blinds: solar gain × 0.3 when the sun on the window exceeds 250 W/m^2
- sDA300/50% = share of the floor area with at least 300 lux for 50% of the occupied hours; ASE1000,250 = share of the area with 1,000 lux of direct sun for more than 250 hours

References:

- ASHRAE Standard 55: thermal environmental conditions for human occupancy (adaptive comfort model) - https://www.ashrae.org/technical-resources/standards-and-guidelines
- ABNT NBR 15575 (Residential buildings - performance) and NBR 15220 (Thermal performance of buildings) - https://www.abntcatalogo.com.br
- EN 17037: Daylight in buildings - https://standards.cencenelec.eu
- ANSI/IES LM-83-23: sDA and ASE - https://webstore.ansi.org/standards/iesna/ansiieslm8323
- Embree ray-tracing kernels (Intel) - https://github.com/embree/embree
- IfcOpenShell: open-source IFC toolkit - https://ifcopenshell.org
- pvlib python: a Python package for modeling solar energy systems (Holmgren, Hansen, Mikofski, JOSS 2018) - https://joss.theoj.org/papers/10.21105/joss.00884
- Numba: JIT compiler for Python - https://numba.pydata.org

EnergyPlus + AirflowNetwork

Used for: Cross-check of comfort and energy of the upgrade packages (section 7.4) and the whole NBR 15575-1:2021 simulation: PHFT, hottest and coldest operative temperature, thermal load, air changes through the openings (section 6.4).

Parameters:

- One thermal zone per room plus a ventilated attic zone; walls shared between rooms; conduction transfer functions for every layer
- Windows as simple glazing systems; real eaves as shading surfaces
- AirflowNetwork: every window, open door and crack is a flow path; wind pressure coefficients from the OpenFOAM CFD (24 directions), wind speed at the opening height from the power-law profile (exponent 0.22)
- NBR 15575 run: occupancy, lighting and equipment from the tables of the standard; windows open only while the room is occupied, the room air is at least 19 °C and warmer than outdoors; bathroom windows open
- Thermal-load run: windows closed, ideal air conditioner at 23 °C (cooling) and 21 °C (heating) while occupied
- Operative-temperature band 18-26 °C because the annual mean temperature of the weather file is below 25 °C
- Openings (SimpleOpening): discharge coefficient 0.6, opening factor = openable fraction of the window (0.5 sliding), closed-leak coefficient 0.00063 kg/s (windows) and 0.0024 kg/s (doors) at 1 Pa, exponent 0.63 / 0.59; attic eave openings 0.4% of the floor area (crack 0.0019 kg/s per m^2 of floor, exponent 0.65)
- Ventilation study (section 6.4): six full-year runs of the same model with floor-plan changes (doors closed, grille above the doors, casement windows 90% open, 0.2 m^2 ceiling grille per room with eave openings 5× larger, and the last two together) plus a run with the windows closed; the flow through each opening is read hour by hour (AFN Linkage Node Volume Flow Rate)
- Wind sectors of the flow arrows: eight sectors of 45° centred on N, NE, ... NW, using the wind direction of the weather file; hot hours = windows open and mean operative temperature of the bedrooms and living room above 26 °C
- Moisture and mould screening (section 9.4): two full-year free-running runs (windows opened by the occupants, and windows closed), hourly relative humidity and temperature of every zone air and the inside face temperature of its opaque walls and ceiling; the only moisture source is the occupants (bedrooms and living room)

Formulas:

- PHFT = occupied hours with 18 °C ≤ operative temperature ≤ 26 °C ÷ occupied hours × 100

- Minimum level: PHFT(real) > 0.9 × PHFT(reference); T(max,real) ≤ T(max,ref) + 2 °C; T(min,real) ≥ T(min,ref) − 1 °C
- ΔPHFT = PHFT(real) − PHFT(reference); the intermediate and superior levels need ΔPHFT ≥ 45 − 0.58 × PHFT(reference)
- Load reduction = (load of the reference − load of the real house) ÷ load of the reference
- Airflow through an opening (AirflowNetwork, large vertical opening): driven by the pressure difference ΔP = wind pressure ($C_p \times \frac{1}{2}\rho U^2$) ± stack pressure (ρ g Δh ΔT/T) between the two sides
- Air changes per hour (ACH) = air volume entering through the exterior openings in one hour ÷ room volume
- Flow through an opening: Q = Cd × A × opening factor × √(2 ΔP / ρ), with ΔP from the wind and stack pressures above; a crack follows Q = C × ΔP^n
- Heat carried out by ventilation = ρ cp Σ V(out) × (T(zone) − T(outdoor)), with ρ cp = 1.2 × 1005 J/(m³·K), summed over every exterior opening and ceiling-to-attic path, hour by hour (kWh per year)
- Exhaust effectiveness η = (T(closed) − T(ventilated)) ÷ (T(closed) − T(outdoor)), the operative temperatures of a room in the occupied hours when the closed house is above 26 °C and more than 1 °C above outdoors; 0 = ventilation does not cool, 1 = the room follows the outdoor air
- Degree-hours = Σ max(T(operative) − 26 °C, 0) over the occupied hours
- Surface relative humidity RH(s) = min[100, RH(air) × es(T air) ÷ es(T surface)], es(T) = 6.112 × exp(17.62 T ÷ (243.12 + T)) hPa (Magnus); mould screening: 30-day running average of RH(s) below 80% while the 30-day average of T(s) is between 5 and 40 °C (ASHRAE 160 simplified test)

References:

- EnergyPlus (US Department of Energy, NREL) - https://energyplus.net
- ANSI/ASHRAE Standard 140: Standard method of test for the evaluation of building energy analysis computer programs - https://www.ashrae.org/technical-resources/bookstore/standard-140
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- ABNT NBR 15575-1:2021 section 11.4 (thermal performance by simulation) and its public amendment draft (LabEEE, March 2021) - https://labeee.ufsc.br/sites/default/files/documents/P_ABNTNBR15575_1_2020CNGPR_PosCN_SiteLabEEE.pdf
- ANSI/ASHRAE Standard 160-2009: Criteria for moisture-control design analysis in buildings (30-day running average of the surface relative humidity below 80% at 5-40 C; later addenda propose a mould index) - https://www.ashrae.org/technical-resources/standards-and-guidelines
- WHO guidelines for indoor air quality: dampness and mould (2009) - https://www.who.int/publications/i/item/WHO-EURO-2009-4256-44019-62079-eng

Radiance

Used for: Daylight cross-check of the fast engine: sDA300/50%, ASE1000,250, mean lux and the LEED daylight points (section 8.1).

Parameters:

- Two-phase daylight coefficients (rfluxmtx), 4 light bounces
- Perez all-weather sky built from the weather file (gendaymtx), 2,305 sky patches, 8,760 hours
- Blind rule per room: blinds closed (5% transmittance) when more than 2% of the room gets 1,000 lux of direct sun
- Glass transmittance 0.80 for the whole opening; no furniture, frames, neighbours or trees

Formulas:

- Illuminance at a sensor = Σ (daylight coefficient × sky patch luminance), summed over the 2,305 patches for every hour
- sDA and ASE as defined by IES LM-83 (same definitions as the fast engine)

References:

- Radiance Synthetic Imaging System (Lawrence Berkeley National Laboratory) - https://www.radiance-online.org
- CIE 171:2006 Test cases to assess the accuracy of lighting computer programs - https://cie.co.at/publications/test-cases-assess-accuracy-lighting-computer-programs
- Perez et al. (1993): all-weather model for sky luminance distribution - https://doi.org/10.1016/0038-092X(93)90017-I
- ANSI/IES LM-83-23: sDA and ASE - https://webstore.ansi.org/standards/iesna/ansiieslm8323

OpenFOAM (CFD)

Used for: Wind pressure coefficients (Cp) in front of every window, door and eave opening, for 24 wind directions; they feed the EnergyPlus AirflowNetwork (section 6.4 and 6.5).

Parameters:

- Steady incompressible RANS (simpleFoam), standard k-epsilon turbulence model, wall functions
- Atmospheric boundary-layer inlet: 10 m/s at 10 m, roughness length z0 = 0.3 m (suburban)
- Domain 6H upstream, 15H downstream, 5H to each side, 6H above (H = building height); snappyHexMesh, 0.15 m cells on the house, coarser around the lot walls and neighbours
- Surroundings: lot walls and neighbouring buildings as closed boxes with heights estimated from photographs
- Convergence: residuals p 1e-5, U 1e-6, k and epsilon 1e-5; pressure is the mean of the last 20 iterations

Formulas:

- Cp = p ÷ (½ ρ $U_{EP}(z)^2$), with $U_{EP}(z) = U10 \times (z \div 10)^{0.22}$ the wind profile EnergyPlus uses
- Cp(EnergyPlus) = Cp(CFD) × $(U_{CFD}(z) \div U_{EP}(z))^2$, the correction for the different wind profiles (factor 0.39 to 0.79 with height)
- Reynolds-averaged momentum: $\nabla\cdot(U\,U) = -\nabla p/\rho + \nabla\cdot[(\nu + \nu_t)(\nabla U + \nabla U^T)]$, with $\nu_t = C\mu\, k^2 \div \varepsilon$ and Cμ = 0.09

References:

- OpenFOAM v2312 (OpenCFD / ESI Group), computational wind engineering with RANS k-epsilon - https://www.openfoam.com
- EN 1991-1-4: Eurocode 1, wind actions (table 7.1, pressure coefficients of vertical walls of rectangular buildings) - https://eurocodes.jrc.ec.europa.eu
- EnergyPlus AirflowNetwork and surface-averaged wind pressure coefficients of Swami and Chandra (1987), EnergyPlus Engineering Reference - https://energyplus.net/documentation

Biophilia analysis (ray cast and plan isovist, this project)

Used for: Views out of each room (sky, vegetation, ground, distance), the Biophilic Design Matrix of 14 patterns, the Biophilic Healing Index and the suggestions (section 9 and the Biophilia tab).

Parameters:

- Eye heights 1.2 m (seated) and 1.6 m (standing), 1.0 m behind the window plane; field of view 84° × 56°, 400 × 256 rays per view; the opening is the glass rectangle of the window
- Classes: sky (no hit above the horizon), vegetation (trees of the IFC, boxes for trunk and crown), ground (plane at z = 0), built (walls, roofs, neighbours); trees are NOT in the base-case engines, only here

- Three-layer test: sky, vegetation (the middle layer; a wall, a street or neighbouring houses are not nature) and ground each at least 5% of the opening; green view = vegetation at least 15% of the opening
- Focal distance = median distance to the first obstacle (built or vegetation) in the horizontal band of ±5° of the opening, measured from the eye (1 m behind the window); compared with 6 m. One storey with the floor at z = 0.1 m is assumed
- Plan isovist: 72 directions at 1.4 m from a 0.6 m grid inside each room; refuge = a point with at least 60% of the directions blocked within 2 m and a sight line of 6 m or more
- Matrix column from the counted rooms (occupied and not service): none = not included; a weak form only = only in detail; up to half = in a few areas; more than half = house; low-confidence proxies are capped at in a few areas
- Healing index: ten components 0, 1 or 2; sunlight from the direct-sun hours on the windows of the counted rooms (0: no room gets 0.05 h/day; 1: some do; 2: at least half get 0.5 h/day or more from two or more compass quadrants), life from the trees and the green views, curves from the curved elements of the IFC; the others come from the owner (project.json, biophilia)
- Planted-wall scenario: an idealised slab of foliage 0.4 m thick and 4 m wide (1.8 m high; 3.4 m high as an upper bound), 0.2 m in front of the obstacle of each window whose nearest obstacle is under 4 m away and covers 60% or more of the opening; measured only on the views, never in the base-case engines

Formulas:

- BHI = Light + Color + Gravity + Fractals + Curves + Detail + Water + Life + Representations of nature + Organized complexity, each 0 (none), 1 (some) or 2 (a large amount); $0 \le BHI \le 20$ (Salingaros 2019)
- Opening share of a class = rays of the class that cross the window rectangle ÷ rays that cross it
- Ray through the opening: eye + $t \cdot d$ with t = distance behind the window ÷ $(d \cdot n)$ and the crossing point inside the glass rectangle
- Isovist area $\approx \frac{1}{2} \Sigma r^2 \Delta\theta$ over the 72 sight lines r ($\Delta\theta = 5°$)

References:

- Embree ray-tracing kernels (Intel) - https://github.com/embree/embree
- Browning, Ryan and Clancy (2014): 14 Patterns of Biophilic Design, Terrapin Bright Green - https://www.terrapinbrightgreen.com/reports/14-patterns/
- Salingaros (2019): The biophilic healing index predicts effects of the built environment on our wellbeing, Journal of Biourbanism 8 - https://www.biourbanism.org/the-biophilic-healing-index-predicts-effects-of-the-built-environment-on-our-wellbeing/
- Salingaros: Neuroscience experiments to verify the geometry of healing environments, proposing a Biophilic Healing Index (book chapter, Routledge) - https://www.routledge.com
- UGREEN: Medimos a Biofilia Dessa Mansão Antes da Obra (video; source of the seven-column rubric and the three-layer view rule) - https://www.youtube.com/watch?v=jMehPp3SbCo
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- Lavdas and Salingaros (2022): Architectural beauty: developing a measurable and objective scale, Challenges 13(2), 56 - https://doi.org/10.3390/challe13020056

Software used

Python 3.11.9 · NumPy 2.4.6 · pandas 3.0.6 · pvlib 0.16.1 · trimesh 5.1.1 · Embree (embreex) 4.4.0 · IfcOpenShell 0.9.0 · Shapely 2.2.0 · Numba 0.68.0 · Matplotlib 3.11.2 · SciPy 1.17.1 · EnergyPlus 26.2.0 · Radiance 6.0.2 · OpenFOAM v2312 (Docker)

Calculation steps and how reliable each is

Step	How it is calculated	Reliability
3D model and orientation	The building is read from an IFC file (IfcOpenShell). Rooms are the IfcSpace volumes; walls, slabs, roofs, windows and doors become triangle meshes. The model is rotated so that true north comes from the IFC True North (plus an optional correction). Why: Exact as far as the model is exact: sizes, window positions and materials are estimates unless the model comes from a survey.	High
Sun position and shadows	Solar position for each of the 8,760 hours with pvlib (NREL SPA-based algorithm). From a grid of sensor points (0.5 m, at 0.76 m height) in every room a ray is traced towards the sun against all opaque elements with Embree (a ray-tracing library): blocked = shadow; a ray through a window is attenuated by the window transmittance. Why: Sun geometry is deterministic; the result is only as good as the geometry (neighbours and trees count only if they are in the model).	High
Weather	A typical meteorological year of hourly data (EnergyPlus .epw such as TMYx from climate.onebuilding.org, or a typical year from PVGIS): temperature, humidity, direct/diffuse/global radiation, wind speed and direction. Why: A typical year is not a forecast: real years are hotter or colder. Wind data are for open terrain at 10 m.	Medium
Daylight (lux and lumens)	Illuminance on the work plane = direct sun (ray traced) + sky light (isotropic sky, sky-view factor by ray tracing) + an inter-reflected component (split-flux method, mean reflectance 0.55). Radiation is converted to lux with luminous efficacies of 100 lm/W (direct) and 115 lm/W (diffuse). Lamp lumens use the lumen method: lux x area / (utilisation 0.45 x maintenance 0.8). Metrics: NBR 15575 centre-of-room lux, sDA300/50% and ASE1000,250 (IES LM-83), EN 17037 sunlight hours. Why: The isotropic sky ignores the brighter sky near the sun and split-flux overestimates light deep in the rooms: the Radiance cross-check gives an area-weighted sDA of 6% against 41% here. Use it only for quick comparisons.	Medium
Reference engine for daylight: Radiance	The same rooms and sensor points are recalculated with Radiance (Lawrence Berkeley National Laboratory), a backward ray tracer validated against the CIE 171:2006 test cases and the engine behind ClimateStudio and Honeybee. Method: two-phase daylight coefficients (rfluxmtx, 4 light bounces) multiplied by a Perez all-weather sky built from the weather file (gendaymtx, 2,305 patches) for all 8,760 hours; sDA300/50% and ASE1000,250 as in IES LM-83, with the blind rule approximated per room (blinds closed, 5% transmittance, when more than 2% of the room gets 1,000 lux of direct sun). A sensor above the roof checks the sky: it reproduces the weather file's illuminance within about 2%. Why: Validated engine; what is still missing in the run is furniture, frames, the glass-door lattice and trees (the neighbours' boxes are included), and the blind rule is simplified. Prefer these numbers over the fast daylight engine.	High
Reference engine for heat and energy: EnergyPlus	The retrofit packages are recalculated with EnergyPlus (US Department of Energy), the heat-balance engine validated by ASHRAE Standard 140 and used inside DesignBuilder and OpenStudio. One zone per room plus a ventilated attic zone above it; walls shared between rooms; conduction transfer functions for every layer; windows as simple glazing systems; the real eaves as shading surfaces; window ventilation by wind and buoyancy (windows 50% open when the outside air is cooler and above 18 C); an ideal-loads air conditioner with the same set point, schedules and COP as the fast engine. Comfort comes from a free-floating run (ASHRAE 55 adaptive), energy from a second run with the air conditioner. Why: Validated engine, but the model is simplified: flat roof with the same attic volume, no internal gains, no door opening, no neighbours' shade, window glass properties assumed, not calibrated with measurements. Compare options; do not treat the bill as a forecast.	Medium
NBR 15575-1:2021 by simulation and cross ventilation (EnergyPlus + AirflowNetwork)	A real model and a reference model of the house are simulated for the 8,760 hours with EnergyPlus. Natural ventilation is solved by the AirflowNetwork: windows, open doors and cracks are flow paths driven by wind pressure (Swami and Chandra facade coefficients for a low-rise building, local wind speed at the opening height), stack effect and opening height; windows open only while the room is occupied, when the room is at least 19 C and warmer than outdoors; bathroom windows stay open. Occupancy, lighting and equipment follow the standard's tables; a second run with windows closed and an ideal air conditioner (23 C cooling / 21 C heating while occupied) gives the thermal load. PHFT, hottest and coldest operative temperature and load reduction give the minimum, intermediate and superior levels. Why: Follows section 11.4 of NBR 15575-1:2021 (full text, tables checked against the public amendment draft); the bioclimatic zone of Maringa in NBR 15220-3:2024 comes from a non-official article; our assumptions are listed in the report (transient rooms, open doors, ground temperature). The wind pressure coefficients come from the OpenFOAM CFD of this house with its lot walls and neighbours (heights estimated), not from a wind-tunnel test. Not an official compliance report.	Medium
Air circulation, floor-plan changes and hot-air exhaust (EnergyPlus + AirflowNetwork)	The same airflow network is run for the whole year with several floor-plan changes (doors closed, grille above the doors, casement windows, ceiling exhaust). The flow through each window, door, grille and ceiling opening is read hour by hour and averaged for the hours with open windows, the hot hours and each wind direction, which gives the arrows on the plan. Exhaust effectiveness compares the operative temperature of each room with a run with the windows closed and with the outdoor temperature; the heat carried out by the air is rho cp V (T zone - T outdoor). Why: The flows are averages between rooms (one air node per room, no velocity field and no air-age or stagnant-zone calculation); the wind pressure comes from the CFD of the outside only, with closed openings, and the grille, ceiling opening and casement fraction are idealised sizes, not products. Good for ranking plan changes (differences between variants are more reliable than absolute flows); not a replacement for an interior CFD or a smoke test.	Medium
Wind pressure on the facades: OpenFOAM (CFD)	For each of 24 wind directions (every 15 degrees) the house (a closed envelope rebuilt from the IFC roofs, eaves and porches) is placed in a computational domain (6H upstream, 15H downstream, 5H to the sides, 6H above, H = building height) and meshed with snappyHexMesh (1.2 to 2.0 million cells, 0.15 m on the house, coarser around the lot walls and neighbours). OpenFOAM v2312 solves the steady Reynolds-averaged Navier-Stokes equations with the standard k-epsilon turbulence model, an atmospheric-boundary-layer inlet (10 m/s at 10 m, roughness length 0.3 m) and rough-ground wall functions. The lot walls and the neighbouring buildings are included as closed boxes (heights estimated from photographs). The pressure 0.3 m in front of every window, door and eave opening, converted to a coefficient of the dynamic pressure of the EnergyPlus wind profile (power law, suburban terrain) at that height, gives the wind pressure coefficients handed to the EnergyPlus AirflowNetwork. A numerical verification (three meshes, roughness, domain size, a box compared with the Eurocode) gives the uncertainty of each coefficient. It runs in a Docker container. Why: RANS k-epsilon is the usual engineering model for mean wind loads on buildings and OpenFOAM is validated for it, but this run uses the real roof shape, lot walls and neighbours whose heights are estimates from photographs, no trees, a single approach roughness and a mesh of moderate size; no wind-tunnel check was made.	Medium
Indoor temperature (thermal model)	Each room is a lumped heat-balance model solved hour by hour (Python, compiled with Numba): wall, ceiling and floor layers as heat-storage nodes (resistances and capacities from the layer conductivity, density, specific heat and thickness, ISO 6946 / NBR 15220-2 style), sol-air temperature on the outside surfaces, a ventilated attic, windows (U-value, solar gain), ventilation (infiltration when closed, cross-ventilation by wind when open, rooms linked through doors) and thermal mass. Operative temperature = mean of air and surface temperature. Comfort uses the ASHRAE 55 adaptive model (80% and 90% acceptability) and the NBR 15575 hottest/coldest-day checks. Why: A simplified one-zone-per-room model, not calibrated with measurements. It is cross-checked against EnergyPlus for the retrofit packages (see the EnergyPlus step): the engines agree on the ranking of measures but EnergyPlus gives about half the air-conditioning energy. Use it for quick comparisons, not as an exact forecast.	Medium
Air-conditioning / heating energy and cost	When the room is above the cooling set point (or below the heating set point) during its schedule, the model adds the heat flow needed to hold the set point; electricity = heat / COP (cooling) or fuel = heat / efficiency (heating), cost = energy x tariff. Windows are closed while the equipment runs. Sensible heat only; no people, appliances or lighting heat. Why: Real bills differ: latent load, occupants' habits, equipment efficiency and the tariff (an assumption) all matter. Treat the cost as an order of magnitude and as a way to compare options.	Low
Wind and cross-ventilation	Wind roses from the weather file; each window is classified windward / leeward / side-on for each hour; rooms whose connected windows face both the wind and the lee are given more air changes. No pressure calculation (the CFD and the AirflowNetwork are used only in section 6.4). Why: A rule-based estimate: wind at the house is lower and more turbulent than at the weather station; no pressure-driven flow calculation.	Low
Upgrade scenarios	Each upgrade changes the base model and the whole year is simulated again: roof insulation adds a conduction layer (thickness / conductivity) on the attic side of the ceiling; draught sealing lowers the closed-house air change rate; glazing changes the window U-value, solar gain factor and visible transmittance; a fixed brise cuts direct sun by the shadow depth = overhang x tan(profile angle); movable blinds multiply the solar gain by 0.3 whenever the sun on the window is above 250 W/m2 during the day. Why: Good for ranking options (differences between scenarios are more reliable than absolute values). Material properties come from one catalogue (materials/catalog.json) with a source and a confidence for every value, shown next to the value; U-value, heat capacity, thermal lag and solar factor of each construction are computed with the NBR 15220-2 formulas, and the wall and roof limits depend on the bioclimatic zone (NBR 15575-4/-5). Shading is analytic, not ray traced.	Medium
Ratings and LEED pre-assessment	Numbers are compared with ABNT norms where they exist (NBR 15575, NBR 15220, NBR 5413, NBR 16401), international standards otherwise (EN 17037, IES LM-83, ASHRAE 55, WHO) and a few rules of thumb that are labelled as such. The LEED v4.1 daylight points follow Option 1 (sDA300/50% and ASE1000,250). Why: Several norm values were read from public summaries or amendment drafts, not the full ABNT texts; the LEED result is an indicative pre-assessment, not a certification.	Medium
Views out of every room (ray cast)	From an eye inside each room (seated 1.2 m and standing 1.6 m, 1 m behind the window, looking straight out) one ray per pixel is traced through the window opening and classified by what it meets first: sky, vegetation (the trees of the IFC), ground or built (walls, roofs, neighbours). The shares of the opening, the distance to the nearest obstacle in the horizontal band (median) and a test of three layers (sky, middle ground, ground, each at least 5% of the	Medium

	opening) are reported, and the coloured renders are drawn from the same classes. A plan isovist (72 sight lines at 1.4 m from a 0.6 m grid of the room) gives the longest sight line and the number of separate groups of long views. Why: Geometry is as good as the model: the trees are boxes at positions estimated from a street photo, neighbours and walls have estimated heights, the glass is treated as fully transparent and the view is one fixed direction. The 6 m distance and the three-layer test are rules of this tool taken from a video summary, not a published standard.	
Moisture and mould screening (EnergyPlus)	Two full-year free-running EnergyPlus runs of the real house (windows opened by the occupants, and windows closed) give, for every room, the air relative humidity, the air temperature and the inside surface temperature of its opaque walls and ceiling. The surface relative humidity is RH(air) x es(T air) / es(T surface) at the coldest surface (es by the Magnus formula). The simplified test of ASHRAE Standard 160 is applied: the 30-day running average of the surface relative humidity must stay below 80% while the 30-day average temperature is between 5 and 40 C; the share of the year above it and the highest 30-day average are reported next to the direct sun and the cross-ventilation of the room. Why: A screening, not a diagnosis. In the model people are the only moisture source (bedrooms and living room), the kitchen and bathrooms have none, walls store no moisture, there is no rain or rising damp, and the surface temperature is that of the coldest flat wall or ceiling, not of corners and thermal bridges. ASHRAE addenda replace the flat 80% limit by a mould-index model because it is conservative (check the current edition), and the WHO (2009) gives no numerical limit. The 'closed' case is a limit case (windows shut all year, no air conditioning), not how the house is used; the 30-day average wraps around the year.	Low
Biophilia: design matrix and healing index	The 14 patterns of biophilic design (Browning, Ryan and Clancy 2014) are placed in seven columns (not applicable, not included, only in detail, in a few areas, house, house + 1 building, all buildings), a rubric taken from the UGREEN video. A column is chosen from the share of the occupied, non-service rooms that meet a strong criterion (more than half = house; up to half = in a few areas); a weak form alone gives only in detail. Patterns that no engine can measure (water, non-visual connection, material, complexity, risk) are not evaluated unless the owner declares them. Salingaros' Biophilic Healing Index sums ten components rated 0, 1 or 2 (0 to 20); only sunlight, life and curves are estimated from the model, the others need an observer's rating. No aggregate of the matrix is invented. Why: Indicative only. The thresholds are this tool's own; the matrix has no official score; the Salingaros index is an observer's estimate with equal weights and the author calls its link with healing a conjecture, not verified. Patterns marked as proxies (non-rhythmic stimuli, natural systems, mystery) use weak indicators and are capped below 'house'. Not clinically validated; potted plants do not purify the air in real rooms.	Low

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- PVGIS typical meteorological year (European Commission JRC) - https://re.jrc.ec.europa.eu/pvg_tools/en/
- Climate.OneBuilding.Org: TMYx weather files - https://climate.onebuilding.org
- Embree ray-tracing kernels (Intel) - https://github.com/embree/embree
- IfcOpenShell: open-source IFC toolkit - https://ifcopenshell.org
- Numba: JIT compiler for Python - https://numba.pydata.org
- ASHRAE Standard 55: thermal environmental conditions for human occupancy (adaptive comfort model) - https://www.ashrae.org/technical-resources/standards-and-guidelines
- ABNT NBR 15575 (Residential buildings - performance) and NBR 15220 (Thermal performance of buildings) - https://www.abntcatalogo.com.br
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- DATec 14 (Saint-Gobain, light steel frame, 2013) and DATec 020-E (Tecverde, wood frame): SiNAT technical assessments of PBQP-H (layers, aged absorptances, wool conductivity; ministry site) - https://www.gov.br/cidades
- Oliveira and Alves (2019 workshop paper; 2021 article, PARC Pesquisa em Arquitetura e Construcao): cast-in-place concrete walls in Uberlandia - https://doi.org/10.20396/parc.v12i00.8661094
- Bezerra et al. (UPE, ENTAC): light steel frame versus masonry mini-walls, OSB and cement board properties (conference site) - https://eventos.antac.org.br/index.php/entac
- Oliveira, Aquino, Siqueira and Avila, Revista DELOS v.19 n.82 (2026): BIM and EnergyPlus comparison of masonry and light steel frame with and without insulation - https://doi.org/10.55905/rdelosv19.n82-051

11. Assumptions, limits and sources

- **Dimensions** come from the 3D model supplied: the approved municipal drawings (2008 house 1:50, 2012 back extension 1:75, Prefeitura de Maringá); the house as built may differ in details and this is not a survey.
- **Materials**: the standard Brazilian 8-hole ceramic block with plaster (not confirmed) and a tiled roof over a slab ceiling (the drawings say concrete tiles, "telha de cimento", over a slab, "forro laje"; the model uses the clay-tile values, which are thermally close). Neighbouring buildings, street trees and curtains are not modelled unless they are in the model.
- **Doors** are assumed open for light and air. Furniture, people, appliances and lighting heat are not included (empty house).
- The comfort model is a simplified hourly heat balance, not calibrated against measurements: use it to compare rooms and options. EnergyPlus would be the next level of rigour.
- **Wind** data are for open terrain at 10 m; inside a plot with walls and neighbours the speeds are lower. The fast engine's wind rule ignores this; the pressure-based ventilation of section 6.4 includes the walls and neighbours through the CFD, with estimated heights.
- **Daylight** uses an isotropic sky; real partly-cloudy skies are brighter near the sun.
- Rating scales marked *Rule of thumb* are ours; the ones marked norm come from the documents listed below. Several norm values were read from public summaries and amendment drafts: confirm them in the full ABNT texts before using them in a legal, contractual or financing decision.

Sources

- ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements
- ABNT NBR 15575-1:2021 + amendment (thermal performance by simulation vs. a reference house)
- ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas
- ABNT NBR 15575-5:2021 amendment draft: simplified method for roofs
- ABNT NBR 15220-3: Brazilian bioclimatic zoning and building guidelines (zones and limits depend on the bioclimatic zone of the project)
- ABNT NBR 5413:1992 (withdrawn in 2013 but still the usual Brazilian reference for HOME lighting levels)
- ABNT NBR ISO/CIE 8995-1:2013 (workplace lighting; reading/writing tasks 300-500 lx)
- EN 17037:2018 Daylight in buildings (UK National Annex dwelling targets 100/150/200 lx; sunlight 1.5/3/4 h)
- IES LM-83-12 (sDA and ASE), as used by LEED v4.1
- ASHRAE 55 adaptive comfort model (80% / 90% acceptability)
- ABNT NBR 16401-2 (air conditioning, indoor comfort): summer zone about 22.5-25.5 C at 65% RH and 23-26 C at 35% RH
- Passive House Institute criterion: heating demand of at most 15 kWh per m2 per year
- USGBC LEED v4.1, Daylight credit, Option 1 (simulation): sDA300/50% average of 40% / 55% / 75% = 1 / 2 / 3 points; ASE1000,250 above 10% needs a glare strategy (thresholds read on the LEEDuser v4.1 credit page)
- ANSI/IES LM-83-23 (published Feb 2023): spatial daylight autonomy (sDA) and annual sunlight exposure (ASE); LEED v4.1 still cites the 2012 edition, same two metrics
- WHO Housing and Health Guidelines 2018 (indoor temperature at least 18 C)
- WMO Beaufort wind scale
- Rule of thumb of this study (no norm gives a number)

Run it again after changing the model

Edit the 3D model (house.ifc), then double-click **run_all.bat** in the House3D folder. It redoes the simulation, this PDF and the data for the 3D viewer (viewer / open_viewer.bat).